

Malaysian Consensus Guidelines on

Antiretroviral Therapy

2026



MINISTRY OF HEALTH MALAYSIA

TABLE OF CONTENTS

Acknowledgement	4
Editorial committee	5
Contributors	5
Reviewers	6
List of abbreviations	7
Chapter 1 Introduction	9
1.1 What is new in this guideline?	
Chapter 2 Assessment of Adults with HIV Infection	11
2.1 Initial assessment	
2.2 Monitoring patients on antiretroviral therapy (ART)	
Chapter 3 Principles of antiretroviral therapy (ART) initiation	18
3.1 Counselling, linkage and retention in care	
3.2 Rapid initiation of antiretroviral therapy (ART)	
Chapter 4 Starting antiretroviral therapy (ART)	27
4.1 Principles of selecting ART for first-line regimens	
4.2 Preferred and alternative options for first-line ART	
4.3 Considerations prior to starting treatment	
4.4 Timing of ART initiation in opportunistic infections	
4.5 Primary prophylaxis	
4.6 Vaccinations	
Chapter 5 Monitoring while on antiretroviral therapy (ART)	41
5.1 Adverse events of ART	
5.2 Dose adjustment of ART for impaired renal and liver functions	
5.3 Common ART drug interactions	
5.4 Dolutegravir (DTG)-related monitoring	
Chapter 6 Management of treatment failure	44
6.1 Approach to the patient with detectable viral load	
6.2 Risk factors for treatment failure	
6.3 Assessment of treatment failure	
6.4 Viral resistance testing	
6.5 Second-line ART regimen	
6.6 Treatment-experienced patients with limited or no therapeutic option	
Chapter 7 Managing HIV with co-infections	51
7.1 HIV and TB co-infection	
7.2 HIV and hepatitis B virus (HBV) co-infection	
7.3 HIV and hepatitis C virus (HCV) co-infection	

TABLE OF CONTENTS [con't]

Chapter 8	Prevention and management of co-morbidities in PLWH	63
	8.1 Cardiovascular disease	
	8.2 Dyslipidaemia	
	8.3 Diabetes mellitus	
	8.4 Hypertension	
	8.5 Metabolic dysfunction-associated steatotic liver disease (MASLD)	
	8.6 Obesity	
	8.7 Chronic kidney disease	
	8.8 Osteoporosis / Osteopaenia	
	8.9 Cancer screening	
	8.10 Mental health and psychosocial wellbeing	
Chapter 9	Managing special populations living with HIV	104
	9.1 Prevention of mother-to-child transmission	
	9.2 HIV treatment among adolescents	
	9.3 Women's health in HIV	
	9.4 HIV among transgender populations	
	9.5 Ageing with HIV	
	9.6 HIV treatment among PLWH who use substances	
Chapter 10	HIV prevention	136
	10.1 Pre-exposure prophylaxis (PrEP)	
	10.2 Occupational post-exposure prophylaxis (PEP)	
	10.3 Non-occupational post-exposure prophylaxis (nPEP)	

ANNEXES

Annexe	3.1	Initial ART counselling guide	147
Annexe	4.1	Special considerations when initiating ART	149
Annexe	4.2	Recommended vaccines for adults living with HIV	151
Annexe	5.1	Common and/or severe adverse events of ART	153
Annexe	5.2	ART dosing recommendations in adults with renal or hepatic impairment	166
Annexe	5.3	Common ART drug interactions	170
Annexe	8.1	Patient Health Questionnaire (PHQ)	199
Annexe	8.2	Generalized Anxiety Disorder Questionnaire (GAD)	201
Annexe	9.1	Intrapartum zidovudine prophylaxis	203
Annexe	9.2	ART and contraceptive drug-drug interactions	204
Annexe	9.3	Menopause Rating Scale (MRS)	206

Acknowledgement

The Malaysian Consensus Guidelines on Antiretroviral Therapy 2026 were developed through the collective efforts, expertise and commitment of healthcare professionals and stakeholders dedicated to improving the quality of HIV care in Malaysia.

The Malaysian Society for HIV Medicine (MASHM), gratefully acknowledges the contributions of the members of the guideline development committee, expert reviewers, clinicians, pharmacists and researchers who generously shared their knowledge, experience and insights throughout the guideline development process. Their collective efforts have helped ensure that the guidelines reflect current scientific evidence, advances in antiretroviral therapy and the evolving needs of people living with HIV in Malaysia.

Although Malaysia has made significant progress over the years in the management of HIV, challenges remain, particularly among individuals at risk who may be reluctant to seek HIV testing and treatment due to fear of stigma and discrimination. Strengthening and expanding HIV services, particularly within primary care settings, through accessible and confidential HIV testing and timely linkage to treatment is essential. These revised guidelines are intended to assist health care providers in playing a crucial role in the treatment and prevention of HIV infection.

We would also like to extend our appreciation to the Ministry of Health Malaysia and its various programmes for their continued support of the national HIV response and for their invaluable contributions to the advancement of HIV prevention, treatment and care.

Finally, we sincerely thank all contributors for their time, dedication and commitment to making this revision of guidelines a success. It is our sincere hope that these revised guidelines will be beneficial to all health care providers who are involved in the management of HIV in Malaysia.

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ABBREVIATIONS

1HP	1-month daily regimen of isoniazid plus rifapentine
2DR	two-drug regimen
3HP	3-month once-weekly regimen of isoniazid plus rifapentine
3HR	3-month daily regimen of isoniazid plus rifampicin
3TC	lamivudine
4R	4-month daily regimen of rifampicin
6-9H	6- to 9-month daily regimen of isoniazid
ABC	abacavir
AIDS	acquired immunodeficiency syndrome
ALT	alanine aminotransferase
anti-HBc	hepatitis B core antibody
anti-HBs	hepatitis B surface antibody
anti-HCV	hepatitis C antibody
APRI	aspartate aminotransferase-to-platelet ratio index
ART	antiretroviral therapy
ARV	antiretroviral drugs
AST	aspartate aminotransferase
ATV/r	atazanavir/ritonavir
AZT	zidovudine
BMI	body mass index
CAB-LA	long-acting cabotegravir
CAB/RPV	cabotegravir/rilpivirine
CMV	cytomegalovirus
CNS	central nervous system
CrCl	creatinine clearance
CVD	cardiovascular disease
DAA	direct-acting antiviral
DCV	daclatasvir
DDI	drug-drug interaction
DM	diabetes mellitus
DMPA	depot medroxyprogesterone acetate
DRV/r	darunavir/ritonavir
DTG	dolutegravir
DXA	dual-energy X-ray absorptiometry
EFV	efavirenz
eGFR	estimated glomerular filtration rate
ESRD	end-stage renal disease
FIB-4	Fibrosis-4 index
FTC	emtricitabine
G6PD	glucose-6-phosphate dehydrogenase
GAHT	gender-affirming hormone therapy
HAV	hepatitis A virus
HbA1c	glycated haemoglobin
HBeAg	hepatitis B e antigen
HBsAg	hepatitis B surface antigen
HBV	hepatitis B virus
HBV DNA	hepatitis B virus deoxyribonucleic acid
HCC	hepatocellular carcinoma
HCV	hepatitis C virus
HDL-C	high-density lipoprotein cholesterol
HIV	human immunodeficiency virus
HIVAN	HIV-associated nephropathy
HPV	human papillomavirus
HTE	heavily treatment-experienced

ICOPE	Integrated Care for Older People
ID	infectious diseases
IGRA	interferon-gamma release assay
INSTI	integrase strand transfer inhibitor
IRIS	immune reconstitution inflammatory syndrome
LDL-C	low-density lipoprotein cholesterol
LLV	low-level viraemia
LPV/r	lopinavir/ritonavir
LSM	liver stiffness measurement
LTBI	latent tuberculosis infection
LTFU	lost to follow-up
MAC	Mycobacterium avium complex
MASH	metabolic dysfunction-associated steatohepatitis
MASLD	metabolic dysfunction-associated steatotic liver disease
MHT	menopausal hormone therapy
MMR	measles, mumps and rubella
MoCA	Montreal Cognitive Assessment
MSM	men who have sex with men
NNRTI	non-nucleoside reverse transcriptase inhibitor
nPEP	non-occupational post-exposure prophylaxis
NRTI	nucleoside reverse transcriptase inhibitor
NVP	nevirapine
OI	opportunistic infection
PEP	post-exposure prophylaxis
PI	protease inhibitor
PLWH	people living with HIV
PMTCT	prevention of mother-to-child transmission
PrEP	pre-exposure prophylaxis
PWID	people who inject drugs
RAL	raltegravir
RBV	ribavirin
RDV	ravidasvir
RNA	ribonucleic acid
RPV	rilpivirine
RTV	ritonavir
SJS	Stevens-Johnson syndrome
SMX	sulfamethoxazole
SOF	sofosbuvir
TAF	tenofovir alafenamide
TB	tuberculosis
TB-IRIS	tuberculosis-associated immune reconstitution inflammatory syndrome
TDF	tenofovir disoproxil fumarate
TEN	toxic epidermal necrolysis
TG	triglyceride
TLD	tenofovir disoproxil fumarate/lamivudine/dolutegravir
TMP	trimethoprim
TMP/SMX	trimethoprim/sulfamethoxazole
TPT	tuberculosis preventive treatment
U=U	undetectable equals untransmittable
UPCR	urine protein-to-creatinine ratio
VEL	velpatasvir
VF	virologic failure
VL	viral load
WHO	World Health Organization
ZDV	zidovudine

INTRODUCTION

Globally, an estimated 40.8 million (37.0 million - 45.6 million) people are living with HIV¹. In Malaysia, there are approximately 84,000 to 85,000 people living with HIV (PLWH)². While the country achieved a remarkable 50% drop in new cases between the year 2000 and 2009, the trajectory shifted from year 2010 to 2025, plateauing at an average of 3,169 new cases annually (with 3,004 preliminary cases recorded in 2025)².

Since the advent of combination Antiretroviral Therapy (ART) consisting of three or more antiretroviral drugs that act on different targets in the virus in 1996³, widespread access across government tertiary centres has slashed historic HIV-related mortality by over 50%, successfully transforming a once-fatal diagnosis into a highly manageable chronic condition⁴.

1.1 What is new in this guideline?

To align with the latest clinical evidence, this updated guideline introduces several paradigm shifts in Malaysian HIV care:

- **Universal rapid initiation:** Formalised “Rapid ART” protocols across all clinical stages.
- **Preferred dolutegravir (DTG)-based regimen:** Solidified DTG as the gold-standard first-line agent, complete with explicit metabolic, neuropsychiatric and pregnancy monitoring protocols.
- **De-escalation of routine monitoring:** Rationalised CD4 cell count testing, removing the need for routine counts once a patient is virologically suppressed with a CD4 count greater than 200 cells/mm³.
- **Treatment simplification:** Introduction of evidence-backed Two-Drug Regimens (2DR) and strategic Nucleoside Reverse Transcriptase Inhibitor (NRTI) “recycling” for second-line regimen.
- **Cardiometabolic and ageing focus:** Mandatory 10-year cardiovascular disease (CVD) risk scoring and structured screening for age-related comorbidities.
- **Person-centred PMTCT programme:** Updated vertical transmission prevention, incorporating empathetic, structured breastfeeding protocols and high-risk infant prophylaxis.

1.2 Assessment of adults living with HIV

ARVs cannot eradicate HIV from the human body nor cure HIV infection. The goals and benefits of ART include:

- **Confirmatory testing:** All referred cases of HIV infection need confirmatory testing upon entry of care.
- **Prophylaxis history:** Universal screening for prior exposure to PrEP or PEP to contextualise potential drug resistance.
- **Active TB rule-out:** Immediate symptom screening for active TB. If active TB is clinically ruled out, immediate initiation of TB Preventive Therapy (TPT) must be considered.
- **Baseline monitoring:** Establishment of structured viral load, CD4 cell count, organ function baseline investigations and necessary imaging to guide long-term treatment success.

1.3 Foundation of effective HIV management

- **Rapid ART initiation:** Direct offer to start ART rapidly following diagnosis, irrespective of clinical staging or CD4 counts.
- **Opportunistic infection (OI) timing:** For most OIs, ART should be initiated early within 2 weeks, except in Cryptococcal meningitis and TB meningitis.
- **Holistic support:** Integration of psychosocial interventions, adherence counseling and peer-led service-level support.

1.4 Evolution of HIV treatment

- **Pre-ART considerations:** Pre-ART characteristics and presence of specific comorbidities prior selection.
- **First-line selection:** DTG-based regimens are established as the preferred first-line regimen due to high genetic barrier to resistance and excellent tolerability profile.
- **Regimen optimisation:** Clear criteria for shifting stable patients to simplified 2DR to minimize long-term drug toxicity.
- **Strategic NRTI recycling:** Endorsement of recycling NRTIs into second-line regimens, optimising procurement costs in resource-limited settings without compromising viral suppression.

1.5 Monitoring

Table 1.1: Monitoring parameters and clinical considerations

Parameter	Clinical Considerations
HIV viral load	Primary marker for assessing treatment response and confirming sustained viral suppression
CD4 cell count	Assesses immune status and recovery Routine monitoring may be discontinued in patients with sustained viral suppression and stable CD4 counts > 200 cells/mm ³

1.6 Management of co-morbidities, ageing and PMTCT

- **Cardiovascular care:** Annual 10-year CVD risk estimation scoring (e.g., Framingham or ACC/AHA calculators) alongside aggressive cardiometabolic health tracking.
- **Non-Communicable Diseases (NCDs):** Structured screening paths for diabetes mellitus, chronic kidney disease, osteoporosis/bone density loss, and HIV-associated neurocognitive disorders (HAND).
- **Advanced PMTCT:** Closing the vertical transmission gap via structured, person-centred breastfeeding guidance alongside enhanced triple-drug prophylaxis customised for high-risk infants.

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All adults with HIV infection entering into care should have a complete medical history, physical examination, baseline laboratory evaluation and counselling about the HIV infection.¹

The goals of the initial evaluation are:

1. To confirm the diagnosis of HIV infection
2. To identify risk factors related to HIV acquisition
3. To obtain appropriate baseline historical data and a comprehensive medical review of symptoms
4. To perform a thorough physical examination looking for clinical manifestations of HIV infection and evaluation of other co-infections
5. To ensure patient's understanding of HIV infection and its mode of transmission
6. To assess social support and coping strategies
7. To initiate medical care

All newly diagnosed patients should be asked about any prior use of ART for PrEP and PEP.² For treatment-experienced patients who present to a new health care provider, obtain a complete ART history and review of past medical records.² All patients entering care should be screened for symptoms of active TB and considered for TPT if active TB has been ruled out.⁶

Table 2.1 • History taking for PLWH²

	Symptoms / Components	Significance
History of presenting complaint	• Constitutional symptoms: fever, loss of weight and loss of appetite • CNS: headache and dizziness • GIT: oral thrush, odynophagia, diarrhoea and rectal bleeding • Respiratory: chronic cough and shortness of breath especially on exertion. • Skin lesions: suggestive of bacterial, fungal, Mycobacterium, scabies, viral, malignancy or mites. • Soft tissue swelling: lymphadenopathy, tumour or abscess. • Genitourinary: urethral discharge or genital ulcers suggestive of STIs. • Eye symptoms: blurring of vision, diplopia, floaters (suggestive of CMV, toxoplasmosis)	Diagnosis of opportunistic infection

	Symptoms / Components	Significance
Screening for TB symptoms with W4SS ⁶	• Prolonged cough, prolonged fever, loss of weight and night sweats	To screen for TB and investigate if symptoms present or to counsel for TPT if active TB has been ruled out
Drug history	• Current medications and dosages • Alternative medications/ supplements • Smoking, alcohol and recreational drugs use	Allergy Potential drug-drug interaction
ART history	• Use of PrEP or PEP • Treatment experienced: past ART history and known virological response/resistance	To decide ART regimen
Past and current medical, surgical, obstetrics and gynaecology history	• Infections: TB, hepatitis, herpes, varicella, sexually transmitted infections (syphilis, gonorrhoea and chlamydia), and previous opportunistic infections. • Comorbidities: Diabetes mellitus, ischaemic heart disease, hypertension, renal disease and dyslipidaemia. • Relevant history: Previous surgeries and obstetric and gynaecological conditions. • Treatment history: Previous and ongoing treatments, including completion status.	Risk of worsening condition due to ART, concomitant drug-drug interaction and co-infection
	• Vaccination history • Last negative HIV test • Blood donation within a year of diagnosis	Vaccination counselling Alert the local blood bank
Psychosocial history	• Circle of confidentiality • Partner and children • Support network • Occupation, housing and education • Mental health issues • Substance use disorders	To identify and address potential barriers to care

	Symptoms / Components	Significance
Sexual and reproductive history	• Sexual history and practices • Safer sex and risk reduction • Partner status and disclosure issue • Participation in chemsex • Last menstrual period (for women)	Counselling for transmission prevention measures e.g. PMTCT, PrEP or nPEP
Family history	• History of DM/HPT/cardiovascular diseases	To evaluate the risk in patient

2.2 Monitoring patients on Antiretroviral Therapy (ART)

Effective monitoring of individuals on ART is essential to ensure sustained viral suppression, prevent treatment failure and support long-term health outcomes.

2.2.1 Viral load

Viral load (VL) testing is the preferred method for assessing treatment efficacy and identifying early signs of treatment failure. HIV VL should be tested:

- Just before initiation of ART
- Every 4 to 6 months after initiation of ART to assess treatment response and for early detection of treatment failure
- Every 6 to 12 months in patients who have achieved virological suppression for ≥ 1 year
- Before changing treatment regimens

Effective therapy should generally result in a 10-fold decrease in HIV-1 RNA copies/mL in the first month and suppression to less than 200 copies/mL by 6 months.

2.2.2 CD4 cell count

Although the role of CD4 count monitoring has been reduced with the expansion of VL testing, CD4 counts remain useful in specific contexts:

- A baseline CD4 count is recommended at or before ART initiation to assess the need for opportunistic infection (OI) prophylaxis.
- In patients starting ART with advanced HIV disease ($CD4 < 200$ cells/mm³), CD4 monitoring should continue every 6 months until $CD4 \geq 200$ cells/mm³.
- When the CD4 reaches 200 to 350 cells/mm³, test the CD4 annually.
- Once the CD4 is more than 350 on 2 occasions 1 year apart, no further CD4 count is required unless treatment failure is suspected.

Table 2.2 • Important Laboratory Investigations²

Evaluation	Investigations	Specific tests	When: Entry to care	When: At follow-up on ART	Comments
HIV disease	HIV confirmatory test		✓		All referred cases of HIV infection need a confirmatory test
	Plasma HIV RNA	HIV viral load	✓	Every 4 to 6 months after ART initiation Annually if stable Refer section 2.1	
	CD4		✓	Every 4 to 6 months after ART initiation Refer section 2.1	Determine the need for OI prophylaxis
	HIV resistance test		✓ *		*If transmitted drug resistance is suspected (e.g. acquisition from a source with known or suspected resistance), or if the patient has current or previous exposure to long-acting injectable PrEP.
Co-infections	Syphilis serology	RPR/ VDRL TPPA/ TPHA TP Ab (EIA, CIA)	✓	Annual screening if at risk	
	Hepatitis A serology#	Hep A IgG	At risk group		# If test is available in the facility At risk group: MSM
	Hepatitis B serology	Hepatitis B surface antigen (HbsAg) Hepatitis B surface antibody (HbsAb)#	✓	Annual screening if at risk	# Vaccinate if non-immune

Evaluation	Investigations	Specific tests	When: Entry to care	When: At follow-up on ART	Comments
Co-infections	Hepatitis C serology	HCV antibody	✓	Annual screening if at risk	
	Chest X-Ray		✓	When clinically indicated	All new patients should have a baseline CXR, to rule out active TB
CVS	ECG		✓ *	When clinically indicated	*If patient has other risk factors for IHD
Haematology	Full blood count (FBC)		✓	Every 4 to 6 months	*If initiated on AZT, FBC should be repeated at 4, 8 and 12 weeks of initiation or if symptomatic of anaemia
Renal function	Renal Profile UFEME		✓	At 2 - 4 weeks, and 12 weeks after initiation of TDF, and then 6 monthly if stable	TDF may cause renal tubular dysfunction Routine monitoring of calculated creatinine clearance should be performed for all patients on TDF during follow up DTG may cause benign raise in creatinine, alternative monitoring may be needed to distinguish from true renal pathology More frequent monitoring maybe needed if patient has risks factors e.g. CKD, DM, hypertension
Liver function	ALP, AST, ALT, Bilirubin, Albumin		✓	Every 4 to 6 months	NRTI and NNRTI can cause hepatotoxicity If on NVP, ALT needs to be monitored more frequently at baseline, 2, 4, 12 weeks and then every 3 to 6 months Obtain ALT in patients with new onset rash

Evaluation	Investigations	Specific tests	When: Entry to care	When: At follow-up on ART	Comments
Metabolic	Fasting lipid profile		✓	Every 6 to 12 months	EFV, NRTIs, PIs (with the exception of unboosted atazanavir), can cause insulin resistance and dyslipidaemia For those >40 years old, determine CVD risk score
	Fasting blood sugar		✓	Every 6 to 12 months	
Others	Serum cryptococcal antigen#				To identify asymptomatic patients who need pre-emptive treatment for cryptococcal disease #If CD4 < 100 cells/mm ³
	HLA-B*5701				To determine the risk of hypersensitivity reaction in selected patients initiated on ABC Discuss with ID physician
	Urine pregnancy test (UPT)		✓		Assess contraceptive use and whether the last menstrual period occurred as expected If menstruation is delayed or missed, perform UPT If not pregnant, assess fertility intentions and contraceptive needs
	Cervical/ PAP smear for women		✓	Annually	After 3 consecutive annual normal PAP smears, repeat screening every 3 years

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3.1 Counselling, linkage, and retention in care

This section outlines the essential components of patient management for PLWH, with emphasis on counselling and sustained engagement in care. These elements are integral to the HIV care continuum and are critical for achieving durable viral suppression.

Effective long-term management of HIV is underpinned by several key elements:

- Appropriate prescription of antiretroviral therapy (ART) regimens
- Comprehensive patient support, including psychosocial and service-level interventions
- Sustained adherence to care and treatment.

Together, these components support lifelong HIV treatment and optimal clinical outcomes.

This guideline section provides medical practitioners with practical guidance to establish and sustain an effective treatment trajectory, with emphasis on:

- Pre-ART and adherence counselling, to ensure patients are informed, prepared, and supported for lifelong treatment
- Clear pathways for prompt linkage to HIV care following diagnosis
- Patient-centred models of care to support long-term retention and continuity of care

These strategies aim to achieve sustained viral suppression, immune recovery, and prevention of HIV transmission, consistent with the principle of **Undetectable = Untransmittable (U=U)**.

3.1.1 Pre-Antiretroviral Therapy (ART) counselling

The primary goal of pre-ART counselling is to ensure the patient is informed, prepared, and committed to lifelong treatment adherence.

Key concepts: HIV and ART

- Provide clear education on HIV, including the nature of the virus, modes of transmission, and the role of antiretroviral therapy (ART) in suppressing viral replication and enabling immune recovery.
- Emphasise that ART is not curative, but is a highly effective, lifelong treatment that enables long-term viral suppression and prevention of disease progression.

Treatment rationale

- Emphasise that maintaining an undetectable viral load through effective ART improves individual clinical outcomes
- Explain the U=U principle, whereby sustained viral suppression eliminates the risk of sexual transmission of HIV.

Laboratory results

- Explain baseline laboratory results, including CD4 count and HIV viral load, and their role in assessing immune status and monitoring treatment response.
- Clarify treatment targets, particularly achieving viral suppression, typically defined as an HIV viral load of <200 copies/mL.

Psychosocial readiness

- Assess the patient's readiness to adhere to daily, lifelong medication.
- Identify potential barriers (stigma, cost, transport, homelessness) and help develop tailored coping strategies.

Disclosure and support

- Discuss the benefits and risks of disclosure to partners, family, and friends.
- Partner notification is mandatory.
- Encourage identifying a supportive network (treatment buddy) who can assist with adherence.

Future parenting and pregnancy planning

- Discuss reproductive intentions, including timing of pregnancy and available fertility options where relevant.
- Address prevention of HIV transmission, including sustained viral suppression, partner protection for serodiscordant couples and maternal-to-child transmission prevention.
- Support psychosocial preparation, including emotional readiness, disclosure considerations and access to ongoing counselling and support services

Future follow up and monitoring plans

- Explain the expected clinical, virological, and immunological response to ART, including symptom improvement, viral load decline, and CD4 recovery over time.
- Emphasise the importance of regular follow-up and laboratory monitoring to assess treatment response, detect treatment failure early and optimise ongoing patient care.

3.1.2 ART initiation and adherence

This focuses on the practical aspects of starting and maintaining the prescribed ART regimen.

Regimen details

- Explain the prescribed ART regimen, including drug names, pill burden, dosing schedule, and administration instructions (e.g. timing, with or without food)
- Promote the use of adherence aids, such as reminders, pill boxes, or linking doses to daily routines

Managing adverse events

- Prepare patients for common anticipated early adverse events, such as nausea and fatigue, by providing reassurance, practical management strategies and guidance on when to seek medical review.
- Guide patients to differentiate between mild, transient adverse events and symptoms requiring immediate medical attention.
- Discuss the possibility of immune reconstitution inflammatory syndrome (IRIS) after ART initiation.

Drug-drug interactions

- Review all concurrent medications, including prescribed drugs, over-the-counter products, supplements and illicit substances.
- Advise patients to consult their prescriber before starting any new medication, including over-the-counter or herbal products.

Supply management

- Educate patients on medication refills, maintaining adequate supply and the importance of avoiding treatment interruption.

The critical importance of adherence

- Adherence to ART is essential for sustained viral suppression, reducing HIV transmission and preventing drug resistance.
- Effective adherence interventions should be comprehensive, patient-centred and tailored to individual socioeconomic, behavioural and care-related barriers.
- Specific groups at risk of suboptimal adherence include:
 - Individuals with limited family or social support
 - People who inject drugs
 - Adolescents and young adults
 - Pregnant women
 - Individuals with underlying psychiatric or mental health conditions
- Counselling on adherence should be individualised, non-judgemental, ongoing and should emphasise that high and sustained adherence is critical for:
 - Consistent and correct dosing of antiretroviral therapy
 - Achievement and maintenance of viral suppression, including an undetectable viral load
 - Optimal long-term clinical outcomes, including immune recovery and overall wellbeing
- Counselling should also inform patients that suboptimal adherence and missed doses increase the risk of viral rebound, development of drug resistance, treatment failure, and poorer long-term health outcomes.

Table 3.1 • Common barriers and strategies to improve adherence

Barrier category	Common barriers	Strategies to improve adherence
Therapy/ Medication	High pill burden	Simplify the regimen (e.g., switch to single-pill, once-daily regimens)
	Complex dosing schedule (e.g., specific food requirements, multiple doses per day)	Standardise dosing (e.g., once-daily dosing, minimal food restrictions)
	Experiencing or fearing medication side effects	Choose drugs with minimal side effects and provide proactive symptom management and counselling
Behavioural and psychosocial	Forgetting to take medication because of changes in daily routine or inadequate planning	Implement reminder systems (e.g., alarms, mobile apps, pillboxes, visual cues)
	Depression, anxiety, or substance use disorder	Integrate mental health and substance abuse treatment with HIV care
	Treatment fatigue or feeling overwhelmed by the daily commitment	Provide continuous, nonjudgmental adherence counselling using motivational interviewing techniques
	Lack of understanding of the importance of adherence (e.g., relationship between adherence and viral suppression)	Provide thorough patient education on the purpose of ART and the risk of resistance if non-adherent
Social and environmental	Fear of disclosure, stigma, or discrimination leading to hiding medication	Maintain patient confidentiality and link patients to support groups or peer educators to reduce isolation and shame
	Unstable housing, food insecurity, or lack of transportation	Address social determinants of health (SDOH) by connecting patients to housing, food assistance, and transportation programmes
	Lack of social support from family or friends	Encourage identification of a trusted “treatment supporter” and provide family counselling where appropriate

Barrier category	Common barriers	Strategies to improve adherence
Healthcare system	Suboptimal communication or lack of trust between patients and healthcare providers	Foster a collaborative, empathetic, and respectful provider-patient relationship; ensure open discussion of adherence challenges
	Long clinic waiting times, distance to the clinic, or logistical difficulty in obtaining refills	Decentralise drug delivery (e.g., multi-month dispensing, community pick-up points) and streamline clinic visits
	Intermittent drug shortages	Improve supply chain management and use standardised regimens to minimise the risk of shortages

3.1.3 Key steps in linkage to care

Effective linkage to care following HIV diagnosis is essential to ensure timely initiation of treatment and sustained engagement in care. The following steps should be implemented to facilitate successful linkage:

Step 1 Communicating the HIV test result

- Confirm HIV diagnosis following a reactive rapid HIV test using a supplementary test (e.g. a different rapid HIV Ab/Ag test $\pm$ HIV PCR).
- Additional confirmatory testing is particularly important in asymptomatic individuals or those who deny high-risk exposure.
- Results should be communicated clearly, sensitively, and confidentially.

Step 2 Initial counselling and assessment of social concerns

- Provide essential information on HIV, modes of transmission, and the importance of reducing risk behaviours.
- Explain the need for ongoing, regular HIV care, even in individuals who feel clinically well at diagnosis.
- Address individual needs and concerns, including emotional support, follow-up plans, and disclosure to partners.
- Emphasise confidentiality of HIV test results, while explaining that notification to the Ministry of Health Malaysia is mandatory under the Prevention and Control of Infectious Diseases Act 1988 (Act 342), and that the patient may be contacted by the District Health Office
- Discuss partner notification for HIV to ensure sexual or needle-sharing partners are informed of potential exposure and offered counselling, testing, treatment, and appropriate health services. Partner notification may be conducted through:
 - Self-referral (where the patient informs partners directly)
 - Dual referral (with support from a healthcare worker), or;

- Provider referral (where a healthcare worker notifies partners on the patient's behalf).
- Reinforce the importance of continued HIV care despite absence of symptoms.
- Provide written educational materials where available.

Step 3 Linkage to HIV care services

- Identify the nearest clinic or hospital
- providing HIV services, considering patient accessibility and convenience.
- Arrange timely appointments with referral clinicians or counsellor nurses to minimise waiting times.
- Confirm and communicate referral requirements, including referral documentation and baseline laboratory investigations prior to specialist review.
- Provide additional support, where feasible, such as transport assistance and appointment reminders, to promote attendance.
- Consider virtual or differentiated care models for stable patients, where appropriate.

Step 4 Track referrals

- Track all referrals to confirm patient attendance at the receiving clinic.
- Establish a follow-up strategy for patients who fail to attend scheduled appointments.
- Implement a reminder or alert system to prompt the referring doctor to verify follow-up after a predetermined period if no feedback is received from the receiving clinic.

Step 5 Referral to peer support groups and Non-Governmental Organisations (NGOs)

- Refer patients to peer support groups or NGOs, where available, to provide psychosocial support, peer mentoring, and trust-building.
- Utilise trained peers or NGO workers to support patient understanding of HIV care and assist with navigation of health and social services.

Step 6 Linkage to related health and support services

- Link patients to relevant clinical services, as indicated, including:
 - Sexual and reproductive health or genitourinary clinics for sexually transmitted infections.
 - Maternal and child health services for pregnant women living with HIV and infants born to mothers with HIV.
 - Tuberculosis services (e.g. chest clinics) for suspected or confirmed TB co-infection.
- Link patients to additional support services, where required, including:
 - Substance use treatment services, such as methadone maintenance programme.
 - Social welfare services or shelter homes for individuals with limited social support.

3.1.4 Retention in care

Retention in care refers to continuous engagement in HIV medical care over the course of a lifetime, which is critical for long-term health and preventing onward transmission.

Differentiated service delivery

- Implement flexible, patient-centred models of care, especially for patients who are stable, virally suppressed, and adherent (e.g., multi-month dispensing, fast-track clinic visits).

Proactive monitoring and tracing

- **Monitor:** Track patient appointments. Flag any patient who misses a scheduled clinic visit or laboratory test.
- **Trace:** Implement a standardised protocol for contact/tracing of patients who are lost to follow-up (LTFU). This involves sequential attempts via phone, SMS, WhatsApp, email and if necessary, community health worker outreach.

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3.2 Rapid initiation of Antiretroviral Therapy (ART)

All individuals living with HIV should be offered ART regardless of CD4 count or clinical stage. Rapid initiation of ART, defined as commencing treatment within seven days of HIV diagnosis, should be provided following clinical assessment and evaluation of patient readiness.

Rapid ART initiation leads to faster viral suppression, improved retention in care, reduced risk of HIV transmission and better survival outcomes. It is strongly recommended for individuals who are ready and willing to commence treatment, particularly those with advanced HIV disease (e.g., low CD4 count, symptomatic disease, or high risk of disease progression).

Prior to initiating ART, screening for opportunistic infections (OIs) and baseline investigations should be conducted; however, these should not unnecessarily delay treatment if no contraindications exist.

Key considerations for rapid ART initiation:

- **Patient readiness**

Ensure the patient is adequately informed about HIV infection and ART, and is willing to initiate and adhere to treatment.

- **Priority groups**

Rapid ART initiation should be prioritised for individuals with:

- Advanced HIV disease
- Pregnancy or breastfeeding
- Acute HIV infection

- **Situations where ART initiation may be deferred**

Temporary delay of ART may be considered in the following circumstances:

- Suspected or confirmed central nervous system opportunistic infections, particularly cryptococcal meningitis or tuberculosis meningitis, due to the risk of immune reconstitution inflammatory syndrome (IRIS).
- Severe co-morbid conditions requiring stabilisation prior to ART initiation (e.g., uncontrolled renal or hepatic disease that may affect ART selection).
- Significant psychosocial or structural barriers that may compromise adherence and require urgent intervention before treatment initiation.

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4.1 Principles of selecting ART for 1st line regimen

The following factors should be considered to determine which ART regimen is the best for a particular patient:

- Co-morbid and organ dysfunction (e.g. renal insufficiency, liver impairment, cardiovascular disease, metabolic syndrome, bone disease, anaemia, psychiatric illness)
- Co-infections in particular with Hepatitis B
- On-going treatment for active opportunistic infection (e.g. TB due to potential drug-drug interactions)
- Pregnancy and women of child-bearing age
- Seroconversion while on PrEP
- Impact of the regimen itself (e.g. pill burden, pill size, potential for drug interactions, anticipated adverse events, food or fasting requirements, swallowing difficulty, and considerations in adolescents)
- Results of baseline HIV drug resistance testing whenever available
- Convenience of the regimen, taking into account patient factors such as occupation and available social support to enhance adherence

4.1.1 Antiretroviral options

Antiretroviral therapy (ART) comprises drugs from different antiretroviral classes that target various stages of the HIV life cycle. The antiretroviral drugs currently available in Malaysia are summarised in Tables 4.1 and 4.2.

Fixed dose combinations (FDCs) consist of two or more antiretroviral drugs combined into a single tablet. FDCs reduce pill burden and overall treatment cost. Simplified dosing regimens improve adherence and contribute to durable virological suppression.^{3,4}

Treatment selection should be individualised according to clinical factors, comorbidities, potential drug interactions, tolerability and patient preferences.

Table 4.1 • Antiretroviral drugs currently available in Malaysia

Class	Abbreviation
Nucleoside reverse transcriptase inhibitors (NRTI)	
Abacavir	ABC
Emtricitabine	FTC
Lamivudine	3TC
Tenofovir disoproxil fumarate	TDF
Tenofovir alafenamide	TAF
Zidovudine	AZT or ZDV
Non-nucleoside reverse transcriptase inhibitors (NNRTI)	
Efavirenz	EFV
Nevirapine	NVP
Rilpivirine	RPV
Protease Inhibitors (PI)	
Atazanavir	ATV
Darunavir	DRV
Lopinavir	LPV
Ritonavir	RTV
Integrase Strand Transfer Inhibitors (INSTI)	
Dolutegravir	DTG
Cabotegravir	CAB

Table 4.2 • Fixed dose combinations registered in Malaysia

Fixed Dose Combinations (FDC)
Abacavir/Lamivudine (ABC/3TC)
Zidovudine/Lamivudine (AZT/3TC)
Tenofovir disoproxil fumarate/Emtricitabine (TDF/FTC)
Tenofovir disoproxil fumarate/Emtricitabine/Efavirenz (TDF/FTC/EFV)
Tenofovir disoproxil fumarate /Lamivudine/Dolutegravir (TDF/3TC/DTG)
Tenofovir alafenamide/Emtricitabine (TAF/FTC)
Lopinavir/Ritonavir (LPV/r)
Atazanavir/Ritonavir (ATV/r)
Darunavir/Ritonavir (DRV/r)

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4.2 Preferred and alternative options for first-line ART

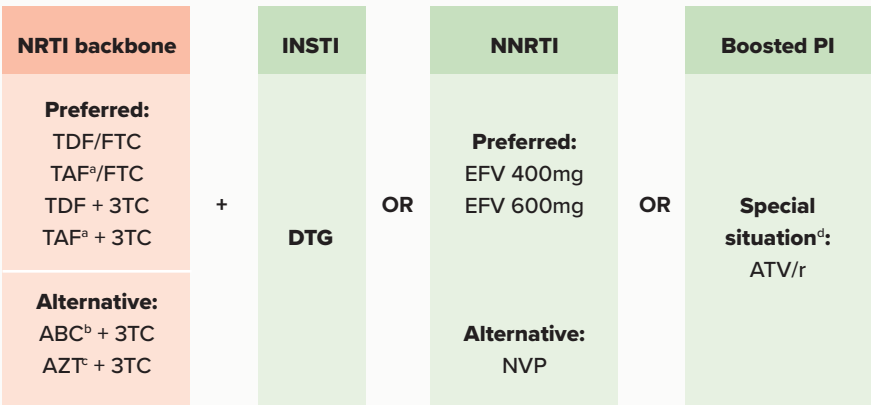
The primary goals of initial ART in treatment-naïve PLWH are to achieve sustained viral suppression, restore immune function, reduce HIV-associated morbidity and prevent onward sexual transmission (Undetectable = Untransmittable [U=U]).

ART for treatment-naïve individuals generally comprises two nucleoside reverse transcriptase inhibitors (NRTIs) combined with a third active antiretroviral drug from one of the following classes:

- Integrase strand transfer inhibitor (INSTI)
- Non-nucleoside reverse transcriptase inhibitor (NNRTI)
- Protease inhibitor (PI) with a pharmacokinetic booster, such as ritonavir

The recommended first-line ART options are summarised in Figure 4.1.

Figure 4.1 • Preferred and alternative ART for patients who are ART naïve



^a TAF has fewer bone and kidney toxicities than TDF. However, TDF is associated with lower lipid levels.

^b Perform HLA-B*5701 testing before initiating ABC; do not start ABC if the result is positive. If using ABC/3TC in a patient with VL>100,000 copies/mL, combination with DTG is preferred over NNRTI.

^c AZT should not be initiated in patients with baseline Hb < 8.0g/dL.

^d Boosted PI should only be chosen in special situations and should be discussed with an ID Physician.

Two-Drug Regimen (2DR)

While standard first-line ART comprises three active drugs, the 2DR of dolutegravir (DTG) plus lamivudine (3TC) avoids tenofovir-associated renal and bone toxicity and may be considered for selected treatment-naïve PLWH.

However, this regimen is not recommended for individuals with:

- HIV RNA >500,000 copies/mL
- CD4 count <200 cells/mm³
- HBV co-infection (HBsAg positive)

Note: HIV genotypic resistance testing is preferred but not mandatory due to limited availability; if ordered, do not delay 2DR initiation while awaiting results.

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4.3 Considerations prior to starting treatment

ART is recommended for all PLWH, regardless of CD4 count. In the absence of opportunistic infections, ART should be commenced as soon as possible, ideally within 2-4 weeks of diagnosis.¹ For same-day ART initiation, its advantages and disadvantages should be discussed with each individual, including the benefits and harms of same-day ART. Once initiated, ART should be continued indefinitely without interruption, unless severe toxicity occurs.

The selection of an initial ART regimen should be guided by the regimen's efficacy, barrier to resistance, potential adverse events profile, convenience, co-morbidities, concomitant medications, the potential for drug-drug interactions, social factors and availability. Refer to Annexe 4.1 for the summary of special considerations when initiating ART.

4.3.1 Nucleoside Reverse Transcriptase Inhibitors (NRTI)

FDA-approved NRTIs include ABC, TDF, TAF, AZT, 3TC and FTC. TDF and AZT are generally comparable in terms of efficacy; however, some studies have shown TDF-based therapy to be more effective and associated with fewer side effects than AZT.

TAF and TDF are two approved tenofovir formulations that can be used in patients with Hepatitis B co-infection. However, TDF has been strongly associated with bone loss and renal toxicities. TDF should be avoided in individuals with chronic kidney disease with CrCl <50 ml/min (especially when used with Ritonavir). TDF should be avoided in the setting of osteoporosis, a history of fragility fracture, and/or severe osteopaenia (T score < -2.0).^{2,5}

TAF is a prodrug of tenofovir that yields relatively lower plasma levels and high intracellular levels of tenofovir; and it is less likely to cause renal and bone toxicities than TDF. Nevertheless, TAF should not be co-administered with rifamycins as it would reduce the drug levels. Conversely, levels of fasting low-density lipoprotein (LDL) cholesterol, high-density lipoprotein (HDL) cholesterol and triglycerides increased more in the TAF group than in the TDF group, with no change in the total cholesterol to HDL ratio.

ABC may be considered in special circumstances where the preferred regimens (TDF or TAF-based) are not suitable because of toxicities or anticipated drug-drug interactions. However, ABC is not recommended in PLWH where HIV viral load is >100,000 copies/ml unless the third agent is either a DTG or PI⁶⁻⁸. It is contraindicated if HLA-B*5701 is positive as it poses a risk of hypersensitivity. ABC needs to be used with caution in persons with high CVD risk.

4.3.2 Integrase Strand Transfer Inhibitors (INSTI)

INSTI is now the recommended 1st line regimen (in combination with 2 NRTIs) for most PLWH. Combination of DTG with NRTIs are more favourable when compared with efavirenz-based regimens in the context of virological success, infrequent treatment-limiting adverse events and the tolerability of the medication.^{6, 10-12}

4.3.3 Non-Nucleoside Reverse Transcriptase Inhibitors (NNRTI)

NNRTI-based regimens have become less favourable as an initial therapy in most PLWH, but selected regimens may be effective in some circumstances. Emergence of drug resistance in the context of virological failure has been reported with all NNRTIs, which have a lower barrier to resistance compared with INSTIs or boosted PIs.

NVP and EFV have comparable clinical efficacy when used in combination ART. However, NVP is associated with a higher risk of rash, Stevens-Johnson Syndrome, and hepatotoxicity compared to EFV. In case of severe hepatotoxicity or skin reactions, NVP should be permanently discontinued. NVP must be avoided in women with a CD4 count >250 cells/mm³ and in men with a baseline CD4 count >400 cells/mm³ due to a significant increase in the incidence of symptomatic hepatic events. Lead in dosing of 2 weeks for NVP should be practiced to decrease the risk of hepatitis and rash.

EFV is the NNRTI of choice in individuals with TB/HIV co-infection who are receiving rifampicin-based TB treatment. EFV is associated with neuropsychiatric side effects, cognitive impairment, depression with suicidality and potential drug-to-drug interactions¹⁷. In the ENCORE 1 Study, EFV 400mg daily was better tolerated than the standard dose of EFV 600mg daily¹⁸. Regimens containing EFV 400mg were comparable to EFV 600mg in terms of viral suppression and mortality. EFV 400mg can be co-administered with rifampicin-containing anti-TB treatment, with co-administration well tolerated and plasma concentrations maintained above levels considered effective.

NNRTIs have a low genetic barrier to resistance and long half-lives. Abrupt discontinuation of NNRTI without maintaining the NRTI backbone will increase the risk of NNRTI resistance due to its long half-life. Hence, when NNRTI is stopped due to an adverse event, the backbone NRTI should be continued for at least 1 week before stopping all drugs.

4.3.4 Protease Inhibitors (PI)

PI including DRV/r, LPV/r and ATV/r can be considered in the following clinical scenarios:

- Unable to tolerate the adverse events of INSTI or NNRTI
- Resistant testing indicating resistance to the standard first-line regimen

Some observational cohorts have demonstrated an association between PI, including DRV/r, with an increased risk of CV events¹⁹⁻²⁰. If ATV/r is used, watch out for hyperbilirubinemia.

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4.4 Timing of ART initiation in opportunistic infections

The timing of ART initiation in PLWH with opportunistic infections should balance the benefits of early viral suppression and immune recovery against the risk of immune reconstitution inflammatory syndrome (IRIS) and treatment-related complications. Timing should therefore be individualised according to the type of opportunistic infection, clinical response to treatment and relevant monitoring parameters, as summarised in Table 4.3.

Table 4.3 • Timing of ART initiation in selected opportunistic infections

Condition	When to start ART	Parameters to assess before initiation
Active tuberculosis (excluding <i>tuberculous meningitis</i>)	Within 2 weeks after starting TB treatment Initiation within 2 weeks is particularly recommended if the CD4 < 50 cells/mm ³	Clinical stability, tolerability of TB treatment and potential drug–drug interactions
Tuberculous meningitis	Generally 2–4 weeks after starting TB treatment Start high-dose corticosteroids immediately at diagnosis	Clinical improvement and improving cerebrospinal fluid (CSF) parameters
Cryptococcal meningitis	Within 2–4 weeks after starting antifungal therapy: • At 2 weeks: if the specified clinical criteria are met • At 4 weeks: if the criteria for earlier initiation are not met	For initiation at 2 weeks: clinical improvement, controlled intracranial pressure, negative CSF cultures and the ability to closely monitor/treat for increased intracranial pressure and cerebral oedema.

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4.5 Primary prophylaxis

Opportunistic infections (OIs) commonly occur in individuals with undiagnosed HIV, those diagnosed late, or those with known HIV who are not engaged in care. This overview outlines the prophylaxis for common OIs in PLWH, which may be either primary (to prevent the first episode of an OI) or secondary (to prevent recurrence). For recommendations on secondary prophylaxis, refer to the latest edition of the National Antimicrobial Guideline (NAG) under Infections in Immunocompromised Patients: Opportunistic Infections in Patients with Human Immunodeficiency Virus (HIV).

4.5.1 *Pneumocystis jirovecii* Pneumonia (PJP)

Trimethoprim/sulfamethoxazole (TMP/SMX; Bactrim) is recommended for PJP prophylaxis to all susceptible individuals as it has been shown to decrease the risk of PJP by ninefold in this population.

The indications for initiating, discontinuing and restarting prophylaxis, together with the recommended regimens, are summarised in Table 4.4.

Table 4.4 • Primary prophylaxis for *Pneumocystis jirovecii* pneumonia (PJP)

When to start	What to start	When to stop
CD4 count < 200 cells/mm ³ Or CD4 percentage of total lymphocytes <14% Or CD4 200-250 cells/mm ³ if ART initiation is delayed	Preferred: TMP/SMX (80/400mg) 1-2 tablets PO daily Alternative: Dapsone ^a 100mg tablet PO daily Or Aerosolized Pentamidine 300mg monthly via ultrasonic nebulizer	CD4 count increased to > 200 cells/mm ³ for ≥ 3 months in response to ART Or CD4 count is 100–200 cells/mm ³ and HIV viral load remains suppressed for ≥ 6 months in response to ART Note: For PLWH who develop PJP at a CD4 count >200 cells/mm³ while not receiving ART, discontinuation of prophylaxis may be considered once sustained viral suppression has been achieved for at least 6 months
Restarting primary prophylaxis CD4 count <100 cells/mm ³ regardless of the HIV viral load Or CD4 100-200 cells/mm ³ and HIV viral load is not suppressed		

^a Individuals should be tested for glucose-6-phosphate dehydrogenase (G6PD) deficiency before administration of dapsone. An alternative agent should be used if the patient is found to have G6PD deficiency.

4.5.1.1 Managing Trimethoprim/sulfamethoxazole adverse events

- For non-life-threatening adverse events, TMP/SMX should be continued whenever clinically feasible.
- If TMP/SMX is discontinued because of a mild adverse event, reintroduction should be considered once the reaction has resolved. This may involve gradual dose escalation through desensitisation or administration at a reduced dose (Table 4.5).
- TMP/SMX should be permanently discontinued without rechallenge following life-threatening adverse events, including suspected or confirmed Stevens–Johnson syndrome (SJS) or toxic epidermal necrolysis (TEN).

Table 4.5 • Protocol for TMP/SMX desensitisation

Step	Dose of TMP + SMX	Formulation dosing
Day 1	16mg TMP + 80mg SMX	2 mL oral suspension
Day 2	32mg TMP + 160mg SMX	4 mL oral suspension
Day 3	48mg TMP + 240mg SMX	6 mL oral suspension
Day 4	64mg TMP + 320mg SMX	8 mL oral suspension
Day 5	80mg TMP + 400mg SMX	One SS tablet
Day 6	160mg TMP + 800mg SMX	Two SS tablets

Abbreviation: TMP - Trimethoprim, SMX - Sulfamethoxazole, SS - single strength

Note: TMP/SMX oral suspension contains 8mg TMP and 40mg SMX per mL. One SS tablet contains 80mg TMP and 400mg SMX.

4.5.2 *Toxoplasma gondii* encephalitis

PLWH should have serum for *Toxoplasma gondii* IgG antibody done as soon as possible after the initial diagnosis of HIV if their CD4 count is less than 200 cells/mm³

Table 4.6 • Primary prophylaxis for *Toxoplasma Gondii*

When to start	What to start	When to stop
CD4 count < 100 cells/mm ³ who is also seropositive (IgG) for <i>Toxoplasma gondii</i>	Preferred: TMP/SMX (80/400 mg) 2 tablets PO daily Alternative: TMP/SMX (80/400 mg) 1 tablet PO daily Or	CD4 count > 200 cells/mm³ for > 3 months and sustained HIV viral load suppression in response to ART Or

When to start	What to start	When to stop
	Dapsone 50 mg tablet PO daily PLUS Pyrimethamine 50 mg once weekly PLUS Folinic acid 30 mg PO once weekly	CD4 count between 100-200 cells/mm ³ and HIV viral load remains suppressed for ≥ 6 months in response to ART
Restarting primary prophylaxis CD4 count <100 cells/mm ³ regardless of the HIV viral load Or CD4 100-200 cells/mm ³ and HIV viral load is not suppressed		

^a Individuals should be tested for glucose-6-phosphate dehydrogenase (G6PD) deficiency before administration of dapsone. An alternative agent should be used if the patient is found to have G6PD deficiency.

4.5.3 Cryptococcal meningitis / Cryptococcosis (localised non-meningeal disease)

Routine primary prophylaxis in the absence of a positive serum Cryptococcal Ag test is not recommended due to the low risk of disease, lack of survival benefit with prophylaxis, possible drug interactions and potential development of antifungal drug resistance. The chief means of prevention is the use of ART to optimise immune function.

4.5.4 Histoplasmosis

Primary prophylaxis is only indicated if CD4 count < 150 cells/mm³ AND the individual is at high risk of occupational exposure or living in a hyperendemic rate of histoplasmosis (>10 cases/100 person-years).

4.5.5 Disseminated *Mycobacterium Avium* Complex (MAC) Disease

Primary prophylaxis for MAC is not required for PLWH if ART is started promptly (within 2 weeks from presentation), regardless of the individuals' CD4 cell count.

4.5.6 Cytomegalovirus (CMV) disease

Primary prophylaxis to prevent CMV disease is not routinely recommended.

4.5.7 Talaromycosis

Primary prophylaxis is recommended for people who reside in hyperendemic regions (northern Thailand, Vietnam and Southern China).

4.5.8 *Rhodococcus* infections

Primary prophylaxis to prevent *Rhodococcus* infection is not routinely recommended.

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4.6 Vaccinations

4.6.1 Introduction

Vaccination is an essential component of comprehensive care for adults living with HIV. Immunodeficiency increases susceptibility to vaccine-preventable diseases and may lead to more severe illness, complications, prolonged hospitalisation, and higher mortality compared with the general population.

Effective antiretroviral therapy (ART) improves immune response to vaccines; however, vaccination should not be delayed solely to wait for immune reconstitution — especially in individuals at ongoing risk of exposure. Wherever possible, vaccination status should be reviewed during routine HIV care visits, hospital admissions, pregnancy, travel consultations, pre-exposure prophylaxis follow-up, and opportunistic infection recovery.

4.6.2 General principles of vaccination in HIV

- Inactivated, conjugate, toxoid, and recombinant vaccines are safe regardless of CD4 count.
- Live vaccines may be used only if CD4 ≥ 200 cells/mm³, stable on ART and when benefits outweigh risks.
- Vaccine immune responses may be reduced in patients with:
 - CD4 < 200 cells/mm³
 - Unsuppressed HIV viral load
 - Older age
 - Severe comorbidities
- Post-vaccination serologic testing may be considered for specific vaccines (e.g., HAV, HBV) to verify immune response

4.6.3 Timing of vaccination

- Do not delay vaccination waiting for immune recovery unless specifically indicated (e.g. live vaccine with low CD4).
- Vaccination may be offered regardless of ART timing, including newly diagnosed individuals.
- Individuals with recent opportunistic infection may still receive inactivated vaccines unless clinically unstable; decisions about live vaccines should always consider CD4 and immunologic status.

4.6.4 Special population

Vaccination recommendations for PLWH in special populations should be individualised according to immune status, clinical circumstances, pregnancy, travel plans and risk of exposure, as summarised in Table 4.7 below.

Table 4.7 • Vaccination considerations for special populations living with HIV

Population	Considerations
Pregnancy	Use inactivated vaccines only (e.g. Tdap, influenza) Live vaccines are contraindicated
Post-exposure or outbreak settings	Accelerated schedules may be used (e.g. MMR, varicella if CD4 $\geq$ 200 cells/mm ³) Shared decision-making advised
Travel	Review destination-specific vaccines (e.g. yellow fever, typhoid, meningococcal, hepatitis A) Certain live vaccines may be contraindicated; expert consultation advised
Asplenia or complement deficiency	Prioritize conjugate vaccines (e.g. meningococcal ACWY, Hib) in addition to standard immunisations

4.6.5 Recommended vaccines for adults PLWH

Recommended vaccines and schedules for adults living with HIV are summarised in Annexe 4.2.

4.6.6 Implementation strategies

- Integrate vaccination status review into routine HIV care pathway, ART initiation and opportunistic infection follow-up.
- Use electronic medical record (EMR) alerts or care checklists to prompt vaccination — especially pneumococcal, HBV, HPV, influenza and COVID-19.
- Provide patient-held vaccine cards / passports to track vaccinations over time, boosters and travel-related needs.
- Involve nurses, pharmacists and case managers in vaccine delivery, documentation, and reminders — important in busy HIV clinics.
- For live vaccines (MMR, varicella) — ensure CD4, viral load, ART adherence are reviewed before vaccination; consult infectious diseases / HIV specialist if CD4 low or unstable.

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5.1 Adverse events of ARTs

Refer to Annexe 5.1 for common and/or severe adverse events of ART

5.2 Dose adjustment of ART for renal and hepatic impairment

Refer to Annexe 5.2 for ART dosing recommendations in adults with renal or hepatic impairment

5.3 Common ART drug interactions

Clinically significant drug–drug interactions remain a major consideration in antiretroviral therapy, as they may reduce treatment efficacy or increase toxicity. Before initiating, switching, or adding any medication, including over-the-counter products, herbal supplements, and traditional remedies, potential interactions with antiretrovirals should be carefully reviewed.

For comprehensive and regularly updated information, healthcare professionals should refer to validated interaction resources such as the University of Liverpool HIV Drug Interactions tool (HIV iChart mobile application or www.hiv-druginteractions.org).

5.3.1 Nucleoside Reverse Transcriptase Inhibitors (NRTI) drug interactions

Clinically relevant NRTI interactions are listed in Annexe 5.3A; refer to the annexe for the full interaction table.

5.3.2 Non-Nucleoside Reverse Transcriptase Inhibitors (NNRTI) drug interactions

Clinically relevant NNRTI interactions are listed in Annexe 5.3B; refer to the annexe for the full interaction table.

5.3.3 Protease Inhibitor (PI) drug interactions

Clinically relevant PI interactions are listed in Annexe 5.3C; refer to the annexe for the full interaction table.

5.3.4 Integrase Inhibitor (INSTI) drug interactions

Clinically relevant INSTI interactions are listed in Annexe 5.3D; refer to the annexe for the full interaction table.

5.4 Dolutegravir (DTG)-related monitoring

5.4.1 Caution and monitoring

- Monitor creatinine 1-3 months post initiation of DTG.
- A rise in creatinine is expected with DTG due to the inhibition of tubular secretion of creatinine from proximal renal tubule, but no effect on glomerular filtration and usually does not progress after 1st month and should plateau. This increase is not associated with actual kidney damage or a reduction in GFR, and does not indicate impaired kidney function.
 - Creatinine rise < 25% or <30 mmol/L is still acceptable - Likely DTG related
 - Creatinine rise > 25% or >30 mmol/L - To workout for possible renal disease, stop TLD and consider other regimes
- Monitor liver function test
- Monitor for any allergic reaction such as rash

5.4.2 Preferred monitoring schedule

Table 5.1 • The recommended monitoring parameters for DTG-based regimens

Parameters	Frequency of monitoring
HIV VL	After ART initiation or switching due to virological failure: Repeat VL after 4 months. After switching for other indications: Repeat VL after 6 months Once stable virological suppression is achieved: Monitor VL every 6–12 months, depending on feasibility
CD4	CD4 < 200 cells/mm ³ : 3 to 6 monthly CD4 200-350 cells/mm ³ : annually CD4 count is >350 cells/mm ³ on two occasions at least one year apart, no further CD4 monitoring is required
FBC	Every 4 to 6 months
Renal function	At week 4 then week 12; then 4 to 6 monthly if stable
Liver function	At week 4 then every 4 to 6 months
FSL / FBS	Every 6 to 12 months

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6.1 Approach to the patient with detectable viral load

6.1.1 Viral “blips”

Defined as isolated transient rises in viral load to above detectable level while on treatment after having achieved prior viral suppression and is followed by re-suppression. The levels generally, do not exceed 200 copies/mL. It may reflect technical variations in laboratory assay performance, or biological events associated with viral replication (immunisation, other viral infection). Isolated “blips” are not associated with subsequent virologic failure.

6.1.2 Low-level viraemia (LLV)

LLV is defined as a persistently low, but detectable viral load, typically within the range of >50 to 1000 copies/mL.

While the exact threshold at which LLV becomes predictive of disease progression varies across studies, there is clear evidence that incomplete viral suppression poses significant risks of :

- **Virological and immunological failure:** It can lead to the accumulation of resistance mutations, resulting in an increase in viral replication, a reduction in CD4 cell counts, and a heightened risk of virologic progression and clinical deterioration.
- **Compromised future treatment:** Due to increasing drug resistance.

Management recommendations

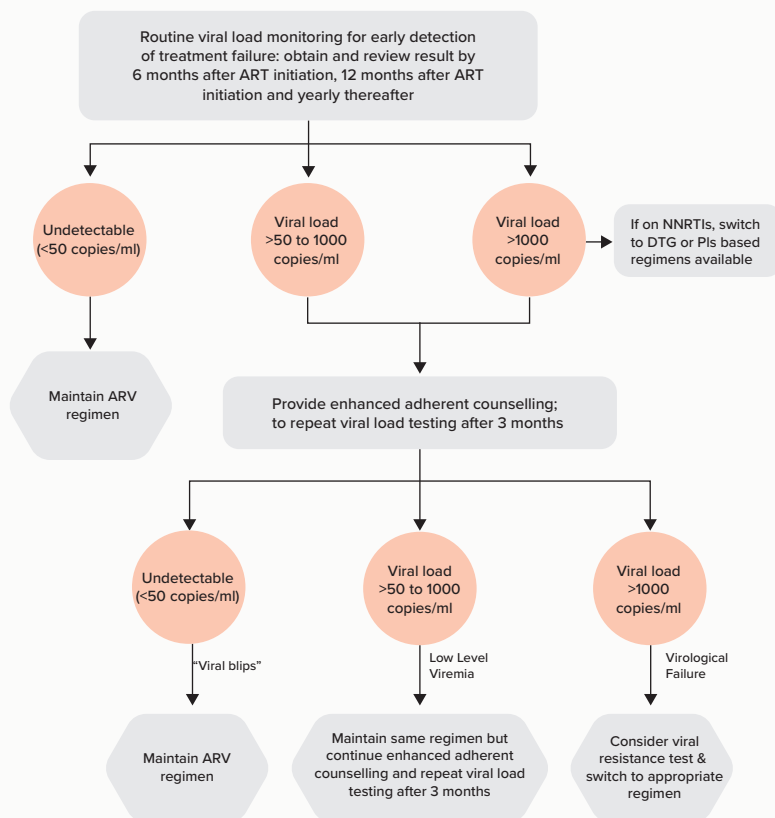
Patients experiencing LLV require:

1. **Reinforced adherence:** Provide enhanced adherence counselling on the importance of treatment adherence.
2. **Close monitoring:** Close surveillance with 2 to 4 monthly viral load testing. (expedite viral load results availability)
3. **Treatment switch:** Any indication of HIV disease progression or immunological failure should immediately prompt clinicians to switch the patient to 2nd line therapy.

6.1.3 Virologic failure (VF)

WHO defines virologic failure as a persistently detectable viral load above 1000 copies/mL. This diagnosis requires two consecutive viral load measurements, taken 3 months apart with adherence support provided between the measurements, after at least six months of starting a new ART regimen.

Figure 6.1 • Approach to managing detectable HIV viral load



Adapted from the WHO consolidated guidelines on HIV prevention, testing, treatment, service delivery and monitoring (2021)

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6.2 Risk factors for treatment failure

Treatment failure may result from interacting viral, drug-related, patient and programme factors. Early identification of these factors allows timely intervention to optimise treatment outcomes, as summarised in Table 6.1.

Table 6.1 • Risk factors for HIV treatment failure

Factor category	Associated risk factors
Viral factors	● Higher baseline HIV RNA level ● Lower pre-treatment or nadir CD4 T-cell count ● Prior AIDS diagnosis ● Presence of pre-existing drug resistance mutations ● Prior treatment failure, with development of drug resistance or cross resistance
Drug related factors	● Previous ART history using less potent regimens or low genetic barrier to resistance ● Drug side effects and toxicity ● Suboptimal pharmacokinetics (variable absorption, metabolism, food/fasting requirements, adverse drug-drug interactions with concomitant medications)
Patient factors	● Poor adherence to ART and missed clinic visits ● Co-morbidities (e.g. depression, active substance use) ● HIV/TB co-infection¹, presence of opportunistic infections during ART initiation², undernutrition²
Programme factors	● Drug supply factors ● Retention of patient to treatment programme

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6.3 Assessment of treatment failure

If treatment failure is suspected or confirmed, a thorough assessment should be done promptly because of increased risk of HIV disease progression.

Assessment of a patient with suspected or confirmed virologic failure should include:

- ART history
- Trends in HIV viral load and CD4 counts
- Occurrence of HIV-related clinical events (WHO stage 1 to 4)
- Physical examination to assess signs of clinical progression
- Results of prior resistance testing (if available)
- Factors potentially contributing to reduced plasma drug levels:

Table 6.2 • Causes and management of reduced antiretroviral drug exposure

Contributing factor	Assessment and management considerations
Poor or incomplete adherence	Defined as less than 90% to 95% of doses taken, is one of the most common reasons for virologic failure¹ Identify and address the underlying cause(s) of non-adherence (e.g. poor access to medication, mental health disorders, active substance use) Simplify the regimen if possible (e.g. decrease pill count or dosing frequency)
Incorrect dosing/frequency	Cause(s) include: ● Patient's educational level and health literacy ● Neurocognitive impairment
Drug intolerance	Patients who experience adverse effects may not adhere to their ART regimen Management strategies include: ● Symptomatic treatment (e.g. antiemetics, anti-diarrhoea) ● Changing one drug to another within the same class ● Changing drug classes (e.g. from an NNRTI to an INSTI/PI)
Pharmacokinetics	Cause(s) include: ● Suboptimal pharmacokinetics (e.g. variable absorption, metabolism) ● Food requirement/ fasting ● Drug-drug interaction: concomitant medications may reduce concentrations of ART

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6.4 Viral resistance testing

Genotypic, rather than phenotypic testing is the preferred resistance testing to detect the presence of drug resistance mutations in relevant viral genes. The result is useful in guiding ART regimen selection. In the management of virologic failure, viral resistance testing is recommended.

Viral resistance testing should only be done when the viral load is more than 1000 copies/ml and the results must be interpreted in the context of the patient's treatment history.

Drug-resistance testing should be performed while the person is taking the failing ART regimen or within 4 weeks of discontinuing ART regimen¹. If more than 4 weeks have elapsed since ART regimens were discontinued, resistance testing may not detect previously selected resistance mutations due to lack of drug-selective pressure.

6.4.1 First line treatment failure with no resistance mutation detected

The patient's medication history and adherence should be reassessed in cases of virological failure where no detectable resistant mutations are identified through viral resistance testing. The treating physician should assess potential contributing factors, such as non-adherence, drug–drug interactions or inadequate selective drug pressure at the time the test was conducted.

Once adherence issues have been appropriately addressed, 1st line ART may be either reinitiated or continued. However, the individual should be monitored carefully and repeat viral load testing performed after 3 months². If there is inadequate virological response, resistance testing should be performed to detect any archived resistant mutations. If resistance mutations are found, switch to a suppressive regimen based on drug and genotype history³.

6.5 Second line ART regimen

Table 6.3 summarises the preferred and alternative second-line ART regimens, taking into consideration the failing first-line regimen.

Table 6.3 • Recommended second line regimen⁴

Failing first line regimen	Preferred second line regimen	Alternative second line regimen
TDF + 3TC (or FTC) + DTG	TDF + 3TC (or FTC) + DRV/r	TDF + 3TC (or FTC) + ATV/r (or LPV/r)
TDF + 3TC (or FTC) + EFV	TDF + 3TC (or FTC) + DTG	TDF + 3TC (or FTC) + DRV/r (or ATV/r) (or LPV/r) AZT (or ABC) + 3TC + DRV/r (or ATV/r) (or LPV/r)
AZT + 3TC + EFV	TDF + 3TC (or FTC) + DTG	TDF + 3TC (or FTC) + DRV/r (or ATV/r) (or LPV/r) ABC + 3TC + DRV/r (or ATV/r) (or LPV/r)
ABC + 3TC + DTG	ABC (or TAF) + 3TC (or FTC) + DRV/r	TAF (or ABC) + 3TC (or FTC) + ATV/r (or LPV/r) AZT + 3TC + ATV/r (or LPV/r)

Note: Substitute TDF with TAF if CrCl < 50 mL/min

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6.6 Treatment experienced patients with limited or no therapeutic option

In a patient with a failing ART regimen with no other ART option, the decision whether to continue the failing regimen or not will be based on cost and side effects of the drugs in the failing regimen.

Continuing the failing ART regimen rather than stopping is beneficial provided that the patient has not developed any side effects and is clinically well. This has to be balanced with the fact that there is accumulation of mutations in the long term which may negatively impact future treatment options should they become available. Hence if a potentially viable ART regimen becomes available, it must be commenced as soon as possible.

6.6.1 Management of heavily treatment-experienced (HTE) PLWH

HTE patients are defined as individuals with two or fewer antiretroviral (ARV) drug classes available for use, with limited fully active ARV agents within each class.

Management focuses on optimising virologic control while preserving future options.

ART should be continued with regimens aimed at maintaining CD4 counts, delaying disease progression, preserving future treatment options and minimising toxicity.

Drug resistance testing should be performed while the patient is on the failing regimen to guide subsequent therapy. When constructing a new regimen, all prior genotypic, phenotypic and tropism data must be reviewed, as resistance is cumulative.^{1,3}

A new regimen should ideally include at least two fully active agents, preferably with one having a high barrier to resistance (e.g. DTG or boosted PI). Adding a single active drug to a failing regimen is strongly discouraged due to rapid resistance development.^{1,2}

If full suppression is not achievable due to extensive resistance, ART should be continued to maintain CD4 counts, delay disease progression, minimize toxicity, and preserve future treatment options. In such cases, regimens may include partially active NRTIs with close monitoring or prioritise three fully active agents if no high-barrier drug is available.^{1,2}

Dual therapy with DTG and boosted DRV may be considered if both remain active.^{1,2}

Novel agents with unique mechanisms such as ibalizumab, fostemsavir and lenacapavir provide additional options for constructing effective regimens in multidrug-resistant cases.^{3,4}

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7.1 HIV and TB co-infection

7.1.1 Introduction

Tuberculosis (TB) remains the leading opportunistic infection and causes high mortality among PLWH. They are 14 times more likely to develop active TB disease than people without HIV.¹ HIV-associated immunosuppression accelerates progression from TB infection to TB disease, while TB itself enhances HIV replication and systemic immune activation, creating a cycle that worsens clinical outcomes. The disruption of CD4⁺ T-cell-mediated immunity in HIV impairs containment of *M. tuberculosis*, leading to higher bacillary burden, atypical radiographic presentations and increased rates of extrapulmonary and disseminated disease². Because symptoms may be subtle or non-specific, diagnosis is often delayed.

7.1.2 Latent TB

A state of persistent immune response to stimulation by *M. tuberculosis* antigens with no evidence of clinically manifest active TB disease. Most infected people have no signs or symptoms of TB but are at risk for TB disease. This was previously referred to as latent TB infection (LTBI).

PLWH who are unlikely to have active TB should receive TB preventive treatment (TPT) shown in Table 7.1, regardless of CD4 count.

Presence of any of the following screening questions for active TB should prompt evaluation for TB and other opportunistic infection(s)¹:

- Current cough (any duration)
- Fever
- Night sweats
- Weight loss

Chest x-ray may be used to screen for TB disease¹.

Tuberculin skin test (TST) or interferon-gamma release assays (IGRAs) may be used to test for TB infection¹, although they are not required prior to starting latent TB treatment.

Table 7.1 • TB preventive treatment (TPT) regimens^{3,4}

Regimens	Drugs and dosing	Duration
3HR	Daily Isoniazid + Rifampicin + Pyridoxine 20mg	3 months
3HP	Weekly Isoniazid + Rifapentine Dosing: Isoniazid 300mg/Rifapentine 300mg FDC 3 tablets weekly	3 months
4R	Daily Rifampicin	4 months
6-9H	Daily Isoniazid + Pyridoxine 20mg	6 – 9 months
1HP	Daily Isoniazid 300mg + Rifapentine 600mg + Pyridoxine 20mg Dosing: Isoniazid 300 mg/Rifapentine 300mg FDC 1 tablet daily + Rifapentine 300mg daily + Pyridoxine 20mg daily	1 month

7.1.3 TB disease

Molecular rapid diagnostics such as Xpert MTB/RIF assay, is recommended to be used to screen for TB disease, especially if PLWH is hospitalised for HIV-related reasons¹.

Urine LF-LAM may be used to assist in the diagnosis of active TB disease in PLWH if:

- There is/are signs and symptoms of TB (pulmonary and/or extrapulmonary) or
- Seriously ill or
- CD4 cell count of less than 100 cells/mm³ (irrespective of signs and symptoms of TB)¹

Choice of anti-TB is similar to non-HIV infected patients³.

Regime for TB treatment as shown in Table 7.2:

- 4-drug regimen: Isoniazid (H), Rifampicin (R), Ethambutol (E), Pyrazinamide (Z) or a fixed-drug combination (FDC) regimen based on weight
- The regimen stated below is the minimum duration of treatment. The duration of treatment may be prolonged, depending on immune reconstitution and clinical response.

Table 7.2 • TB treatment regimens

TB		Regimens ³
Drug-susceptible pulmonary TB		2HREZ/4HR ● 2 months intensive: HREZ + Pyridoxine, followed by ● 4 months maintenance: HR + Pyridoxine
Drug-susceptible extra-pulmonary TB (EPTB)	TB meningitis	2HREZ/10HR ● 2 months intensive: HREZ + Pyridoxine, followed by ● 10 months maintenance: HR + Pyridoxine
	TB bone or joint	2HREZ/4-7HR ● 2 months intensive: HREZ + Pyridoxine, followed by ● 4-7 months maintenance: HR + Pyridoxine
	Other EPTB (e.g. TB lymphadenitis)	2HREZ/4HR ● 2 months intensive: HREZ + Pyridoxine, followed by ● 4 months maintenance: HR + Pyridoxine

7.1.4 Timing of ART initiation in TB co-infection

For the recommended timing of ART initiation in PLWH with active tuberculosis, refer to Section 4.4.

7.1.5 Preferred ART and antituberculosis regimen

Selection of ART and anti-TB regimens should account for potential drug–drug interactions. Preferred regimens and required adjustments are summarised in Table 7.3. Refer to Annex 4.1 for special considerations when initiating ART.

Table 7.3 • Anti-TB regimen selection according to ART regimen³

Current ART regimen	Preferred anti-TB regimen
NNRTI-based ART	Standard anti-TB regimen
PI-based ART	Rifabutin based anti-TB regimen Do not co-administer with rifapentine or rifampicin
INSTI-based ART	Rifabutin based anti-TB regimen is preferred If rifampicin or rifapentine based regimen is used, the INSTI dose must be adjusted: ● Dolutegravir 50mg BD

7.1.6 TB-Immune Reconstitution Inflammatory Syndrome (TB-IRIS)

Definition of paradoxical TB-IRIS: worsening of symptoms of a known disease, either at a new body site or at the original body site.⁵

Risk factors:

- Low baseline CD4 cell count
- Extensive or disseminated TB
- Early initiation of ART
- Rapid immunological and virological responses to ART

Clinical features of TB-IRIS:

- Fever
- Increased or new lymphadenopathy
- New or worsening pulmonary lesions
- Pleural and pericardial effusions
- Ascites
- Psoas abscess
- Cutaneous lesions
- New or expanding CNS tuberculomas

Management of TB-IRIS:

- Corticosteroids may be used. The effective dose and duration have scarce evidence.
- Concurrent initiation of ART and prophylactic prednisolone in PLWH with CD4 cell count < 100 cells/mm³, who started anti-TB treatment within 30 days prior to ART, may reduce risk of TB-IRIS by 30%.
- The suggested oral prednisolone dose is 40 mg daily for 2 weeks, followed by 20 mg daily for 2 weeks ⁵.

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7.2 HIV and Hepatitis B co-infection

Chronic Hepatitis B virus (HBV) infection is common among PLWH and accelerates progression to cirrhosis, end stage liver disease and hepatocellular carcinoma.¹ Globally, about 5-20% of PLWH are co-infected with HBV, representing more than 3 million people, particularly in sub-Saharan Africa and Southeast Asia.²

Management of HIV/HBV co-infection should therefore prioritise^{3,4,5}:

- Universal HBV serological testing at HIV diagnosis
- Integration of HBV-active antiretroviral therapy for all co-infected persons, regardless of ALT or HBV DNA levels
- Appropriate hepatocellular carcinoma (HCC) and variceal surveillance for those with cirrhosis or other risk factors.

7.2.1 Effects of HIV on HBV disease progression

Compared to HBV mono-infection, HIV/HBV coinfection is associated with:

- Lower probability of spontaneous clearance of acute HBV infection
- Higher HBV replication with paradoxically lower transaminases levels
- More rapid decline in Hepatitis B surface antibody (anti-HBs)
- More episodes of reactivation
- Lower seroconversion rates from HBeAg to anti-HBe antibody
- Less necroinflammation on liver biopsy, but accelerated progression to fibrosis and cirrhosis.⁶
- Higher liver-related mortality.⁷
- Decreased treatment response compared with persons without HIV coinfection

7.2.2 Effects of Antiretroviral Therapy (ART) on HBV

It is not uncommon to see elevations in transaminase levels after initiation of ART. The rises in transaminases are due to immune restoration disease with hepatic flares and/or toxicity of ART.⁸

7.2.3 Goals of therapy

The goal of therapy for chronic HBV infection is to improve quality of life and survival of infected patients by preventing progression of the diseases to cirrhosis, end stage liver disease and eventually reducing the risk of hepatocellular carcinoma and liver-related mortality.

The following surrogate markers are instrumental in defining treatment success:

- Persistent suppression of HBV DNA (preferable undetected HBV DNA) is the primary goal of antiviral therapy.
- HBsAg loss is the ultimate goal therapy.
- Normalisation of ALT is an additional endpoint.

Additional goals of antiviral therapy in HBeAg positive patients is seroconversion from HBeAg to anti-HBe antibody.

7.2.4 Pre-treatment assessment

All HIV/HBV coinfecting patients should undergo:

- History and physical examination: treatment history, hepatotoxic medications and evidence of advanced liver disease
- Laboratory tests: full blood count, renal profile, liver function test, coagulation profile
- Virological tests: HBeAg, anti-HBe antibody $\pm$ HBV-DNA viral load by PCR (Quantitative), anti-HCV
- Vaccination status: check HAV IgG/Total
- Liver disease staging: stage of liver fibrosis via non-invasive methods (serum biomarkers such as AST to platelet ratio index (APRI), Fibrosis-4 (FIB-4) scoring or ultrasound-based technology such as liver elastography (FibroScan). Liver elastography is generally the preferred non-invasive method. Consider liver biopsy if it is deemed necessary

7.2.5 Treatment recommendations for HIV/HBV co-infection

- Immediate ART initiation is recommended in all HIV/HBV co-infected individuals regardless of CD4 cell count or ALT.^{3,9}
- All HIV/HBV co-infected individuals should receive ART that includes two HBV-active agents - tenofovir: either tenofovir disoproxil fumarate (TDF) or tenofovir alafenamide (TAF) plus lamivudine (3TC) or emtricitabine (FTC). Avoid use of 3TC/FTC as the sole HBV-active drug.^{3,9,10}
- For HIV/HBV co-infected persons with CKD, tenofovir-inclusive regimens may require dose adjustment. TDF/FTC inclusive regimens require dose adjustment if creatinine clearance < 50 ml/min, and TAF/FTC inclusive regimens are not recommended in patients with a creatinine clearance < 30 ml/min unless patient is on haemodialysis, where it still may be used.⁹⁻¹¹
- Switching from TDF to TAF is recommended if side effects like renal insufficiency or bone mineral disorders occur.⁹
- Refer to Annexe 4.1 for special considerations when initiating ART, Annexe 5.2 for ART dosing recommendations in adults with renal or hepatic impairment, and Section 4.2 for available ART options.
- If TDF or TAF is strictly contraindicated, entecavir should be used in addition to a fully suppressive ARV regimen. Entecavir has weak activity against HIV. Using entecavir for HBV treatment without ART may result in selection of the M184V mutation that confers HIV resistance to 3TC and FTC.¹⁰
- All patients with chronic HBV should be evaluated to assess the severity of HBV infection. The presence or absence of clinical evidence for cirrhosis should be assessed to guide management and surveillance.
- Patients with chronic HBV should be advised to receive Hepatitis A vaccination (if anti-HAV IgG/Total is negative), abstain from alcohol and counsel on prevention methods that protect against both HBV and HIV transmission.

- The duration of treatment of HBV/HIV co-infection is lifelong.^{3,12}
- Non adherence or discontinuation of anti-HBV treatment is not recommended as it may result in transaminase flares and hepatic damage.¹⁴
- When ART regimens are altered (e.g. due to HIV resistance), drugs that are effective against HBV should not be discontinued without substituting another drug that has activity against HBV.¹⁴
- TDF-based regimens remain safe and effective in pregnancy and prevent mother-to-child transmission of both HIV and HBV.³
- HBV DNA should be repeated every 3-6 months initially to ensure effective HBV suppression, which is defined as HBV DNA below the level of detection. Once suppression is achieved and stable, monitoring can be performed every 6-12 months. It is important to monitor HBV DNA levels in HBV/HIV co-infected person because elevation of HBV DNA predates ALT abnormalities, thus if only ALT level is monitored, it tends to miss the early phase of uncontrolled HBV DNA replication.¹¹
- In patients who are HBeAg-positive, HBeAg and anti-HBe antibody should be monitored annually to determine if seroconversion has occurred. More frequent monitoring may be considered if ALT fluctuations occur.^{9,12}
- Role of Hepatitis B core Ab (HBcAb): HBV reactivation has been reported in people with HIV and prior exposure to HBV (positive HBcAb, negative HBsAg) when HBV active drugs are withdrawn as part of an ART regimen. Regardless of Hepatitis B surface antibody level, people with HIV and prior HBV exposure are at low risk of reactivation and the associated hepatitis, which can result in serious hepatocellular injury.¹³ A monitoring strategy (e.g. ALT monitoring every 1 to 3 months for 6 months) is considered a safe and effective way to assess HBV reactivation in low-risk people.⁵

7.2.6 Hepatocellular carcinoma (HCC) and variceal surveillance

- **HCC surveillance:**¹⁵
 - Ultrasound of liver +/- Alpha-fetoprotein every 6 months for:
 - All with cirrhosis
 - Males > 40 years and females >50 years
 - Those with a first-degree family history of HCC
- **Oesophageal varices surveillance:**
 - Surveillance of oesophageal varices via upper endoscopy should be offered for all patients with cirrhosis at diagnosis, unless they meet non-invasive criteria to defer endoscopy.¹⁵
 - Patients with liver stiffness measurement (LSM) ≤ 15 kPa and platelet count $>150 \times 10^9/L$ have a very low risk of high-risk varices and can safely avoid endoscopy. These patients should undergo annual reassessment with elastography and platelet count.¹⁶⁻¹⁸
 - For patients with no varices at screening endoscopy, surveillance endoscopy should be repeated at 2- to 3-year intervals, depending on liver disease activity. Shorter intervals (2 years) are advised if liver injury is ongoing; longer intervals (up to 3 years) may be appropriate if disease is stable.¹⁶⁻¹⁸

- For patients with small varices at screening endoscopy, surveillance endoscopy should be repeated at 1- to 2-year interval, tailored based on disease progression risk.¹⁶⁻¹⁸ Non-selective beta blockers such as propranolol or carvedilol should be prescribed for primary prophylaxis of variceal haemorrhage.
- Those patients with variceal bleeding should be co-managed with gastroenterologists/hepatologists, with combined secondary prophylaxis using propranolol or carvedilol and endoscopic band ligation.¹⁸⁻¹⁹

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7.3 HIV and Hepatitis C coinfection

7.3.1 Global burden and risk

The global estimate for HIV/HCV co-infection is 2.3 million people¹. Globally, the co-infection prevalence is 6.2% among PLWH¹. The co-infection prevalence of HCV infection among PLWH are highest in people who inject drugs (82.4%), followed by men who have sex with men (6.4%).¹

7.3.2 Disease progression and impact

HIV/HCV co-infection is a major cause of non-AIDS related morbidity and mortality in PLWH. Treatment for Hepatitis C Virus (HCV) is a priority in HIV/HCV co-infection because HIV significantly accelerates liver fibrosis progression. Effective treatment prevents complications associated with liver cirrhosis and viral eradication substantially reduces the rate of extrahepatic deaths².

7.3.3 Treatment and management recommendations

7.3.3.1 Screening and Initial Assessment

- **Screening:** All PLWH, especially those with HCV risk factors (such as ongoing intravenous drug use (IVDU), chemsex and unprotected anal intercourse), should be screened with anti-HCV antibody test. If positive, proceed with HCV RNA or HCV core antigen test to confirm viremic infection.
- **Initial assessment:** Assessment includes thorough history, clinical examination and standard blood investigations. Liver cirrhosis must not be missed and such patients should be referred to a hepatologist. HCV genotyping is often not required with current pan-genotypic Direct Acting Antivirals (DAA), even in patients with liver cirrhosis for certain DAA regimens.
- **Liver fibrosis assessment:** Non-invasive tests such as the APRI (Aspartate Platelet Ratio Index) or FIB-4 (Fibrosis-4) scores are recommended to:
 - Determine appropriate treatment duration and regimen
 - Decide if additional screening is required for varices (OGDS) or hepatocellular carcinoma (HCC) (6-monthly USS and AFP)
 - Diagnosis cut-offs (for DAA therapy):
 - i) No cirrhosis: No clinical evidence of cirrhosis and Fibroscan ≤ 12.5 kPa or APRI ≤ 1.0
 - ii) Compensated cirrhosis: Clinical evidence of cirrhosis* or Fibroscan > 12.5 kPa or APRI > 1.0 or FIB-4 score > 2.2 , and no decompensation (Child-Turcotte-Pugh class A)

**Clinical evidence of cirrhosis includes: nodular liver margin and/or splenomegaly on imaging, platelet $< 150,000$ or presence of varices.*

Table 7.4 • Formula for APRI and FIB-4 scores

Formula for APRI	Formula for FIB-4
$\text{APRI} = \frac{\frac{\text{AST U/L}}{\text{AST U/L (upper limit normal)}}}{\text{Platelet count } (10^9 \text{ L}^{-1})} \times 100$	$\text{FIB-4} = \frac{\text{age (years)} \times \text{AST U/L}}{\text{Platelets } 10^9 \text{ L}^{-1} \times \sqrt{\text{ALT U/L}}}$
<i>Note: For APRI, AST upper limit normal represents the upper limit of normal for AST in the laboratory where these investigations were undertaken.</i>	

7.3.3.2 Antiretroviral Therapy (ART)

- ART initiation is recommended for all PLWH with HCV co-infection, regardless of CD4 count and should be started as soon as possible, preferably prior to DAA.
- Effective ART, leading to HIV viral suppression is crucial as it slows the rate of HCV disease progression.³

7.3.3.3 HCV Treatment with Direct-Acting Antivirals (DAA)

Table 7.5 • Choice of DAA and duration of treatment according to ART

ART regimen	Choice of DAA and duration of treatment	
	Without cirrhosis	With compensated cirrhosis
NNRTI (EFV/NVP)	SOF 400mg daily + RDV 200mg daily for 12 weeks	SOF 400mg daily + RDV 200mg daily for 24 weeks
INSTI (DTG)	SOF 400mg daily + RDV 200mg daily for 12 weeks (based on limited unpublished PK data)	SOF 400mg daily + RDV 200mg daily for 24 weeks (based on limited unpublished PK data)
	Or	Or
	SOF 400mg daily + DCV 60mg daily for 12 weeks	SOF/VEL 400/100 mg daily for 24 weeks or + RBV for 12 weeks
	Or	Or
	SOF/VEL 400/100 mg daily for 12 weeks	If HCV Genotype available: SOF 400mg daily + DCV 60mg daily ± RBV for 12-24 weeks

ART regimen	Choice of DAA and duration of treatment	
	Without cirrhosis	With compensated cirrhosis
PI (other than ATV/r)	SOF 400mg daily + DCV 60mg daily for 12 weeks Or SOF 400mg daily + RDV 200mg daily for 12 weeks Or SOF/VEL 400/100 mg daily for 12 weeks	SOF 400mg daily + RDV 200mg daily for 24 weeks Or SOF/VEL 400/100 mg daily for 24 weeks or + RBV for 12 weeks Or If HCV Genotype available: SOF 400mg daily + DCV 60mg daily ± RBV for 12-24 weeks
PI (ATV/r)	SOF 400mg daily + RDV 200mg daily for 12 weeks Or SOF/VEL 400/100 mg daily for 24 weeks + RBV for 12 weeks Or SOF 400mg daily +DCV 30mg daily for 12 weeks Or Change HIV regimen to alternative PI or INSTI	Consult

Abbreviation: SOF - Sofosbuvir, VEL - Velpatasvir, RDV - Ravidasvir, RBV - Ribavirin, DCV - Daclatasvir

7.3.3.4 Drug-Drug Interactions (DDIs) and monitoring

- **Drug-Drug Interactions (DDIs):** It is crucial to check for DDIs, especially with ART, using reliable resources and adjust the DAA dose if necessary. Refer to Annexe 5.3 for common ART-drug interactions. This check applies to all concomitant medications before starting DAA and whenever a new medication is introduced.
 - Studies have shown that SOF + RDV generally has no clinically significant impact on the concentrations of ART drugs.

- **Monitoring for ribavirin-free DAA treatment**, monitoring is usually minimal:
 - Scheduled follow-up is advisable at week 4.
 - Routine laboratory monitoring (LFT, serum creatinine) should be limited to weeks 4 and 12 post-DAA treatment.
- **Monitoring for ribavirin-based DAA treatment**, monitoring is usually:
 - Scheduled follow-up is advisable at week 2, 4 and every 4 weeks till completion of treatment .
 - Routine laboratory monitoring (FBC , LFT, serum creatinine) on these visits and 12 post-DAA treatment.
- HCV RNA level should be ordered 12 weeks post-treatment. An undetectable result is defined as sustained virological response (SVR12), which is equivalent to a cure.
- **After achieving SVR12:**
 - PLWH should return to standard of care regardless of cirrhosis status.
 - Patients with liver cirrhosis should continue to have 6-monthly HCC surveillance.
 - The need for subsequent variceal surveillance should be determined in consultation with a gastroenterologist/hepatologist.

7.3.3.5 Re-infection and prevention

- **Harm reduction and counselling:** patients should be linked to harm reduction programmes such as opioid substitution therapy and counselled regarding the risk of HCV re-infection post treatment.
- **Annual (or ideally bi-annual) HCV PCR testing** for re-infection is recommended for high-risk populations after DAA-induced SVR.

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8.1 Cardiovascular disease (CVD)

8.1.1 Background

PLWH have a 1.5- to 2-fold increased risk of developing cardiovascular diseases, including myocardial infarction, stroke, ischaemia, heart failure, and sudden cardiac death compared to people without HIV.^{1,2} The mechanisms of the increase in risk of cardiovascular disease in this population are not completely understood but may relate to traditional risk factors, residual inflammation and persistent immune activation due to HIV virus. The risk is increased in the presence of active viraemia and decreases with ART, but still remains elevated when compared to those without HIV.^{3,4}

Effective ART has increased survival in PLWH, resulting in an increasing number of older people with HIV. Thus, it is not surprising that non-infectious comorbidities and complications, including atherosclerotic cardiovascular diseases (ASCVD) and non-AIDS malignancies, now account for a growing proportion of the causes of death among PLWH.⁴ ASCVD includes history of acute coronary syndromes (ACS), myocardial infarction (MI), stable or unstable angina, coronary or other arterial revascularisation, stroke, transient ischaemic attack (TIA), or peripheral artery disease (PAD).⁵ The TREAT Asia HIV Observational Database projected that incidence of new CVD will double over a 10-year period in people with HIV in Asia between 2019 and 2028.²

Cardiometabolic health encompasses a spectrum of parameters (e.g. blood pressure, body weight, lipids) that influence an individual's likelihood of developing cardiometabolic disease. Primary prevention of CVD in individuals with HIV are still focused on addressing traditional risk factors for CVD. Non-traditional risk factors for CVD, including alcohol use disorders, depression, and hepatitis C virus co-infection, and hepatic steatosis are also more common in people with HIV.² It is thus increasingly important that clinicians and PLWH are aware of this increased risk and how to reduce it through lifestyle modifications, personalized ART choices and the use of risk modified medications (such as statins) in those who warrant it.⁶

8.1.2 Screening/Diagnosis

To monitor cardiometabolic health, it is recommended for healthcare providers to assess metabolic risk factors for all PLWH at first visit (if possible), annually, and after ART initiation and switch (1-3 months after ART initiation or switch). Refer to separate chapter(s) for subsequent monitoring frequency in case of abnormal parameters. These assessments include:

- Weight and body mass index (BMI) and waist circumference
- Blood pressure
- Fasting lipid profile (FLP)
- Fasting blood glucose (FBS)
- Urinalysis (at first visit and when clinically indicated- in patients with CKD or DM)

It is also recommended to assess alcohol and substance use, and smoking. Elevated alanine transaminase levels may suggest metabolic dysfunction-associated steatotic liver disease (MASLD), which may need further evaluation.^{2,6}

Currently there is no scientific consensus regarding the most accurate tool for the assessment of ASCVD risk in PLWH as every risk prediction algorithm presents advantages and drawbacks. In general, a risk estimate score from a population with characteristics similar to those of the patient evaluated should be used.⁸ The 2026 ACC/ AHA Guideline on The Management of Dyslipidemia support using PREVENT-ASCVD equation to estimate 10-year ASCVD risk in all adults aged 30 to 79 without ASCVD or subclinical atherosclerosis and with an LDL-C level 1.8–4.9 mmol/L and in PLWH.⁵ Other available calculators are Pooled Cohort Equations (PCE), Framingham Risk Score (FRS), SCORE2 for individuals <70 years old or SCORE2-OP for individuals ≥ 70 years old, or Data Collection on Adverse Effect of Anti-HIV Drugs (D:A:D) model, which was derived from PLWH data and incorporates HIV-specific factors for 5-year ASCVD risk estimate.^{7,9} PREVENT-ASCVD has been shown to have greater precision and accuracy of risk estimates compared to PCE.⁵

Unfortunately, a Malaysian CV risk score is not yet available. The Malaysian Clinical Practice Guideline on Primary and Secondary Prevention of CVD 2017 recommends the use of FRS for risk stratification.¹⁰ A local study published in 2023, which validated four CV risk scores (FRS, SCORE2, RPCE, and World Health Organization cardiovascular disease (WHO CVD) models, however suggested RPCE to be most clinically useful to predict CVD risk. Both FRS and RPCE showed good discrimination, while other models only showed reasonable discrimination. RPCE was however more sensitive than FRS.¹¹

The presence of HIV-related risk factors justifies a 1.5 to 2-fold upward adjustment of calculated risk (Table 8.1).^{7,8} Decision for primary prevention such as statin initiation are guided by these risk estimates.

Considering the higher risk of ASCVD in PLWH, risk assessment should be routinely implemented in individuals aged 40 and above.⁴ For adults 20 to 39 years of age, it is reasonable to assess traditional ASCVD risk factors at least every 4 to 6 years.⁸

8.1.3 Management

Lifestyle modifications such as diet, obesity management, exercise, smoking cessation, glycaemic control, blood pressure and lipid lowering for those with dyslipidemia are recommended for primary prevention of ASCVD.⁴ Observational studies suggest that smoking cessation results in about 50% less risk of IHD, and this is additive to other interventions.¹²

All PLWH should maintain viral suppression with continuous ART to reduce the risk of ASCVD.⁴ In individuals with high ASCVD risk, it is recommended to avoid LPV/r based regimens. If a boosted PI is the desired option, an ATV-based regimen may have advantages over a DRV-based regimen. ABC based regimen should be avoided, as it has been shown to increase the hazard of major adverse cardiovascular event.^{3,14}

8.1.4 Recommendations

- PLWH have a 1.5- to 2-fold increased risk of developing cardiovascular diseases, including myocardial infarction, stroke, ischaemia, heart failure and sudden cardiac death compared to people without HIV.
- Multifactorial risk factors for ASCVD in PLWH including traditional risk factors of CVD such as smoking, hypertension, dyslipidemia, obesity and residual inflammation and persistent immune activation due to HIV virus, even in individuals effectively treated with ART and side effects of ART.

- Cardiometabolic health monitoring is recommended for all PLWH, such as weight, waist circumference, body mass index (BMI), blood pressure, fasting lipids, fasting glucose, and urinalysis (at first visit, annually, and after ART initiation and switch.
- CVD risk estimates scores such as PREVENT-ASCVD, RPCE, FRS or D:A:D score can be used in PLWH.
- Lifestyle modifications such as diet and obesity management, exercise, smoking cessation, blood pressure, glycemic control, and lipid lowering for those with dyslipidemia. Ensure the patient is virologically suppressed.
- Aspirin is not routinely recommended as primary prevention of CVD among adults >70 years of age or among adults of any age who are at increased risk of bleeding. Individualised and shared decision making for the primary prevention of ASCVD among select adults 40 to 70 years of age who are at higher risk ASCVD risk (10% or greater 10-year cardiovascular disease risk) but not at increased bleeding risk.

Table 8.1 • HIV and non-HIV related risk enhancers for 10-year CVD risk estimates⁵

HIV-related risk enhancers	Non-HIV related risk enhancers
● Prolonged HIV viraemia/delay in ART initiation ● Current or nadir CD4 <350 cells/mm³ ● HIV treatment failure or non-adherence ● HCV co-infection ● Exposure to older antiretroviral (ARV) drugs associated with cardiometabolic toxicity	● Family hx of early ASCVD (men <55 years old, women <65) ● Persistently LDL-C 4.1-4.9mmol/l ● Chronic kidney disease ● Pre-eclampsia, premature menopause ● Chronic inflammatory conditions (rheumatoid arthritis, psoriasis, HIV) ● High risk ethnicity (e.g. South Asian Ancestry, Filipino) ● Triglycerides ≥2.0 mmol/l (non fasting) or 1.7mmol/l if fasting ● High-sensitivity C-reactive protein ≥2.0 mg/dL on >1 occasion ● Elevated lipoprotein (a) ≥50 mg/dL or ≥125 nmol/L ● High polygenic risk

8.2 Dyslipidaemia

A large “randomised controlled trial (The Randomised Trial to Prevent Vascular Events in HIV - REPRIEVE) which enrolled more than 7000 individuals with HIV aged 40-75 years receiving ART around the world concluded that in individuals with HIV with low to moderate traditional risk for cardiovascular disease, pitavastatin 4mg daily is safe, effective and prevents major adverse cardiovascular events (MACE) such as heart attack and stroke by 35% compared to placebo over an average of 5 years follow-up period. The results from this study provide evidence on the use of statin therapy as primary prevention among people with HIV who are on ART with low-to-intermediate ASCVD risk.^{3,4}

In terms of recommendations for PLWH age 40-75 years old, statin recommendations according to CVD risk estimates are as below.^{3,4,5}

- High risk: high intensity statin
- Borderline risk: moderate intensity statin
- Intermediate risk: moderate to high intensity statin
- Low risk: evaluate presence or absence of HIV-related factors that can increase ASCVD risk. If present, consider moderate intensity statin
- With diabetes mellitus: moderate intensity statin and perform further risk assessment to consider using high intensity statin

Refer to Table 8.2 for intensity of statin therapy

Refer to Figure 8.1 for treatment recommendations for statin initiation in PLWH

8.2.1 Monitoring of PLWH on lipid lowering therapy

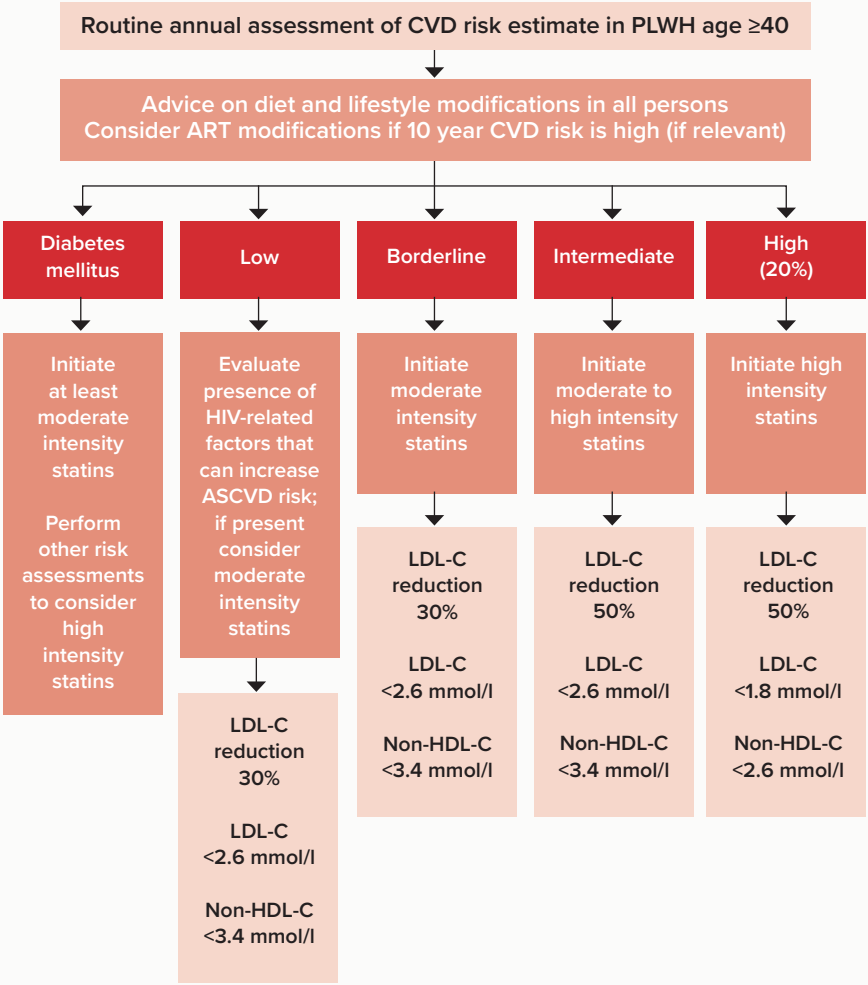
For PLWH starting on lipid lowering therapy, it is recommended for clinicians to perform a lipid profile 4-12 weeks after initiation or dose adjustment, and every 6-12 months thereafter to assess efficacy and adherence to lipid lowering therapy.⁵

Table 8.2 • Intensity of statin therapy

	High intensity	Moderate intensity	Low intensity
LDL-C Lowering	≥50%	30-49%	30-49%
	Atorvastatin 40–80 mg Rosuvastatin 20–40 mg	Atorvastatin 20 mg Rosuvastatin 10 mg Lovastatin 40–80 mg Pravastatin 40–80 mg Simvastatin 20–40 mg Pitavastatin* 4 mg	Pravastatin 10–20 mg Simvastatin 10 mg Lovastatin 20 mg

*Not readily available in Malaysia

Figure 8.1 • Treatment recommendations algorithm for statin initiation in PLWH



Adapted from Panel on Antiretroviral Guidelines for Adults and Adolescents (2024)

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8.3 Diabetes Mellitus (DM)

8.3.1 Introduction

The combination of both HIV and diabetes epidemics represents a synergistic health crisis. Malaysia faces a high background prevalence of diabetes, affecting approximately 20% of the adult population according to national health surveys.¹

For PLWH, this risk is amplified by a complex “triad” of factors: traditional risks inherent to an ageing population, the persistent inflammatory milieu associated with chronic viral infection and the metabolic toxicities of lifelong exposure to ART.² Studies show higher DM prevalence in PLWH (10-25% versus 4-15% in general population) with significant overlap in risk factors like obesity. However, some PLWH develop DM younger and sometimes without obesity.^{14,15} A higher risk of insulin resistance and DM has been described in PLWH on ART compared with non HIV infected patients.³

Additionally, associations between some ART and type 2 DM are more common with older ART such as stavudine, zidovudine, indinavir, and lopinavir and less common with tenofovir, emtricitabine, lamivudine, darunavir, atazanavir and dolutegravir. Coinfection with Hepatitis C virus (HCV) also increases the risk of type 2 DM in PLWH.⁴

Multiple observational studies have associated INSTI, particularly dolutegravir with:

- Weight gain
- Treatment emergent obesity
- Diabetes incident in some cohorts

The coexistence of HIV and DM complicates clinical management and worsens prognosis. Poorly controlled diabetes can impair immune function, potentially hindering CD4 T-cell recovery and increasing susceptibility to infections. Conversely, HIV-associated inflammation accelerates the vascular complications of diabetes. PLWH with diabetes have a higher risk of developing diabetic kidney disease (DKD), which may be compounded by HIV-associated nephropathy (HIVAN) or tenofovir-induced renal toxicity.⁸

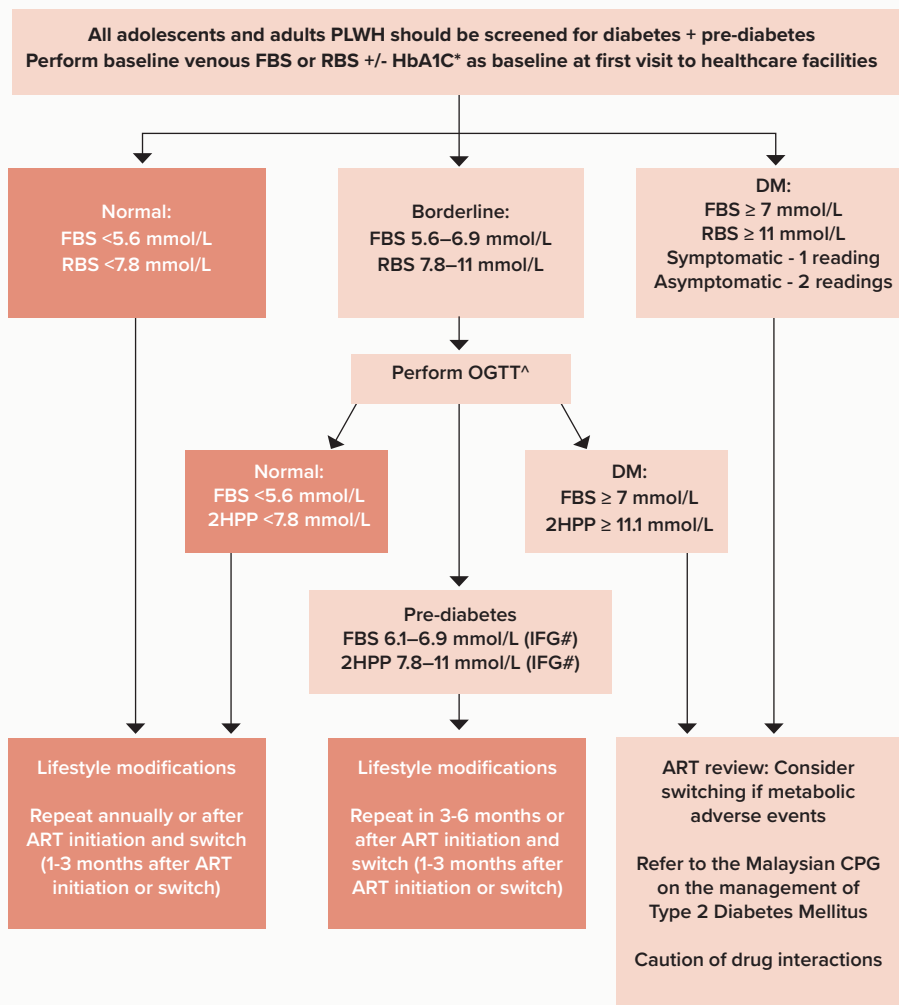
8.3.2 Clinical context of DM in PLWH

The management of DM in PLWH is a dual challenge, it requires the aggressive attainment of metabolic targets while simultaneously maintaining HIV viral suppression. The choice of therapeutic agents must be carefully tailored to account for significant DDIs, renal function status and the potential for cumulative adverse effects (e.g. weight gain, bone density loss). The main goals are to prevent acute hyperglycaemic complications and reduce the long-term risk of microvascular and macrovascular events.⁷

Current national guidelines for the general population should serve as the guiding tool for treatment of diabetes in PLWH. Lifestyle modification is the foundation of management and should be initiated at diagnosis for all patients which include dietary advice, physical activity and smoking cessation.

The selection of glucose-lowering therapy in PLWH is guided by the Malaysian CPG on the Management of Type 2 Diabetes Mellitus but requires specific adjustments for ART drug interactions (refer to Annexe 5.3).

Figure 8.2 • Management algorithm for diabetes mellitus in PLWH



Abbreviations: FBS - fasting blood glucose, RBS - random blood glucose, OGTT - oral glucose tolerance test, IFG - impaired fasting glucose, 2HPP - 2-hour post-prandial blood glucose

*HbA1C—is not recommended as a single screening tool in diagnosis of diabetes in PLWH and should only be used concurrently with FBS as it may underestimate glycemia in PLWH

^ OGTT—load with 75-gram oral anhydrous glucose dissolved in water

Values for IFG may differ, 5.6-6.9 is described in certain guidelines

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8.4 Hypertension

Hypertension is a known risk factor for CVD and consistently identified as the most common co-morbidity among PLWH worldwide. Meta-analyses and large cohort studies indicate that the prevalence of hypertension in PLWH receiving ART ranges widely from 6% to over 50%, depending on the setting and the age structure of the cohort.¹ Crucially, even after adjusting for traditional risk factors such as age, smoking and BMI, PLWH often exhibit a higher risk of hypertension compared to HIV-negative controls, suggesting the presence of HIV-specific pathogenic drivers.²

WHO recommends assessment for major non-communicable diseases such as hypertension at HIV diagnosis or initiation of ART.⁴ A blood pressure check should be performed at least annually.⁵ Screening, diagnosis and management of hypertension in the PLWH are the same as those of for the general population (refer to Figure 8.3).

The Malaysian CPG on Management of Hypertension 5th Edition emphasises opportunistic screening, recommending blood pressure measurement at “every chance encounter” and specifically at every scheduled clinic visit for PLWH.⁶ Given the high rate of undiagnosed disease, this high-frequency screening is critical. The diagnostic criteria differ slightly as in Table 8.3.

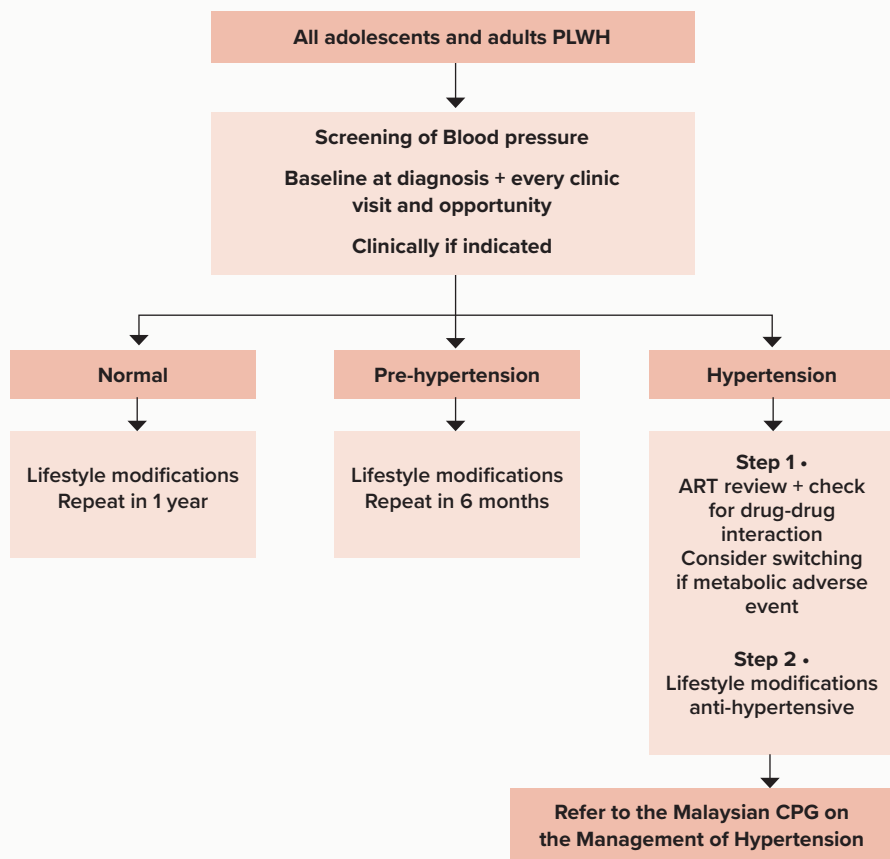
Lifestyle modifications are the first line intervention for early-stage hypertension. As for pharmacological treatment, the choice of antihypertensive agent in PLWH is determined not only by efficacy but also importantly by drug interactions with ART.

The selection of antihypertensive therapy in PLWH is guided by the Malaysian CPG on the Management of Hypertension but requires specific adjustments for ART drug interactions (refer to Annexe 5.3).

Table 8.3 • Diagnostic criteria for hypertension

Feature	Malaysian CPG 2018	EACS Guidelines v12.0 (2023)	US ACC/AHA & DHHS Guidelines
Definition of Hypertension	≥140/90 mmHg	≥140/90 mmHg (Office BP)	≥130/80 mmHg (Stage 1)
Normal BP	<120/80 mmHg	<130/85 mmHg	<120/80 mmHg
Elevated / High Normal	120-139 / 80-89 mmHg (Pre-hypertension)	130-139 / 85-89 mmHg (High Normal)	120-129 / <80 mmHg (Elevated)

Figure 8.3 • Management algorithm for hypertension in PLWH



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8.5 Metabolic dysfunction-associated steatotic liver disease (MASLD)

8.5.1 Introduction

Metabolic dysfunction-associated steatotic liver disease (MASLD) has superseded the term non-alcoholic fatty liver disease (NAFLD) since 2023, reflecting a cardiometabolic diagnostic paradigm.^{1,2} MASLD prevalence in PLWH reaches 38%, substantially higher than the general population, driven by cumulative metabolic risk, HIV-related inflammation and ART effects.^{3,4} MASLD accounts for up to 65% of unexplained liver enzyme elevations in PLWH.⁵ Early identification and non-invasive risk stratification are critical to prevent progression to advanced fibrosis and cirrhosis.

Table 8.4 • MASLD Nomenclature (2023 Update)¹

Previous Term	Current Term	Key Difference
NAFLD	MASLD	Requires ≥1 cardiometabolic risk factor (positive diagnosis)
NASH	MASH (Metabolic Dysfunction-Associated Steatohepatitis)	Steatosis with inflammation/hepatocyte injury ± fibrosis Requires biopsy Stratified by fibrosis stage (F0-F4) Hepatocellular carcinoma (HCC) can occur even in the absence of cirrhosis or histological features of MASH
—	MetALD	MASLD criteria plus moderate alcohol intake (20-50 g/day for women, 30-60 g/day for men)

8.5.1.1 MASLD diagnostic criteria^{6,7}

Presence of hepatic steatosis evidence (criterion 1) PLUS ≥1 cardiometabolic risk factor (criterion 2) WITH criteria 3 and 4 met (all are required).

Criteria 1 • Hepatic steatosis evidence

- Imaging: Ultrasound (first-line), computed tomography (CT), Magnetic Resonance Imaging - Proton Density Fat Fraction (MRI-PDFF), OR
- Transient elastography with CAP ≥248 dB/m (optimal cut-off not established for PLWH; varies by study), OR
- Liver biopsy (if performed) reveals >5% hepatocytes with steatosis (defines presence of steatosis); significant hepatic steatosis is considered when ≥30% of hepatocytes affected.

Criteria 2 • Cardiometabolic risk factor

- To fulfill ≥1 cardiometabolic risk factor (Refer to table 8.5)

Criteria 3 • Alcohol consumption below threshold

- <20 g/day (women), <30 g/day (men)

Criteria 4 • Exclusion of other causes

- Viral hepatitis (HBV, HCV)
- Significant alcohol
- Drug-induced liver injury
- Other chronic liver diseases

Table 8.5 • Cardiometabolic risk factors and its diagnostic criteria

Metabolic Risk Factor	Adult Criteria
Overweight or Obesity	BMI ≥ 25 kg/m ² (≥ 23 kg/m ² Asian ethnicity) or Waist circumference: ≥ 94 cm (men), ≥ 80 cm (women) [Europeans] ≥ 90 cm (men), ≥ 80 cm (women) [South Asians, Chinese, Malaysians] ≥ 85 cm (men), ≥ 90 cm (women) [Japanese]
Dysglycemia or Type 2 Diabetes	Prediabetes: HbA1c 5.7-6.4% or FBS 5.6-6.9 mmol/L or 2HPP OGTT 7.8-11 mmol/L or Type 2 diabetes: HbA1c $\geq 6.5\%$ or FBS ≥ 7.0 mmol/L or or 2HPP OGTT ≥ 11.1 mmol/L or On treatment for type 2 diabetes
Hypertriglyceridemia	TG ≥ 1.7 mmol/L (≥ 150 mg/dl) or On lipid-lowering treatment
Low HDL-cholesterol	HDL-C < 1.0 mmol/L (< 39 mg/dl) in men HDL-C < 1.3 mmol/L (< 50 mg/dl) in women or On lipid-lowering treatment
Hypertension	Blood pressure $\geq 130/85$ mmHg or On treatment for hypertension

MASLD can occur in lean individuals (BMI < 25 kg/m²) with other cardiometabolic risk factors ("lean MASLD"), affecting up to 25% of lean PLWH.^{6,7}

8.5.2 Screening and Diagnosis

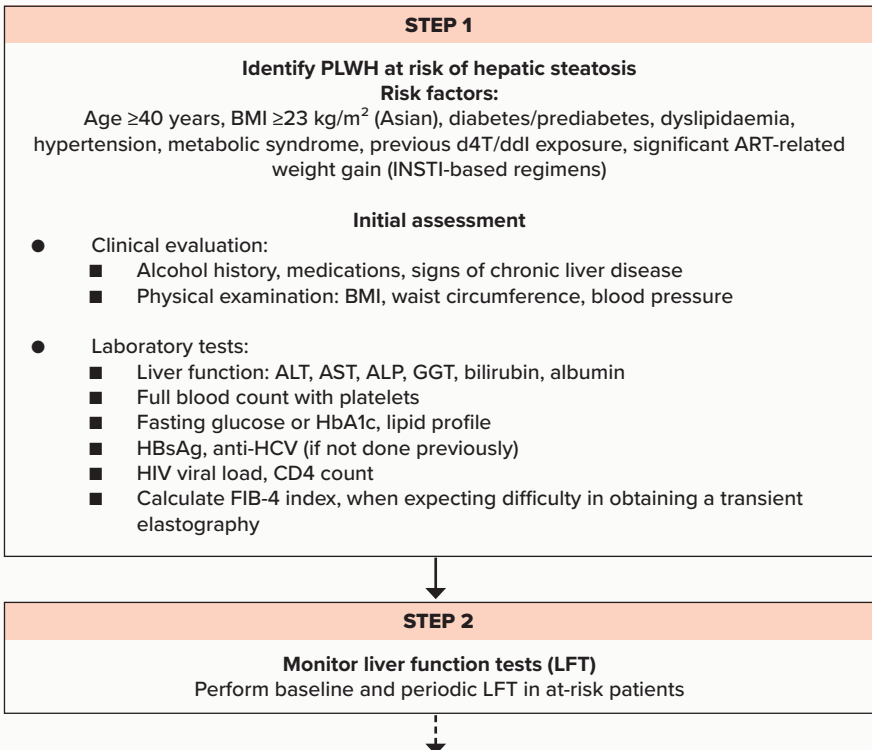
Given the resource constraints in Malaysian healthcare settings, screening should be targeted to PLWH at significant risk of hepatic steatosis, primarily identified by persistent ALT elevation.

Vibration Controlled Transient Elastography (VCTE) such as FibroScan® or sheerwave is a preferred diagnostic modality, potentially available in most state hospitals or accessible via outsourcing to private centres. MASLD screening and management is shown in figure 8.4 and table 8.6.

The primary goal of gastroenterology referral is cirrhosis workup and complication prevention. Gastroenterology provides:

- Confirmation of cirrhosis diagnosis
- HCC surveillance
- Variceal screening and management per Baveno VII guidelines
- Management of cirrhosis complications (ascites, encephalopathy, variceal bleeding)
- Consideration for liver transplant evaluation if appropriate

Figure 8.4 • MASLD screening and management flowchart



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STEP 3

Key screening criterion - persistent ALT elevation

- Persistent ALT $\geq 2 \times$ ULN on at least two occasions separated by ≥ 3 months warrants investigation for steatosis
- Before investigating for steatosis, exclude other causes:
 - Viral hepatitis: Check HBsAg, anti-HCV if not done previously
 - Drug-induced liver injury (DILI): Review all medications including traditional/herbal remedies
 - Autoimmune hepatitis: Consider if other autoimmune features present
 - Alcohol-related liver disease: Detailed alcohol history

↓

STEP 4

Imaging Assessment

Preferred diagnostic modality:

Vibration Controlled Transient Elastography (VCTE) for fibrosis assessment

- Advantages:
 - Simultaneous assessment of steatosis (CAP score) and fibrosis (LSM)
 - Non-invasive, quick (10-15 minutes), no preparation required
 - Potentially available in most state hospitals
 - If unavailable locally, can outsource to private centres via hospital outsourcing programmes
- Interpretation:
 - CAP (Controlled Attenuation Parameter): ≥ 248 dB/m indicates steatosis (optimal cut-off not established for PLWH)
 - S1 (mild) : 248-267 dB/m
 - S2 (moderate) : 268-279 dB/m
 - S3 (severe) : ≥ 280 dB/m
 - LSM (Liver Stiffness Measurement): Refer Table 8.6 for fibrosis staging
- Confirms steatosis; cannot assess fibrosis stage
- If steatosis confirmed, calculate FIB-4 for fibrosis risk estimation
- Consider arranging referral: To centre with FibroScan capability for definitive fibrosis staging

↓

STEP 5

Interpret FibroScan® results and management based on fibrosis stage

- LSM < 8 kPa (F0-F1): HIV clinic management with lifestyle modification and metabolic optimisation
- LSM ≥ 8 kPa ($\geq$ F2): Gastroenterology referral for advanced fibrosis/cirrhosis workup, HCC surveillance, variceal screening

Table 8.6 • Liver Stiffness Measurement (LSM) interpretation and management

LSM (kPa)	Fibrosis Stage	Management
<8	F0-F1 (minimal/no fibrosis) ≥F2 (clinically significant fibrosis)	Manage in HIV clinic: • Lifestyle modification (weight loss, diet, exercise) • Metabolic risk factor optimization • Review ART for metabolic impact • Repeat Fib-4 scoring or/and VCTE in 2-3 years
8–10	If LSM < 10kPa, excludes compensated advanced chronic liver disease (cACLD) but warrants referral to gastroenterologist/ hepatologist	Refer to gastroenterologist/ hepatologist for: • Assessment for advanced fibrosis/ cirrhosis • Initiation of cirrhosis surveillance if indicated • HCC screening (if cirrhosis) • Variceal screening per Baveno VII criteria
10–15	F3 (advanced fibrosis) If LSM > 10kPa, possible cACLD or clinically significant portal hypertension (CSPH)	Refer to gastroenterologist/ hepatologist (as above)
>15	F4 (cirrhosis)	Early referral to gastroenterologist/ hepatologist (as above) g/day for men)

8.5.3 Management of MASLD

8.5.3.1 Lifestyle modification

Lifestyle modification is the primary treatment for all patients with MASLD, regardless of fibrosis stage.

Table 8.7 • Lifestyle intervention targets for MASLD

Component	Goal/Recommendation	Clinical Target
Weight Loss	5–10% body weight reduction	7% reduces steatosis/inflammation ≥10% can regress fibrosis
Diet	500–1,000 kcal/day deficit Mediterranean diet pattern	Balanced traditional meals, reduce simple sugars and refined carbohydrates
Physical Activity	150–200 min/week moderate-intensity aerobic exercise	3–5 sessions/week; combine with resistance training

8.5.3.2 Pharmacological Management

Pharmacotherapy is reserved for individuals with confirmed MASH (histological), significant fibrosis, or high risk of progression (diabetes, metabolic syndrome, persistently elevated ALT).

Key principles of pharmacological management:

- All MASLD patients: Consider statin therapy for cardiovascular risk reduction (safe in MASLD; avoid only in decompensated cirrhosis). Refer to Section 8.2 for detailed management options.
- Diabetes and MASLD: Prioritise GLP-1 receptor agonists and SGLT2 inhibitors. Refer to Section 8.3 for detailed management options.
- Obesity and MASLD: Refer to Section 8.6 for comprehensive obesity management including pharmacotherapy and bariatric surgery indications

8.5.4 ART considerations in MASLD

ART selection is critical in MASLD management, prioritising regimens with minimal metabolic impact whilst ensuring viral suppression.¹⁰

The metabolic impact of ARTs and the various ART switching strategies for MASLD are shown in Table 8.8 and Table 8.9 respectively.

Table 8.8 • ART metabolic impact and guidance

ART	Metabolic Impact	Actionable Guidance
INSTI (DTG)	Weight gain (higher than NNRTIs)	Monitor weight closely, especially at initiation or switching
TDF	Weight-neutral, metabolically favourable	Prioritise if renal/bone health permits
TAF	Weight gain when combined with INSTIs	Weigh renal/bone benefits against metabolic risk
NNRTI (EFV)	Weight-neutral; dyslipidaemia, CNS effects	Use cautiously; CNS effects may hinder lifestyle adherence
Boosted PI (LPV/r)	Dyslipidaemia, insulin resistance	Switch to metabolically neutral option if MASLD develops/worsens

Table 8.9 • Clinical scenarios and ART switching

Clinical Scenario	Action	Regimen Options	Rationale
Exposure to d4T, ddl, or older PIs	Immediate switch	Option 1 • TDF/FTC/DTG Option 2 • TDF/FTC/EFV	Hepatotoxic, severe metabolic dysfunction
High MASLD risk (starting ART)	Prioritise metabolically favourable regimen	Option 1 • TDF/FTC/EFV Option 2 • TDF/FTC/DTG	Option 1 preferred if CNS effects tolerated; monitor weight with option 2
Significant weight gain (>5-10%) on DTG + TDF/FTC with worsening MASLD	Consider switch or intensify lifestyle	Option 1 • TDF/FTC/EFV Option 2 • DTG/3TC (2DR) Option 3 • Continue DTG + aggressive lifestyle modification resistance	Discuss pros/cons with patient

8.5.5 Recommendations

Screening and diagnosis

- Monitor LFT in PLWH with MASLD risk factors; investigate for steatosis if persistent ALT $\geq 2 \times$ ULN, after excluding other causes.⁵
- Use VCTE as preferred diagnostic modality if available for simultaneous steatosis and fibrosis assessment.
- Calculate FIB-4 as an alternative if VCTE is unavailable and ultrasound confirms steatosis. (Refer Table 8.10)
- Perform VCTE for FIB-4 ≥ 1.3 to guide further management.

Referral and management

- Refer LSM ≥ 8 kPa ($\geq F2$) to gastroenterologist/hepatologist for cirrhosis assessment and complication prevention.
- Manage LSM < 8 kPa (F0-F1) in HIV clinics with lifestyle modification and metabolic optimization.

Lifestyle and pharmacological management

- Target 7–10% body weight loss through caloric restriction and 150–200 minutes/week of moderate-intensity exercise in ALL patients.
- For pharmacotherapy options, refer to obesity, DM and dyslipidaemia chapters

ART management

- Prioritise TDF-based regimens in PLWH at risk of or with established MASLD.
- Monitor weight closely on INSTI-based regimens; consider switching to TDF/FTC/EFV or DTG/3TC if significant weight gain (>5–10%) occurs with worsening MASLD.
- Never delay ART due to MASLD concerns.

Table 8.10 • FIB-4 interpretation and management

FIB-4 Score	Risk Category	Action
<1.3 < 2*	Low risk	Reassess every 1–3 years Lifestyle intervention Manage metabolic risk factors
1.3–2.67	Intermediate risk	Arrange Vibration Controlled Transient Elastography or refer gastroenterologist/ hepatologist
>2.67	High risk	Arrange Vibration Controlled Transient Elastography or refer gastroenterologist/ hepatologist

*For patients ≥65 years, use a cut-off of 2.0 (instead of 1.3) to improve specificity.⁸

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8.6 Obesity

8.6.1 Introduction

Obesity has emerged as a significant co-morbidity in PLWH, driven by improved life expectancy, ART associated weight gain and ageing-related metabolic changes.^{1,3} The prevalence of obesity in PLWH now approaches that of the general population in high-income settings, with INSTI-based regimens, particularly when combined with TAF, associated with substantial weight gain especially during the first 12–24 months of therapy.^{2,4} Weight gain after ART initiation is associated with increased risk of metabolic and cardiovascular complications.

The 2025 Lancet Commission on Clinical Obesity reframes obesity diagnosis around organ dysfunction, distinguishing preclinical obesity (excess adiposity without complications) from clinical obesity (adiposity with organ dysfunction or activity of daily living [ADL] limitation).⁸

8.6.2 Screening and diagnosis

8.6.2.1 Annual screening

Screen all PLWH annually at HIV clinic visits using:

- BMI (Asian cut-offs): Overweight 23.0–27.4 kg/m²; Obesity ≥27.5 kg/m²
- Waist circumference (WC): Men ≥90 cm; Women ≥80 cm

8.6.2.2 Confirm excess adiposity

If BMI ≥23 kg/m² or WC above threshold, add ≥1 adiposity marker:

- Waist-to-hip ratio (WHR) or waist-to-height ratio (WHtR)
- If available: bioelectrical impedance analysis (BIA) or dual-energy X-ray absorptiometry (DXA) for % total body fat; visceral adipose tissue (VAT) via CT/DXA surrogate

8.6.2.3 Assign BMI class and clinical stage

BMI should be classified using Asian-specific thresholds (Table 8.11), followed by clinical staging to distinguish preclinical from clinical obesity based on obesity-related organ dysfunction or limitations in activities of daily living (Table 8.12).

In pregnancy, BMI should be interpreted alongside pregnancy-specific weight recommendations.

In older adults, BMI should be interpreted alongside assessment for sarcopenic obesity (low muscle mass with high fat mass) and body composition assessment (DXA, BIA).

Table 8.11 • BMI classification (Asian thresholds)

BMI (kg/m ²)	Classification
23.0–27.4	Overweight
27.5–32.4	Class I Obesity
32.5–37.4	Class II Obesity
≥37.5	Class III Obesity

Table 8.12 • Lancet Commission obesity staging (2025)⁸

Previous Term	Current Term	Key Difference
Preclinical Obesity	Excess adiposity without organ/tissue dysfunction or ADL limitation	Risk enhancers ^{1,2,10–17} : ● Recent INSTI/TAF exposure (≤12–24 months) ● Low CD4 nadir ● Female sex, Black/Asian ethnicity ● Endocrine medications, sedentary lifestyle
Clinical Obesity	Excess adiposity plus ≥1 organ dysfunction or ADL limitation	Examples (≥1 required): ● Metabolic: Prediabetes/T2DM, dyslipidaemia ● Hepatic: MASLD/MASH (CAP ≥248 dB/m; fibrosis markers) ● Respiratory: Obstructive sleep apnoea (OSA) ● Cardiovascular: Heart failure (HFrEF/HFpEF), hypertension, high MACE risk ● Musculoskeletal: Knee osteoarthritis (KOA) with pain/functional impairment ● Reproductive: PCOS, male hypogonadism ● Renal: Microalbuminuria with reduced eGFR ● ADL: Age-adjusted mobility/self-care limitation

8.6.3 ART management principles

8.6.3.1 Background

ART-associated weight gain, particularly with INSTI-based regimens and TAF, is common but must be balanced against the critical importance of viral suppression.^{1,2,4}

Weight gain carries metabolic and cardiovascular consequences, with even modest BMI increases in those with normal baseline weight associated with increased CVD and diabetes risk.^{5–7}

However, it is essential to distinguish “return to health” weight gain in those presenting with advanced disease (CD4 <200 cells/mm³) or low baseline BMI from unhealthy weight gain in those already at normal or elevated BMI.

8.6.3.2 ART and weight gain risk

ART-associated weight gain varies by regimen and is influenced by patient-related factors. Integrase strand transfer inhibitor (INSTI)- and tenofovir alafenamide (TAF)-based regimens are generally associated with greater weight gain, particularly during the first 12–24 months. Table 8.13 summarises the relative weight-gain risk and clinical considerations of commonly used ART regimens in Malaysia.

Table 8.13 • ART regimens and weight gain risk in Malaysian context

ART Regimen	Availability in Malaysia	Weight Gain Risk	Clinical Considerations
TDF/3TC/DTG or TDF/FTC/DTG	Most common	Moderate	Higher weight gain in first 12–24 months; greater risk in women, Black/Asian ethnicity, low CD4 nadir
TAF/FTC/DTG	Limited availability	Higher vs TDF	Use cautiously if obesity risk factors present
TDF/FTC/EFV	Previous first-line; still used in some settings	Lower vs INSTI	Consider if obesity risk high and CNS effects tolerated
PI-based (LPV/r, DRV/r, ATV/r)	Less common	Moderate metabolic effects	Lipodystrophy/dyslipidaemia concerns > weight gain

8.6.3.3 When to consider ART switching for obesity

Consider ART switch from DTG/TAF-based regimens to TDF-based or EFV-based regimens only if ALL of the following apply¹:

- Rapid weight gain (>5–10% in 6–12 months) in a patient with baseline normal or elevated BMI (i.e., not “return to health” weight gain)
- Development of obesity-related complications (prediabetes/T2DM, hypertension, dyslipidaemia, MASLD)
- No other ART optimisation reasons exist
- Shared decision-making with patient
- Close viral load monitoring post-switch
- Patient understands modest benefit expected from ART switch alone

8.6.4 Principles of management

8.6.4.1 Lifestyle intervention

Prioritise lifestyle and pharmacological interventions for all patients over ART switching. Evidence for the benefit of ART switching for weight management is limited; lifestyle modification and pharmacotherapy are more effective.^{1,10}

Table 8.14 • Lifestyle intervention components

Component	Recommendation	Target
Diet	Mediterranean-style or culturally adapted whole-food pattern	500–750 kcal/day deficit
Physical Activity	Moderate-intensity aerobic exercise + strength training	≥150–300 min/week moderate exercise; 2–3 strength sessions/week
Behavioural Therapy	Motivational interviewing (MI) or cognitive behavioural therapy (CBT); self-monitoring	Address stigma and mental health
12-Week Target	≥5% weight loss OR symptom improvement (OSA/KOA metrics)	

8.6.4.2 Pharmacotherapy

Indications for pharmacotherapy:

- BMI ≥27.5 kg/m² with ≥1 obesity-related comorbidity; or
- BMI ≥30 kg/m²

Referral to a diabetologist, endocrinologist or obesity clinic should be considered if:

- BMI ≥27.5 kg/m² with metabolic complications, such as T2DM, prediabetes, CVD, MASLD or MASH
- BMI ≥30 kg/m² despite intensive lifestyle intervention
- Multiple obesity-related comorbidities requiring specialist assessment
- Access to GLP-1 receptor agonists is required, particularly where cost or availability may be a consideration

Pharmacotherapy should be used as an adjunct to dietary intervention, physical activity and behavioural modification. Treatment selection (Table 8.15) should be individualised according to the patient's clinical condition and approved indications.

Table 8.15 • Pharmacotherapy options for management of obesity in PLWH

Medication	Indications	Malaysian Availability	Key Considerations
GLP-1 receptor agonists (Semaglutide 2.4 mg weekly, Liraglutide 3.0 mg daily, Tirzepatide)	T2DM, prediabetes, CVD, MASLD/MASH, OSA, KOA	Limited by cost	Ideal choice for high-risk patients; administer ≥ 4 hours apart from atazanavir/rilpivirine; minimal ARV interactions
Orlistat 120 mg TDS	First-line practical option	More readily available	ART malabsorption risk—administer ≥ 2 hours apart from ART; monitor viral load/CD4 every 3 months initially
Naltrexone-bupropion	T2DM, CVD (if GLP-1 unavailable)	Variable availability	Monitor BP/mental health; check ART drug interactions
Metformin (off-label)	Prediabetes, insulin resistance	Widely available, low cost	Weight maintenance benefit; minimal ARV interactions

8.6.4.3 Bariatric/Endoscopic procedures

Indications for specialist referral based on Asian BMI thresholds:

- Class III obesity (BMI ≥ 37.5 kg/m²): Eligible for consideration even in the absence of obesity-related comorbidities
- Class II obesity (BMI 32.5–37.4 kg/m²): Eligible for consideration in the presence of ≥ 1 serious obesity-related comorbidity, such as OSA, T2DM, MASLD/MASH with fibrosis or heart failure

Pre-procedure considerations:

- Multidisciplinary assessment and clearance
- Individualised ART pharmacokinetic and administration plan
- Nutritional and psychological evaluation

Post-procedure management:

- Appropriate micronutrient supplementation and long-term monitoring

8.6.4.4 Follow-up and maintenance

Regular follow-up is essential to assess treatment response, monitor obesity-related complications and support long-term weight maintenance. The recommended follow-up frequency, monitoring parameters and management approach according to obesity stage are summarised in Table 8.16.

Table 8.16 • Follow-up monitoring and management according to obesity stage

Component	Preclinical Obesity	Clinical Obesity
Review Frequency	Every 6–12 months	Every 3–6 months (closer during medication titration)
Parameters to Track	All stages monitor: ● Weight/BMI/WC ● HbA1c, FBS/RBS ● Lipids ● ALT/AST ● HIV viral load/CD4 Clinical obesity adds: ● CAP/FIB-4 if MASLD/MASH ● Patient-reported outcomes (sleep, KOA pain, ADL)	
Core Management (All Stages)	Stigma-sensitive counselling Structured lifestyle programme (diet + physical activity + behavioural therapy)	
Additional Actions	Consider early pharmacotherapy if high-risk or rapid gain (>5% in 6–12 months)	If <5–10% weight loss at 3–6 months: ● Intensify lifestyle ● Up-titrate/switch medication ● Re-evaluate surgery ● Address barriers

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8.7 Chronic Kidney Disease

8.7.1 Introduction

Chronic kidney disease (CKD) represents a significant co-morbidity in PLWH, affecting approximately 4.7–9.7% of this population when defined by estimated glomerular filtration rate (eGFR) <60 mL/min/1.73 m².^{1,2} With improved access to ART and increased life expectancy, PLWH are at heightened risk for both HIV-related and age-related kidney diseases.

Table 8.17 • Risk factors for CKD in PLWH^{1,3–12}

Category	Risk Factors
HIV-Specific	● High viral load (poor virologic control) ● Low CD4+ count (<200 cells/mm³) ● Advanced HIV disease ● Co-infections (hepatitis B, hepatitis C, tuberculosis)
Traditional	● Hypertension ● Diabetes mellitus ● Older age (>50 years) ● Cardiovascular disease ● Family history of kidney disease ● Obesity ● Smoking
Medication-Related	● Tenofovir disoproxil fumarate (TDF) exposure ● Protease inhibitors (ritonavir-boosted) ● Nonsteroidal anti-inflammatory drugs (NSAIDs)

8.7.2 Definition and classification of Chronic Kidney Disease (CKD)

Chronic kidney disease (CKD) is defined as abnormalities of kidney structure or function that are present for ≥ 3 months and have implications for health.^{7,9} CKD is diagnosed when either of the following criteria is met:

- eGFR <60 mL/min/1.73 m² for ≥ 3 months; or
- One or more markers of kidney damage persisting for ≥ 3 months, including albuminuria or proteinuria, urinary sediment abnormalities, persistent haematuria of renal origin or structural abnormalities detected by imaging.

Table 8.18 • GFR categories in chronic kidney disease

Stage	eGFR (mL/min/1.73 m ²)	Description of kidney function
1	≥90	Normal or high
2	60–89	Mildly decreased
3a	45–59	Mildly to moderately decreased
3b	30–44	Moderately to severely decreased
4	15–29	Severely decreased
5	<15	Kidney failure

Note: In the absence of other evidence of kidney damage, GFR categories 1 and 2 do not fulfil the criteria for CKD.

8.7.3 Screening and monitoring

8.7.3.1 Screening recommendations

All PLWH should be screened for kidney disease at¹⁰:

- Time of HIV diagnosis
- When ART is initiated or changed
- At least twice yearly in stable patients
- At least annually for urinalysis/proteinuria

More frequent monitoring indicated for:

- Patients with eGFR <60 mL/min/1.73 m²
- Presence of proteinuria or albuminuria
- Use of potentially nephrotoxic medications (TDF, NSAIDs)
- Diabetes, hypertension, or other CKD risk factors

8.7.3.2 Baseline Assessment

Baseline kidney assessment should include the following:

- Serum creatinine and estimated glomerular filtration rate (eGFR): Calculate eGFR using the 2021 race-free CKD-EPI equation. The Cockcroft–Gault equation to be used when creatinine clearance is required for drug dosing.
- Urinalysis:
 - Dipstick testing for protein, blood and glucose
 - If protein is ≥1+, quantify using the urine albumin-to-creatinine ratio (UACR) or urine protein-to-creatinine ratio (UPCR)
 - Repeat testing to confirm persistent proteinuria

Additional investigations, when indicated, include:

- Cystatin C-based eGFR or combined creatinine–cystatin C eGFR, where available, particularly when creatinine-based eGFR may be inaccurate or when greater precision would influence clinical management^{7a}
- Serum electrolytes and phosphate to assess for proximal tubular dysfunction
- Hepatitis B and C serology
- Renal tract imaging to assess for structural or urological abnormalities

8.7.4 Nephrology referral and diagnostic evaluation

Referral to a nephrologist is recommended in the presence of any of the following:⁸

- Sustained, unexplained and clinically significant decline in eGFR (>20% from baseline)
- Progressive proteinuria despite optimised renin–angiotensin system inhibitor therapy, or proteinuria ≥ 1 g/day
- Features suggestive of nephritic syndrome, including haematuria of renal origin with albuminuria/proteinuria, rising blood pressure or declining kidney function
- Suspected acute interstitial nephritis
- Advanced CKD (eGFR < 30 mL/min/1.73 m²)
- Suspected HIV-associated nephropathy (HIVAN)
- Rapidly progressive kidney dysfunction
- Uncertain aetiology of kidney disease

Kidney biopsy should be considered in patients with:

- Suspected HIVAN
- Suspected immune complex-mediated glomerular disease
- Significant proteinuria of uncertain aetiology
- Haematuria accompanied by proteinuria
- Rapidly progressive or otherwise unexplained kidney dysfunction

8.7.5 Antiretroviral therapy considerations

8.7.5.1 General principles

ART should be initiated in all PLWH with CKD or ESRD.^{6,710}

Benefits include reduced mortality, slower CKD progression, though currently underutilised in CKD/ESRD populations.

ART selection should consider:

- Current kidney function
- Risk of nephrotoxicity (refer to Annexe 5.1)
- Drug-drug interactions (refer to Annexe 5.3)
- Dosing adjustments required (refer to Annexe 5.2)

8.7.6 Management of Chronic Kidney Disease

8.7.6.1 General measures

All patients with CKD should receive^{7,9}:

- Lifestyle modifications:
 - Smoking cessation
 - Weight management
 - Regular physical activity
 - Dietary sodium restriction
 - Protein intake moderation (0.8 g/kg/day if eGFR < 30 mL/min/1.73 m²)

- Avoid nephrotoxic agents:
 - NSAIDs (especially with concurrent TDF)
 - Intravenous contrast (if eGFR <30 mL/min/1.73 m², use with caution)
 - Aminoglycosides, amphotericin B
 - Herbal supplements with nephrotoxic potential
- Optimise ART:
 - Achieve viral suppression
 - Switch from TDF to TAF if appropriate
 - Adjust doses for renal function

8.7.6.2 Renin-Angiotensin System (RAS) Blockade

Indications⁷: Hypertension in PLWH with CKD, proteinuria irrespective of blood pressure, and diabetic kidney disease

Agents: Angiotensin-converting enzyme (ACE) inhibitors or angiotensin receptor blockers (ARBs)

Monitoring:

- Check serum creatinine and potassium within 1–2 weeks after treatment initiation or dose escalation
- An increase in serum creatinine of up to 30% from baseline may be acceptable if it subsequently stabilises
- Target blood pressure: <130/80 mmHg

Precautions:

- Monitor for hyperkalaemia and symptomatic hypotension
- Review treatment if serum creatinine increases by >30% after initiation or dose escalation
- ACE inhibitors and ARBs may be continued when eGFR falls below 30 mL/min/1.73 m², with close monitoring and dose adjustment when clinically indicated
- Avoid concurrent use of an ACE inhibitor and an ARB

8.7.7 HIV-related kidney diseases

Table 8.19 • Summary of HIV-related kidney diseases

Type	Clinical Features	Key Risk Factors	Management
HIVAN	Nephrotic-range proteinuria (>3.5 g/day) Rapid ESRD progression Large echogenic kidneys	Advanced HIV (low CD4, high viral load) APOL1 risk alleles West African descent	Immediate ART ACEi/ARB Avoid nephrotoxins Nephrology co-management

Type	Clinical Features	Key Risk Factors	Management
HIVICK	Proteinuria (often subnephrotic) Haematuria Hypocomplementaemia	Immune activation Co-infections (HBV/HCV)	ART Treat infections Consider immunosuppression Nephrology referral
Noncollapsing FSGS	More common in ART era; may represent partially treated HIVAN or secondary FSGS	Ageing Hypertension Diabetes	ART Manage secondary causes
Tubulointerstitial Disease	AKI or CKD with AIN features; may have systemic symptoms	Drugs (antibiotics, NSAIDs, PPIs) DILS, IRIS Opportunistic infections	Remove offending agent ART for DILS Treat infections, corticosteroids in select AIN

Abbreviation: DILS - Diffuse infiltrative lymphocytosis syndrome, AIN - acute interstitial nephritis, IRIS - Immune reconstitution inflammatory syndrome

8.7.8 Important considerations

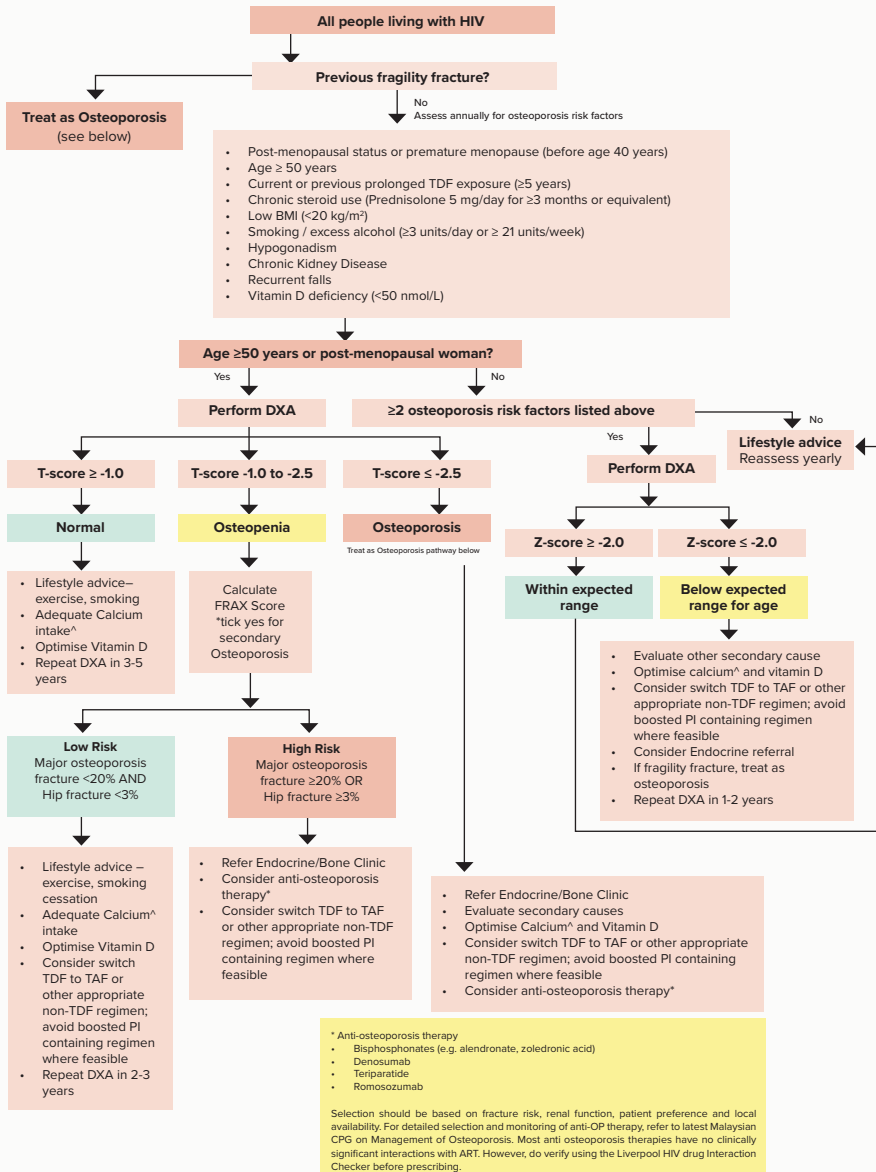
- Implement lifestyle modifications: smoking cessation, weight management, regular physical activity, dietary sodium restriction, protein intake moderation (0.8 g/kg/day if eGFR <30 mL/min/1.73 m²).
- Avoid nephrotoxic agents: NSAIDs (especially with TDF), intravenous contrast (if eGFR <30 mL/min/1.73 m²), aminoglycosides, herbal supplements.⁵
- Initiate ACE inhibitors or ARBs for hypertension in PLWH with CKD, proteinuria (regardless of blood pressure), or diabetic kidney disease; target blood pressure <130/80 mmHg.
- Prescribe statin therapy for primary prevention if indicated by cardiovascular risk score; consider aspirin in patients with diabetes and proteinuria or for secondary prevention.
- Consider SGLT2 inhibitors (dapagliflozin, empagliflozin) for T2DM with CKD (eGFR 20–75 mL/min/1.73 m²), CKD with proteinuria (UPCR >200 mg/g), or heart failure with reduced ejection fraction.
- Consider GLP-1 receptor agonists (semaglutide, dulaglutide, liraglutide) for T2DM with CKD or obesity with CKD.

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8.8 Osteoporosis / Osteopaenia

Figure 8.5 • Bone health and management of osteopaenia/osteoporosis in PLWH



[^] Calcium supplement may reduce dolutegravir absorption and should be administered 2 hours before or 6 hours after dolutegravir if not taken with food

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8.9 Cancer screening

Figure 8.6 • Cancer screening and evaluation in PLWH

All PLWH	
Cervical Cancer	Recommended for all women living with HIV with a cervix. Please refer to section 9.3.3 (Cervical Cancer Prevention and HPV Screening)
Anal Cancer	Recommended for higher-risk PLWH: MSM, transgender women and those with prior HPV-related disease Annual symptom review and digital anorectal examination. Symptom warranting evaluation: Anal bleeding, anal pain or anal mass/lump. Refer suspicious lesions/symptoms for colorectal surgical assessment. Consider referral for anal dysplasia assessment (anal cytology and/or high-resolution anoscopy) where expertise is available
Breast Cancer	Recommended for all women: – Clinical breast examination from age 35 years – Mammography every 2 years for women aged 50-74 years Earlier and/or more frequent screening for increased-risk individuals
Colorectal Cancer	Average-risk adults (no symptoms, no family history) Aged 50-75 years: – Annual Immunochemical Faecal Occult Blood Test (iFOBT) or Faecal Immunochemical Test (FIT) – Positive iFOBT or FIT colonoscopy High Risk (strong family history or genetic syndromes) Age 40 (or 10 years younger than earliest diagnosed 1st degree relative) and above: – Colonoscopy (if normal, 3-5 yearly)
Hepatocellular Cancer	Recommended for PLHIV with chronic Hepatitis B infection, chronic hepatitis C with advanced fibrosis or liver cirrhosis (any cause). Abdominal Ultrasound and Alpha-Fetoprotein every 6 months
Lung Cancer	High risk individuals: age 45-75 years, ≥20 pack-year smoking history. Current smoker or quit within 15 years. Consider annual low-dose CT (LDCT) screening where available.
Prostate Cancer	Men aged 50 years above: Discuss PSA testing and Digital rectal examination using shared decision making. Earlier assessment for higher risk individual (Strong family history or inherited mutations)
Nasopharyngeal Cancer	No routine screening. Prompt referral if: neck mass, persistent nasal obstruction, epistaxis or unilateral hearing loss
Gynaecological Cancers (Ovarian & Endometrial)	No routine screening in asymptomatic average-risk women. Prompt evaluation of Post-menopausal bleeding, Abnormal uterine bleeding, Persistent bloating, Early satiety or Pelvic/abdominal pain

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8.10 Mental health and psychosocial wellbeing

Mental health disorders are among the most significant comorbidities in PLWH, with reported pooled prevalence rates for depression and anxiety at approximately 31% and 29%, respectively; significantly higher than in the general population.^{1,2}

- Impact on outcomes: mental health disorders, particularly depression and substance use have a strong and negative influence on the HIV care continuum.^{3,4}
 - They are associated with suboptimal ART adherence and are strong predictors of virologic non-suppression.^{5,6}
 - PLWH with mental health challenges are significantly less likely to achieve viral suppression compared to those without.^{5,7}
- Contributing factors: the burden is compounded by factors such as HIV-related stigma, social isolation and loneliness, discrimination, poverty, trauma and the side effects of medications.^{2,4}

8.10.1 Integration of systematic mental health screening and care into existing HIV service delivery models

Systematic mental health screening is an essential component of comprehensive and person-centered HIV care.⁴

- Improved HIV outcomes: early identification and management of mental health conditions through screening are essential for improving ART adherence and achieving higher rates of viral suppression, which supports the public health goal of Treatment as Prevention (TasP).^{6,7}
- The World Health Organization (WHO) and other international guidelines strongly support the integration of mental health and psychosocial support into the package of HIV care.⁸

Integrating mental health services into existing HIV service delivery models offers benefits validated by experience in NCD integration.⁴

- Reduced barriers: integrated care reduces barriers like travel, cost and the need for multiple clinic visits, which increases patient access and retention in both HIV and mental health care.^{4,8}
- Decreased stigma: delivering mental health services in the familiar HIV clinic setting is preferred by many PLWH, as it helps mitigate the double-stigma associated with seeking specialised mental health treatment elsewhere.⁴
- Effectiveness: evidence from randomized trials suggests that the integration of mental health and HIV services can lead to a large reduction in moderate-to-severe depression and may improve retention in HIV care and viral suppression.⁴

8.10.2 Key components of mental health screening and management

Routine and systematic screening should be scheduled annually, and use validated, culturally appropriate tools¹⁶. Conditions that should be prioritised for screening include:

- **Depression**
Administer brief screening using Patient Health Questionnaire-2 (PHQ-2) to assess depressive symptoms over the last two weeks, followed by Patient Health Questionnaire-9 (PHQ-9) if needed (refer to Annexe 8.1).^{3,9,10}
- **Anxiety**
Administer brief screening using Generalized Anxiety Disorder-2 (GAD-2) to assess anxiety symptoms over the last two weeks, followed by Generalized Anxiety Disorder-7 (GAD-7) if needed (refer to Annexe 8.2).¹¹
- **Substance use disorder**
Please refer to Section 9.6

8.10.3 Psychological therapies for mental health¹²

Table 8.20 • Recommended psychological interventions for PLWH

Intervention	Recommended for
Behavioural activation	Depression
Relaxation training	Depression
Problem solving treatment	Depression
Cognitive behavioral therapy	Depression Substance use conditions Psychoses
Contingency management therapy	Substance use conditions Psychoses
Family counselling or therapy	Depression
Interpersonal therapy	Depression
Motivational interviewing or motivational enhancement therapy	Substance use conditions
Family-oriented treatment approaches	Depression Anxiety
Common elements treatment approach	Depression Anxiety Substance use conditions

8.10.4 Pharmacotherapy for mental health (Depression)¹⁰

Pharmacotherapy for mental health conditions among PLWH should be guided by established national clinical practice guidelines and delivered as part of a comprehensive and stepped-care approach.

For depressive disorders, antidepressant treatment should be initiated based on symptom severity, functional impairment and clinical risk assessment, with selective serotonin reuptake inhibitors (SSRIs) generally recommended as first-line agents due to their favourable efficacy and tolerability profiles.

Treatment should be regularly reviewed to assess response, adherence, side effects, and potential drug–drug interactions with antiretroviral therapy. Pharmacotherapy is most effective when combined with appropriate psychological and psychosocial interventions, particularly for moderate to severe depression.

Management should follow the *Clinical Practice Guidelines: Management of Major Depressive Disorder* (Second Edition), Ministry of Health Malaysia, with referral to specialist mental health services for complex, treatment-resistant, or high-risk cases.

Table 8.21 • Commonly used antidepressants and their clinical considerations

Antidepressant class	Common examples	Key considerations
Selective serotonin reuptake inhibitors (SSRIs)	Fluoxetine Sertraline Escitalopram Paroxetine Fluvoxamine	Generally recommended as first-line agents due to favourable efficacy and tolerability profiles. Monitor for gastrointestinal side effects, sexual dysfunction, and potential drug–drug interactions.
Serotonin–norepinephrine reuptake inhibitors (SNRIs)	Venlafaxine Desvenlafaxine Duloxetine	Considered in moderate to severe depression. Monitor blood pressure and discontinuation symptoms, particularly with venlafaxine.
Noradrenergic and specific serotonergic antidepressants (NaSSA)	Mirtazapine	May be preferred in patients with prominent insomnia or poor appetite. Associated with sedation and weight gain.
Norepinephrine/dopamine reuptake inhibitors (NDRI)	Bupropion	May be considered in selected patients. Lower risk of sexual dysfunction; contraindicated in patients with seizure disorders.
Multimodal serotonin modulators	Vortioxetine	May be considered as an alternative option in moderate to severe depression, particularly where cognitive symptoms are prominent.

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9.1 Prevention of mother-to-child transmission (PMTCT)

9.1.1 Introduction

Prevention of mother-to-child transmission (PMTCT) of HIV remains one of the most important priorities of Malaysia's national HIV response. With ART and integrated maternal-child health services, HIV transmission from mother to child can be reduced to less than 1%, with the aim to achieve an undetectable viral load.

The HIV prevalence among pregnant women in Malaysia remains low (<0.05%), but ongoing surveillance and high screening coverage (>98%) are maintained to detect new infections early. Vertical transmission rates have declined significantly over the past decade, from >20% before the ART era to <2% in most MOH hospitals with full PMTCT implementations. Most new perinatal HIV cases now occur among women who were diagnosed late in pregnancy or present unbooked during active labour, highlighting the need for continuous strengthening of antenatal screening and linkage to care.

9.1.2 Key recommendations

- Opt-out HIV screening is mandatory for all pregnant mothers in Malaysia.
- Rapid testing kits must be available 24 hours in labour and delivery units.
- Immediate ART initiation is required for confirmed positive mothers with presumptive therapy for infants.
- Preferred choice of ART 1st line regimen is TDF/FTC + DTG or TDF/3TC/DTG
- Documentation and communication between O&G, Paediatric, FMS and ID teams are essential for continuity of care.

- Counselling on PrEP, partner testing and safe infant feeding should be integrated into maternal HIV services.
- PrEP should be part of comprehensive prevention, including:
 - Routine HIV testing
 - STI screening and treatment
 - Condom use
 - Partner engagement
 - Early ART for serodiscordant couples
- Preconception counselling should be offered and documented in the clinical record.
- Hepatitis B screening and vaccination should be completed before conception if non-immune.
- Women planning pregnancy should aim for sustained viral suppression for at least 3-6 months prior to conception.
- HIV-positive couples or serodiscordant partners should be referred to combined HIV-obstetric-fertility clinics (where available).

- Preconception counselling should be documented in the patient record, including viral load status and reproductive plans.
- Partner testing and disclosure counselling are key components of reproductive care.
- Counselling should be patient-centred, respectful of individual values and culturally appropriate.

- Document counselling, adherence review and planned delivery mode in patient's case notes and HIV care record.
- HIV testing must be available in all delivery wards. Women with unknown status in labour should undergo rapid testing.
- IV AZT should be readily available in labour rooms. Refer to Annexe 9.1 for indication and intrapartum management.
- Elective caesarean is at 38 weeks recommended for VL > 50 copies/mL.
- Vaginal delivery acceptable for VL ≤ 50 copies/mL with continued ART.
- Avoid invasive procedures unless obstetrically necessary.

- All newborns should receive post exposure ARV prophylaxis as per paediatric protocol guidelines.
- Routine HCV screening in all HIV-positive pregnancies is recommended.
- Incorporate psychosocial, mental health and adherence support within postpartum visits.
- Formula feeding remains the recommended option to eliminate HIV transmission risk.

Note: Breastfeeding on ART may be considered under shared decision-making, particularly where safe formula feeding is not feasible, provided under these stringent circumstances:

- Maternal viral suppression is sustained.
- ART adherence is reliable.
- Viral load is monitored regularly every 3 months.
- Appropriate infant prophylaxis and testing are ensured.

9.1.3 Pre-pregnancy counseling and care

9.1.3.1 Overview

- All individuals with childbearing potential should receive reproductive health counselling as part of HIV care.
- Discussions should include reproductive desires, contraceptive choices, partner's HIV status and safe conception strategies.
- The aim is to achieve optimal maternal health, prevent sexual and perinatal transmission and support informed, patient-centred decision-making.

9.1.3.2 Partner testing and HIV prevention

- Assess the partner's HIV status at the initial and subsequent visits.
- If the partner's status is unknown, offer or refer for HIV and STI screening.
- For HIV-serodiscordant couples:
 - Discuss risk reduction strategies, ART for the HIV-positive partner with the aim to achieve undetectable viral loads (U=U) and pre-exposure prophylaxis (PrEP) for the HIV-negative partner.

Table 9.1 • HIV management before conception

Goal	Evidence Level	Key Recommendation
Viral suppression	Ensures maternal health and minimises risk of transmission to partner and infant (<1%)	Achieve and maintain undetectable HIV viral load before attempting conception
ART optimisation	ART is safe and beneficial at all stages	Initiate or continue highly effective ART prior to conception
Counselling focus	Empower patients to make informed choices	Explain benefits of viral suppression, early ART and adherence
Infant feeding discussion	Facilitates continuity of care and informed decisions	Begin early counselling on infant feeding options and review throughout pregnancy and again after delivery

9.1.3.3 ART regimen selection in women of childbearing potential

When selecting or evaluating an ART regimen in women who may conceive, consider:

- Effectiveness and potential pharmacokinetic changes during pregnancy (especially in 2nd and 3rd trimesters).
- Hepatitis B status (choose regimen active against both HIV and HBV when co-infected).
- Drug-drug interactions with other chronic medications.
- Maternal and foetal safety, including teratogenicity risk.

9.1.3.4 Reproductive options when one or both partners have HIV

Individuals or couples affected by HIV have the right to make informed reproductive choices. With appropriate care, conception is safe and possible while minimising the risk of HIV transmission to partners and infants.

The goal is to achieve:

- Maternal viral suppression
- No sexual HIV transmission
- Zero perinatal HIV transmission

Table 9.2 • Clinical situations and recommendations

Clinical Situation	Recommendation
Planning for pregnancy	Achieve and maintain sustained viral suppression (at least two undetectable viral loads, ≥ 3 months apart) before attempting conception
Before conception	Both partners should be screened and treated for sexually transmitted infections (STIs) and genital tract infections Rescreen if the conception period is prolonged or if new risk factors develop
When one partner has HIV (serodiscordant couple)	If the partner with HIV is on effective ART with sustained undetectable viral load, timely condomless intercourse for conception carries negligible risk of sexual transmission
When both partners have HIV	Ensure both are on effective ART with sustained suppression and are free from other STIs before conception
Fertility or conception challenges	Refer to an infertility or reproductive medicine specialist for individualised counselling and management

9.1.3.5 Preconception counseling and support

- Discuss timing and method of conception (natural intercourse, assisted reproduction).
- Reinforce ART adherence to achieve undetectable viral load before conception.
- Counsel on risk reduction strategies, including:
 - Timed intercourse during ovulation
 - Consistent adherence to ART
 - Avoidance of unprotected intercourse outside the fertile window
- Screen both partners for hepatitis B, hepatitis C, syphilis, gonorrhoea, chlamydia, and other genital infections.
- Address nutritional status, smoking cessation and alcohol/drug use.
- Begin folic acid supplementation for women intending to conceive.

Key counselling points

- “U=U” (Undetectable = Untransmittable): Achieving undetectable viral load prevents HIV sexual transmission.
- ART is lifelong - continue the same regimen through conception, pregnancy and postpartum unless contraindicated.
- Do not stop ART when pregnancy is confirmed; stopping increases transmission risk.
- Avoid unplanned pregnancy during unsuppressed viral load; provide interim contraception until suppression is achieved.
- Emphasize shared decision-making - involving the patient, partner, and care team in all reproductive plans.

9.1.4 Preferred initial ART regimens in pregnancy

Table 9.3 • Recommended ART regimens

Preferred regimen	TDF/FTC + DTG TDF/3TC/DTG
Alternative regimen	TDF/FTC + EFV TDF/FTC + LPV/r AZT/3TC + EFV

9.1.5 Antenatal monitoring

- HIV RNA viral load:
 - At baseline, four weeks after ART initiation, then every trimester and 36 weeks
- CD4 count: At baseline and as clinically indicated.
- Routine obstetric assessments (BP, urine, growth scan, foetal well-being) as per ANC schedule. There should also be screening for prematurity risk (cervical length measurements).
- Monitor ART toxicities (especially renal, hepatic or haematologic effects).
- Address comorbidities such as weight gain, preeclampsia, anaemia, gestational diabetes and hypertension.

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9.2 HIV treatment among adolescents

9.2.1 Background

Globally, adolescents and young adults represent a growing proportion of PLWH. In Malaysia, almost half (49%) of all new HIV transmissions in 2024 were reported in individuals aged 13-29 years, with most cases reported in young adults aged 20-29 years.¹

Adolescents living with HIV can be categorised into two distinct groups, namely those with perinatally acquired infection (vertical transmission) and those with behaviourally acquired infection (horizontal transmission). Individuals with perinatally acquired infection are usually more treatment-experienced and have undergone a longer duration of disease chronicity compared to those with behaviourally acquired infection.

Perinatally acquired HIV infections have decreased dramatically following the use of antenatal combination antiretroviral therapy (ART) for prevention of maternal-to-child transmission. In contrast, behaviourally acquired HIV infections, particularly via sexual routes, currently make up an increasing proportion of cases involving young adults. This has been attributed to increasing engagement in high-risk sexual activities, low awareness of HIV status and barriers to testing.²

9.2.2 Considerations for adolescents and young adults with HIV

In comparison to older adults, adolescents and young adults with HIV have poorer outcomes on the HIV care continuum.³ Adolescents and young adults face unique developmental, psychosocial and behavioural challenges that influence HIV treatment outcomes. Factors such as stigma, discrimination and fear of rejection can adversely affect HIV treatment and clinical outcomes. Local cultural sensitivities and taboos surrounding sexuality further complicate discussions on risk behaviours and treatment.

Adolescence is a period of developing cognitive process and decision-making capacity, often driven by risk-taking, preoccupation with self-image, and peer pressure, all of which may contribute to adherence barriers. Adolescents with HIV are at risk for neurocognitive impairment and mental health comorbidities, including psychiatric, behavioural and substance use disorders, which can further interfere with cognitive development and adherence to treatment. Addressing mental health issues in adolescents and young adults living with HIV requires a multidisciplinary care approach that integrates medical, psychological and social support to ensure holistic and effective well-being.

The rapid physiologic changes of adolescence, such as puberty and rapid growth, may affect antiretroviral (ARV) drug pharmacokinetics and result in suboptimal drug levels.⁴ Young adults with HIV are also at risk for impaired physical development, such as delayed sexual maturation and impaired bone development.⁵

9.2.3 Antiretroviral therapy in adolescents

All adolescents with HIV should initiate ART as soon as possible after HIV diagnosis, regardless of CD4 count and HIV RNA level.⁶⁻⁸

It is recommended that the selection and dosing of ARVs for adolescents is based on sexual maturity rating (Tanner staging). For pre-pubescent adolescents (Tanner stage I to III), ARV regimens should follow paediatric guidelines while for post-pubescent adolescents (Tanner combination, i.e. DTG plus 2 NRTIs. Alternative regimens include an NNRTI (e.g. efavirenz) or PI (e.g. ritonavir-boosted lopinavir or atazanavir) in combination with 2 NRTIs.

The superior efficacy of DTG compared to NNRTI- or PI-based combination in children and adolescents has been demonstrated by the ODYSSEY trial, where the risk of treatment failure was approximately 40% lower with DTG-based ART than with NNRTI- or PI-based regimens.¹⁰ DTG has also shown good safety, tolerability and predictable pharmacokinetic in adolescents.¹¹

Selection of NRTI for the ARV regimen for post-pubescent adolescents should follow recommendations from adult guidelines for ARV. In view of concerns about potential impact on bone development and renal toxicity, TAF is preferred over TDF in adolescents and young adults under the age of 25 years, prior to peak bone mass accrual.

As adherence to therapy remains a challenge amongst adolescents and young adults on ARVs, treatment should be simplified preferably to a once-daily regimen with fixed dose combinations, whenever available.

Data on the efficacy of two-drug regimen using DTG/3TC in adolescents is limited. The use of two-drug regimen in adolescents is the subject of the ongoing DANCE study, which has reported high efficacy of DTG/3TC as an initial regimen in ART-naïve adolescents with HIV.¹²

Studies on the efficacy of long-acting injectable ARVs in adolescents are still under development. Long-acting injectable CAB/RPV administered to virologically suppressed adolescents with HIV has shown no safety concerns.¹³

9.2.4 Disclosure

Disclosure refers to the process by which children and adolescents are made aware of their HIV status.¹⁴ This process is critical to build health literacy and empower them to be engaged in decision-making about their health. Disclosure has been associated with improved treatment adherence.¹⁵

The WHO recommends that children of school age (6-12 years) should be told their HIV status, and younger children should be informed incrementally to accommodate their cognitive skills and emotional maturity, in preparation for full disclosure.¹⁶ Disclosure of the diagnosis should take into consideration the child's cognitive ability, developmental stage, clinical status, social circumstances, knowledge and coping capacity.

For adolescents with perinatally acquired HIV infection, the process of disclosure would usually have started by the time they enter adolescence. In cases where this has not taken place, disclosure should be achieved by 15 years of age.

Health care professionals need to facilitate the process of disclosure and, when required, play the lead role in the disclosure process. Care providers should continuously assess and provide developmentally appropriate HIV education throughout adolescence.

9.2.5 Transition to adult HIV care

The transition from paediatric or adolescent-focused care to adult HIV services is a highly vulnerable period for the adolescent. Emerging data have shown a heightened risk of attrition with decline in retention and viral suppression for adolescents with HIV after transition to adult services.^{17,18}

The process of health care transition can be challenging for many reasons, such as the presence of developmental or psychosocial delays, attachment to paediatric or adolescent care providers, difficulty in building trust with a new provider, and difficulty in adapting to an adult clinic environment.

Transition is a gradual process that takes several years, starting in early adolescence, culminating in the transfer to adult services and is completed when a young person is well established within adult care. There is currently no definitive evidence-based model for transitioning adolescents to adult HIV care, though research in this area is in progress.

Clinical providers should prepare adolescents for transition long before the actual process occurs, and this can begin when an adolescent is fully aware of their HIV diagnosis.^{19,20} Transitioning to adult care at an older age has been associated with greater satisfaction and better outcomes, including higher rates of viral suppression and retention in care, compared to those who transitioned at a younger age.^{21,22}

An individualised transition plan should be documented and it should address comprehensive care needs of the adolescent, including medical and psychosocial aspects of transitioning. A transition summary should include: diagnoses (past AIDS-defining conditions), medication, allergies, past ART history (including reasons for switches), baseline and acquired resistance, CD4 counts and viral load results, hepatitis status, vaccinations, past medical and presenting history, mental health and neurocognitive function/education, sexual health and relationship education, social and family history including bereavements, social service involvement and smoking, alcohol and drug use.

When appropriate, joint consultations with the paediatric and adult HIV care provider may be considered during the transition period. It is recommended that the paediatric team remains involved after the transfer to adult services until the young person is well established in adult care to monitor engagement.

Assessment of transition readiness using standardised tools, such as the Transition Readiness Assessment Questionnaire (TRAQ)²³ and the HIV adolescent readiness for transition scale (HARTS)¹⁷ have been proposed to be predictive of post-transition outcomes.

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9.3 Women's health in HIV

Women living with HIV have distinct health needs that evolve across the life course, encompassing reproductive health, cancer prevention, menopause, mental health and ageing-related comorbidities. HIV care for women should adopt a life-stage, women-centred approach, integrating these domains into routine HIV services to improve long-term health outcomes and quality of life.

9.3.1 Principles of women-centred HIV care

- Women living with HIV experience distinct biological, hormonal and psychosocial factors that influence disease expression, symptom burden and engagement with healthcare services.^{1,2}
- Care should be individualised where feasible, culturally sensitive and respectful of preferences and autonomy in reproductive and health decision-making.
- Where possible, integration of women's health considerations into routine HIV care may facilitate earlier identification of chronic comorbid conditions.

9.3.2 Reproductive health, fertility and contraception

Women living with HIV should have their fertility intentions and contraceptive needs routinely assessed as part of HIV care, in accordance with their reproductive life stage, health status and personal preferences³. These discussions should be initiated proactively by clinicians and revisited over time, as fertility intentions and contraceptive needs may change across the life course.

Key principles:

- HIV infection does not preclude pregnancy
- Effective ART with sustained viral suppression enables safe conception and prevents vertical transmission
- U=U applies in conception for serodiscordant couples⁴

9.3.2.1 Fertility and referral

Women living with HIV who experience subfertility or infertility should be offered referral for fertility evaluation and treatment where clinically indicated, as HIV infection alone is not a contraindication to assisted reproductive technologies.^{5,6} Referral should be considered in women who have not conceived after 12 months of regular unprotected intercourse, or after 6 months in women aged 35 years or older, or earlier in the presence of known risk factors for infertility.

9.3.2.2 Contraception

Women living with HIV can use all available contraceptive methods, including hormonal contraception, emergency contraception and long-acting reversible contraceptives (LARCs).⁷ The choice of contraception should be made in discussion with the woman and should be acceptable and medically appropriate, taking into account individual risk factors such as thrombosis, cardiovascular disease, stroke risk, migraine with aura and potential interactions with antiretroviral therapy.

When selecting hormonal contraception, clinicians should consider drug-drug interactions (DDIs) with ART⁸:

- Older non-nucleoside reverse transcriptase inhibitors and boosted protease inhibitors may reduce hormonal contraceptive efficacy.
- Integrase strand transfer inhibitor (INSTI)–based regimens have minimal clinically significant interactions with hormonal contraception and are preferred where appropriate.

Where clinically appropriate, selection of antiretroviral regimens with minimal contraceptive interactions may support continuation of a woman's preferred contraceptive method.

Intrauterine devices, including copper and levonorgestrel-releasing systems, are safe and effective for women living with HIV, with no evidence of increased genital tract HIV shedding.

Dual protection (condoms plus effective contraception) should be discussed where appropriate to prevent unintended pregnancy and sexually transmitted infections, even when viral suppression is achieved.

A summary of ART–contraceptive interactions is provided in Annexe 9.2.

9.3.2.3 Pregnancy care

Management of pregnancy, intrapartum care and prevention of mother-to-child transmission should follow Section 9.1.

9.3.3 Cervical cancer prevention and HPV screening

Women living with HIV have a substantially increased risk of persistent HPV infection, cervical intraepithelial neoplasia, and cervical cancer⁹. This elevated risk is not fully mitigated by ART, even with viral suppression.¹⁰

9.3.3.1 HPV vaccination

HPV vaccination is recommended for women living with HIV and should be offered as part of routine care.¹¹

- The nonavalent HPV vaccine (Gardasil 9[®]) is recommended where available as it provides broader coverage of oncogenic HPV types.¹²
- A 3-dose schedule (0, 2, and 6 months) is recommended for women living with HIV, ideally before the age of 26 years, and may be considered up to 45 years based on individual risk and clinical judgement.¹²

HPV vaccination does not replace the need for regular cervical cancer screening.¹¹

9.3.3.2 Screening modality and interval¹¹

- HPV DNA testing is the preferred primary screening modality where available, due to higher sensitivity for detecting high-grade cervical lesions
- Cytology alone (Pap smear) remains an acceptable alternative:
 - In settings where HPV testing is not available, and
 - In women aged <30 years.
- Recommended screening approach
 - Age < 30 years: Cytology screening
 - Age ≥ 30 years:
 - HPV DNA testing every 3 years
 - If HPV DNA testing is unavailable: Cytology annually for 2 years, then every 3 years if results are normal
- Abnormal results:
 - Any positive oncogenic HPV result should prompt direct referral to Gynaecology team for colposcopy.
- Screening cessation/hysterectomy:
 - Lifetime screening for women with HIV
 - Post-hysterectomy screening should follow national recommendations
 - Women with total hysterectomy (cervix removed) for benign disease and no dysplasia/neoplasia do not require further screening
 - Women with subtotal hysterectomy, prior CIN, or other high-risk features require continued follow-up.
- Self-Sampling as HPV testing:
 - Where validated HPV DNA assays are available, either self-collected or clinician-collected samples may be used in women living with HIV. Evidence for self-sampling in this population is less extensive than in the general population; therefore, selection should consider local test validation, patient preference, accessibility and reliable follow-up of positive results.

9.3.4 Menopause and mid-life health

As the population of women living with HIV ages, menopause-related symptoms are increasingly encountered in HIV care settings. Perimenopause refers to the time of menstrual irregularity and symptom transition before the final menstrual period; menopause is diagnosed retrospectively after 12 months of amenorrhoea; postmenopause refers to the period thereafter.

Women with HIV may experience:

- Earlier onset of menopause¹³
- Increased severity of vasomotor, psychological, musculoskeletal, and sexual symptoms¹⁴
- Higher risk of osteoporosis, potentially exacerbated by HIV and ART^{15,16}

9.3.4.1 Clinical approach^{17,18}

- Menstrual history should be documented at baseline and periodic review of women's priorities as they transition through reproductive years, midlife, menopause and later life. Menopausal symptoms should be assessed routinely, particularly from age 35 onwards.
- Menopause in women aged > 45 years is usually a clinical diagnosis after ≥ 12 months of amenorrhoea in those not using hormonal contraception; routine FSH testing is not recommended in this group
- Use a validated symptom assessment tool, such as the Menopause Rating Scale (MRS) to document symptom severity and guide follow-up (Annexe 9.3)¹⁹.
- Exclude alternative or contributing causes of symptoms where relevant, including depression, thyroid disease and ART adverse effects.

9.3.4.2 Management^{14,20}

- Provide lifestyle advice for all women with symptoms, including sleep hygiene, regular physical activity, smoking cessation, reduction of alcohol intake, and weight-bearing exercise. Early menopause (≤ 45 years) or premature ovarian insufficiency (≤ 40 years) should prompt gynaecology specialist referral.
- Menopausal hormone therapy (MHT) may be considered by the treating gynaecology / menopause team after individualised risk assessment. The Malaysian menopause CPG recommends MHT for symptomatic women aged <60 years or within 10 years of menopause, and for prevention of bone loss where appropriate.
- Potential ART-hormone interactions should be reviewed when menopausal therapy is initiated or adjusted. Unlike hormonal contraception, drug-drug interactions with menopausal hormone therapy are less likely to result in clinically significant harm, and management is typically guided by symptom control rather than efficacy.
- Bone health assessment and osteoporosis management should follow Section 8.8.

9.3.5 Ageing, multimorbidity and cancer screening

Women living with HIV are ageing, with an increasing burden of multimorbidity, frailty, metabolic disease, and mental health conditions, often occurring earlier and with greater complexity compared to women without HIV. Comprehensive management of ageing-related conditions is discussed in Section 9.5.

Cancer screening

- Breast cancer is one of the most common cancers among women in Malaysia, and screening practices in women living with HIV are aligned with those for the general population.
- Breast cancer screening should follow national guidelines. Clinical breast examination is recommended from age 35 years, and mammography every 2 years for women aged 50-74 years, with earlier and more frequent screening for those at increased risk (e.g. strong family history, known genetic mutations, or prior chest irradiation).²¹
- Clinicians should promote breast awareness and ensure prompt evaluation of new breast symptoms
- Cervical cancer screening in women living with HIV is addressed in section 9.3.3.
- Colorectal and other age-appropriate cancer screening should follow national guidelines (refer section 8.9).

9.3.6 Additional women-specific considerations

The following issues should be routinely considered in HIV care for women:

- Mental health: Women living with HIV have higher rates of depression, anxiety, trauma, and psychosocial stressors, which may impact adherence, engagement in care, and overall health outcomes². Routine screening is recommended, with referral as appropriate (refer section 8.10).
- Intimate partner violence (IPV) and safeguarding: Consider routine enquiry where appropriate, as IPV may affect treatment adherence, mental health and overall wellbeing.
- Sexual function and genitourinary syndrome of menopause (GSM): vaginal dryness, dyspareunia, and libido concerns should be actively addressed.
- Vaccination: ensure vaccination status is reviewed as part of routine HIV care, including HPV and other recommended vaccines (refer section 4.6).

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9.4 HIV among transgender population

9.4.1 Introduction

Transgender individuals who identify themselves with a gender different from their sex assigned at birth. They are at significant risk for HIV infection¹ and their individual level risk factors include condomless sex (particularly receptive anal sex), coinfection with other sexually transmitted infections, transactional sex, and the shared use of needles for hormone and/or silicone injections. These risks often intersect with other factors such as mental health difficulties, substance use, and many forms of marginalisation and stigmatisation that limit educational and work opportunities.²

The available data in Malaysia consistently show that transwomen (male to female transgender; often classified as “mak nyah”) have high HIV risk. The Malaysian Integrated Bio-Behavioural Surveys (IBBS) conducted in 2009, 2012, 2014 and 2022 consistently reported HIV prevalence between 5 – 10% among them.³ A recent Sarawak IBBS-style survey reported a 10.1% HIV prevalence among these transwomen.⁴

Little is known about female-to-male transgender (transmen) in this country as their number is small and they are less studied. Worldwide, the HIV prevalence among transmen was reported as 2.56% compared to 19.9% among transwomen.⁵

Transgender populations have unique clinical, hormonal, and social factors that can affect both HIV care and overall health.⁶

9.4.2 Clinical and pharmacologic considerations

9.4.2.1 Gender-affirming hormone therapy (GAHT) interactions

- Transwomen often take Oestrogen (Estradiol) and sometimes anti-androgens (Spironolactone, Cyproterone acetate).
- Some antiretroviral (ARV) drugs—especially Efavirenz, Nevirapine, and boosted Protease Inhibitors—can reduce Estradiol levels via liver enzyme induction (CYP3A4).
- This may cause suboptimal feminizing effects, leading to reduced adherence if patients fear their hormones “won’t work.”
- Conversely, oestrogen can modestly affect ARV metabolism, but this rarely leads to clinically significant toxicity.
- Among transmen, the cornerstone of masculinizing hormone therapy is testosterone. No clinically significant interaction has been demonstrated between testosterone and any ARV.

9.4.2.2 Preferred ARV choices

- Integrase inhibitor (INSTI) based regimens are the preferred ARV for transgender patients because they have minimal drug–drug interactions with hormones.
- However, INSTI-based regimens (as well as TAF) are associated with weight gain. Since body image and fat distribution are highly sensitive areas for individuals undergoing GAHT, this metabolic side effect should be actively monitored and discussed with the patient.

9.4.3 Monitoring and laboratory issues

- Baseline and follow-up tests may need to consider the patient's affirmed gender for reference ranges (e.g., haemoglobin, creatinine, liver enzymes).
- Hormone levels (estradiol/testosterone) should ideally be monitored if on GAHT, especially after ARV initiation or changes.
- Routine HIV monitoring (CD4 count, viral load) remains the same as other populations.

9.4.4 Mental health and adherence

- Transgender individuals often face high levels of mental health stressors.
- These contribute to lower ART adherence and poorer retention in care compared to cisgender population.
- Integrated gender-affirming and mental health services improve adherence and outcomes.

9.4.5 Service delivery and environment

- Transwomen globally have substantially higher HIV prevalence and face stigma, criminalisation, violence and health-system discrimination.
- Transgender-inclusive HIV services are crucial — staff should be trained on correct name/pronoun use, confidentiality, and respectful communication.
- Peer navigation and community-led programs significantly improve engagement in care.
- Screening for STIs, mental health conditions, and gender-based violence should be routine.

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9.5 Ageing with HIV

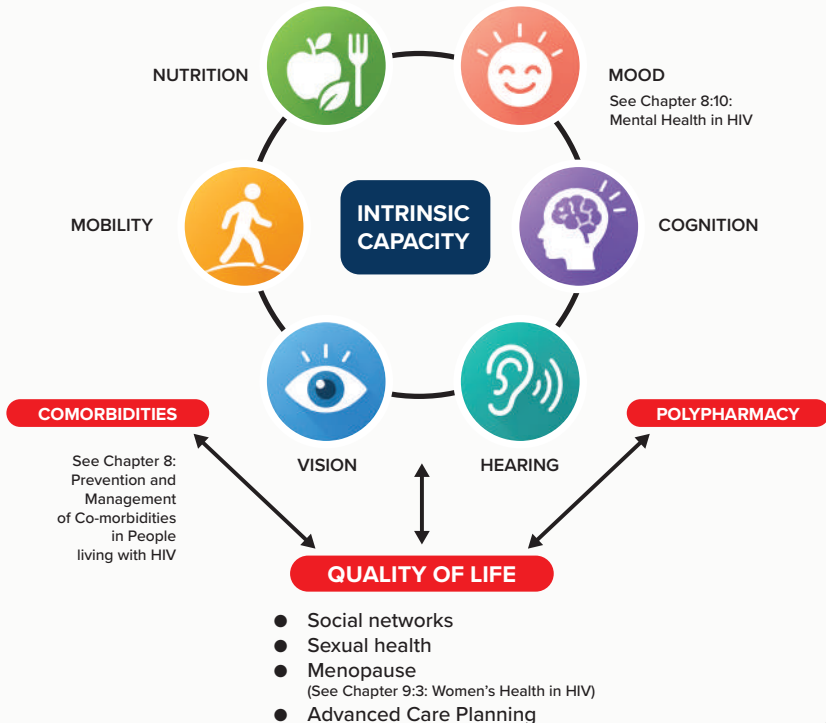
9.5.1 Introduction

In the era of ART, people living with HIV are ageing. As people age with HIV, they experience accelerated comorbidities and geriatric syndromes. The recent Lancet HIV Commission on ageing and HIV emphasises establishing a healthy ageing trajectory from the time of ART initiation, with increasing attention over time to comorbidity control, maintaining function and what matters most to the individual. WHO defines healthy ageing as “developing and maintaining the functional ability that enables wellbeing in older age.” Functional ability refers to an individual’s intrinsic capacity comprising five domains ie. locomotor (mobility), vitality (nutrition), cognitive, psychological and sensory (vision and hearing) and their interaction with the environment.

In order to facilitate healthy ageing and good quality of life, clinicians should proactively prevent or optimise the management of comorbidities, assess for any impairment in an individual’s intrinsic capacity using the Integrated Care for Older People (ICOPE) framework, and minimize polypharmacy where possible.

An older person in the context of HIV is defined as an individual aged 50 and above.

Figure 9.1 • Framework for management of older persons with HIV



9.5.2 Screening

9.5.2.1 Comorbidities

See Chapter 8: Prevention and management of comorbidities in PLWH.

9.5.2.2 Intrinsic capacity

Intrinsic capacity can be assessed using the Integrated Care for Older Persons (ICOPE). The ICOPE is a screening tool meant for use in primary care facilities. Early identification of any loss in intrinsic capacity provides an opportunity for intervention to slow down or reverse the declines identified. These declines often occur before the development of overt frailty which further increases the risk of falls and disability. Where abnormalities are detected, more thorough assessment for that domain is indicated.

Figure 9.2 • ICOPE assessment for intrinsic capacity

Basic assessment for loss of intrinsic capacity

	Filter question If Yes, proceed for in-depth assessment (Step 2)	Tests	Assess fully any domain with a checked circle	Pass
Cognitive decline (Cognition)	Do you have problems with memory or orientation (such as not knowing where you are or what day it is)?	1. Remember three words (use nouns, for example): flower, door, rice. 2. Orientation in time and space: what in the full date today? where are you now (home, clinic, etc)? 3. Recalls the three words?	☐ Wrong to either question or does not know ☐ Cannot recall all three words	☐ Correct to both questions
Limited mobility (Locomotor capacity)		Chair rise test Rise from chair five times without using arms. Did the person complete five chair rises within 14 seconds?	☐ No	☐ Yes
Undernutrition (Vitality)		1. Weight loss Have you unintentionally lost more than 3 kg over the last 3 months? 2. Appetite loss Have you experienced loss of appetite?	☐ Yes ☐ Yes	☐ No to both questions
Vision Impairment (Vision) *with spectacles if normally worn	Do you have problems with your eyes: difficulties in seeing far or near*, eye pain or discomfort?	1. External eye check 2. Visual acuity test using WHO vision screening chart*: • Distance vision (6/12 for each eye) • Near vision (N6 for both eyes)	☐ Fail ☐ Fail	☐ Pass ☐ Pass for both distance and near vision
Hearing Loss (Hearing)	Do you have diabetes, or hypertension, or are currently using steroids or eye medications?	Whisper test or Screening audiometry Digits-triplet-in-noise test	☐ Fail	☐ Pass
Depressive Symptoms (Psychological capacity)	Do you have hearing problem? For those using a hearing aid(s) add "even when using your hearing aid(s)".	Over the past 2 weeks, have you been bothered either of the following: • Feeling down, depressed or hopeless? • Little interest or pleasure in doing thing?	☐ Yes ☐ Yes	☐ No to both questions

9.5.2.3 Management of abnormal ICOPE screening

Table 9.4 • ICOPE domains and its management

ICOPE Domains	Management in Primary Care	Referral
Cognition	See Figure 9.3. For screening and management of abnormal cognitive screening test	
Mobility	• Falls risk assessment and gait observation • Review medications (e.g. sedatives) • Refer to physiotherapy or occupational therapy • Initiate exercise, balance training and fall-prevention advice • Provide walking aids if indicated	Geriatrics / Internal medicine: Recurrent falls Frailty Multi-domain impairment
Nutrition	• Measure BMI and screen for unintentional weight loss • Assess oral / dental health, swallowing, food security and mood disorders • Basic investigations: Full blood count and albumin • Referral to dietician for counselling	Geriatrics / Internal medicine: Unexplained weight loss Sarcopenia Severe malnutrition
Vision	• Screen for cataract, diabetic retinopathy or other eye conditions • Optimise diabetes and hypertension control • Advice on environmental safety (e.g. lighting - falls risk)	Ophthalmology: Visual impairment affecting daily function Suspected retinal disease Cataract CMV retinitis
Hearing	• Ask about hearing difficulties and communication problems • Inspect ears for wax or infection • Counsel on communication strategies and safety	ENT: Need for audiometry or hearing aids Persistent hearing loss
Mood	• See Chapter 8.10 on Mental Health • Assess psychosocial stressors, stigma and social isolation • Provide counselling, psychoeducation and social support • Initiate antidepressants if appropriate and monitor	Psychiatry: Moderate-severe depression Suicidal ideation Treatment-resistant symptoms Comorbid cognitive impairment

In clinics where more resources are available, more in-depth screening can be done as shown in Table 9.5

Table 9.5 • In-depth screening and its assessment tools

Domains	Assessment tools
Cognition	Montreal Cognitive Assessment (MoCA) tool (See Figure 9.3)
Psychological capacity/Mood	Patient Health Questionnaire (PHQ-9) Depression Anxiety and Stress Scale (DASS-21) The Alcohol, Smoking and Substance Involvement Screening Test (ASSIST) Lubben's Social Network Scale (Refer to Section 8.10: Mental Health)

9.5.3 Screening for Cognitive Impairment

9.5.3.1 Montreal Cognitive Assessment (MoCA) Test

The MoCA test is a tool used for early detection of mild cognitive impairment. Training for the use of this tool is available online in English, Bahasa Malaysia, Mandarin and Tamil. The screening and management flow of abnormal cognitive tests is shown in Figure 9.3.

9.5.3.2 What is an adjusted MoCA Score?¹⁰

The normal cut-off for abnormality of the MoCA score is $\leq 26/30$.

An adjusted scoring system validated for the Malaysian population is available, accounting for performance differences depending on age, sex, education and ethnicity.

Adjusted MoCA score calculator:

<https://redcap.um.edu.my/surveys/?s=RPTP4Y47K7JE3C7A>

Table 9.6 • Interpretation of MoCA score

Deficit Score	Impairmen
0	Normal
1	Mild
2	Mild to moderate
3	Moderate
4	Moderate to severe
5	Severe

9.5.4 Polypharmacy

Polypharmacy is defined as a person on 5 medications or more. It often leads to increased risk of drug-drug interactions, reduced antiretroviral adherence and reduced quality of life. As people age with HIV and develop more comorbidities, regular review of their medications is essential to ensure that they are appropriate and safe.

Things to consider during a medication review (Adapted from EACS 2025):

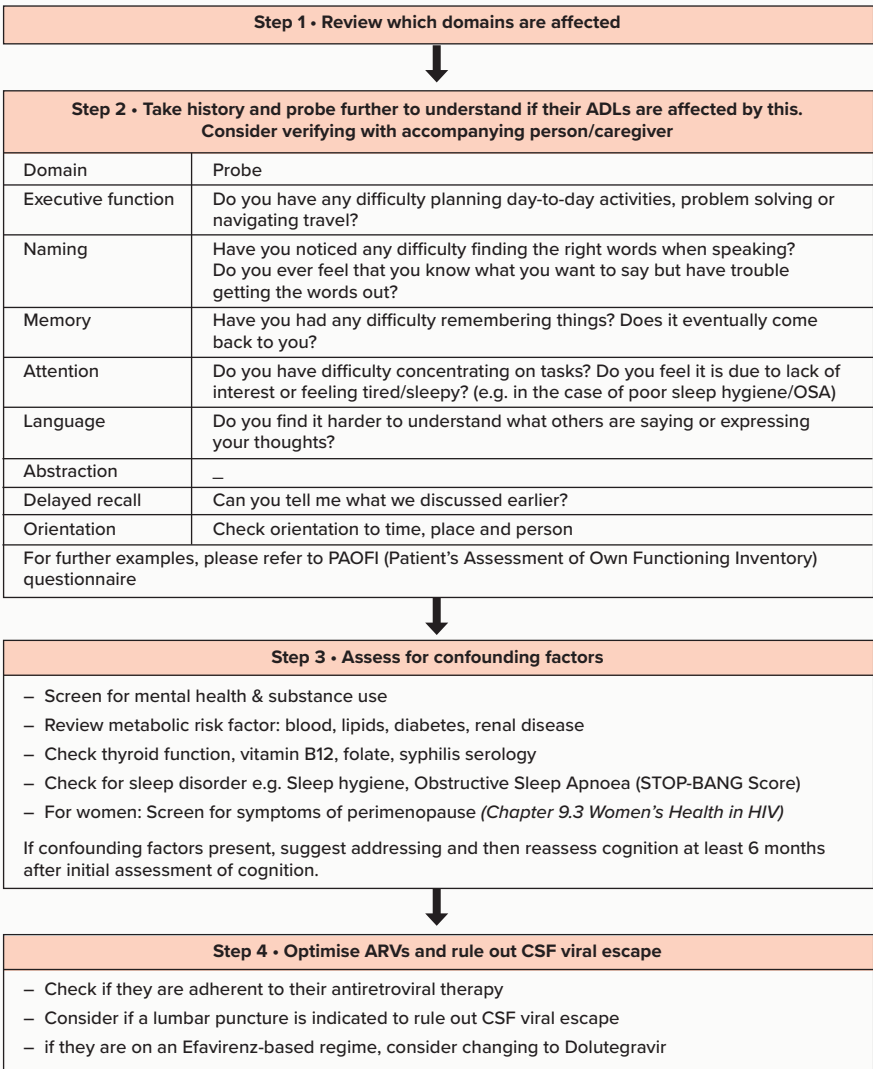
- Is there an indication for the medicine?
- Is this medicine appropriate for older persons with HIV?
- Is the dose correct (e.g. adjusted for renal function)?
- Is there any drug-drug interaction?
- Is the person able to manage his/her own medicines?.

Table 9.7 • Tools for assessing the appropriateness of medications and aid with de-prescribing where possible

Purpose	Tool
Drug-drug interaction	Liverpool HIV Interactions checker https://www.hiv-druginteractions.org/ Refer to Annexe 5.3
Review of potentially inappropriate medication use in older adults	<u>American Geriatrics Society 2023 - Beers Criteria®</u> Note: Mobile phone app is also available with paid subscription <u>STOPP/START version 2</u> Screening Tool for Older People's Prescriptions (STOPP) Screening Tool to Alert to Right Treatment (START)

Figure 9.3 • Screening and management flowchart for abnormal cognitive test

MoCA screening shows impairment





Step 5 • Investigations	
Bloods	Thyroid function test, Vitamin B12, Syphilis
Imaging (if indicated)	Neuroimaging is recommended when cognitive impairment is suspected. Brain MRI is preferred however in resource limited setting; a non-contrast CT brain may be used as an initial screening to exclude gross structural pathology. Indication: Impairment in any domains with instrumental Activities of Daily Living (iADL) affected



Step 6 • Management	
– Optimise metabolic risk factors	
Physical exercise	5 days a week of aerobic exercises (150 minutes/week) with at least 2 days of strength training
Cognitive exercise	Consider brain training mobile apps e.g. Lumosity, NeuroNation, Peak Brain Training
Referral	Consider referral to Neurologist/Geriatician: – If MRI indicated – If adjusted MoCA deficit score* is ≥ 3 – If symptomatic with a MoCA deficit score of 1 or more Consider referral to Neuro Occupational Therapy: – If abnormal MoCA (regardless of domains)
Monitoring/ Follow up	MoCA deficit score 0–1: Every 2 years MoCA score 2: Annual screening MoCA score ≥ 3 : Every 6 months

Developed by Dr Kejal Hasnuk & Dr Wong Pui Li, Universiti Malaya Medical Centre – Sept 2025

9.5.5 Quality of life

A person's quality of life is a measure of their overall wellbeing and a core component in person-centered care. Assessing quality of life enables clinicians to understand what matters most to the individual, their priorities, concerns and tailor care plans to support them.

Table 9.8 • Tools that can be used to guide quality of life assessments

Assessment tool	Tool	Languages
WHOQOL-HIV Bref	Physical, psychological, level of independence, social relationships, environment and personal beliefs	English Bahasa Malaysia
POZQOL (Positive Quality of Life Scale)	Psychological well-being, social connectedness, stigma, health concerns and independence	English Mandarin
HIV PROMS (HIV Patient-Reported Outcome Measures)	Physical, psychological, social and socio-economic factors	English

Other aspects for consideration include:

- Sexual health
- Menopause
- Advance care planning

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9.6 HIV treatment among PLWH who use substances

9.6.1 Introduction

Substance use and substance use disorders (SUDs) are common among people living with HIV (PLWH) and associated with poorer health outcomes. Substance use increases HIV risk through sexual and injecting behaviours, contributes to drug-drug interactions and can lead to adverse events such as hepatotoxicity or overdose. It also interferes with access to HIV care, including testing, linkage to care, ART initiation and long-term adherence.¹ Routine screening for SUDs should therefore be integrated into HIV clinical care.

People who inject drugs (PWID) remain a key population disproportionately affected by HIV. Sharing of needles, syringes, and other injecting equipment is the primary driver of transmission in this group. Globally, the risk of HIV acquisition is 14 times higher among PWID than in the general population, and in 2023 nearly one in eight PWID (1.7 million, or 12%) were living with HIV.² In Malaysia, although HIV transmission has shifted predominantly to sexual routes due to successful harm-reduction programmes, many PWID continue to experience barriers to HIV care, including co-existing mental health disorders, challenges with adherence, and social disadvantages such as homelessness, poverty, criminalisation, and stigma. Integrating ART with harm reduction services such as opioid agonist therapy, needle and syringe programmes, and psychosocial support has been shown to improve adherence and treatment outcomes.

Sexualised drug use (SDU) or “chemsex,” involves the use of psychoactive substances in sexual contexts to enhance or prolong sexual experiences. While initially described among gay, bisexual, and other men who have sex with men (GBMSM), SDU is increasingly reported among transgender women, sex workers, and other groups, each with distinct motivations. Whilst chemsex is typically associated with drugs like mephedrone, gamma-hydroxybutyrate/gamma-butyrolactone (GHB/GBL) and crystal methamphetamine, other drugs such as ecstasy and ketamine can also be used. Furthermore, other substances may be used secondary to these drugs to improve performance and facilitate sex and certain sexual practices, including PDE5 inhibitors and amyl nitrite (poppers). These substances may be smoked, snorted, injected (slamming) or taken orally or used rectally. SDU increases the risk of HIV and STIs due to prolonged sexual activity, multiple partners, and condomless sex,³ while also contributing to dependence, overdose, and mental health complications such as anxiety, depression, or psychosis.⁴

Recognising and addressing substance use including injecting drug use and SUDs within HIV clinical care is essential for effective treatment and prevention. Healthcare providers should routinely inquire about substance use in a respectful, confidential, and non-judgemental manner. PLWH who use substances should be assessed for co-occurring mental health disorders and comorbid conditions such as depression, hepatitis B or C, tuberculosis, and sexually transmitted infections. A compassionate, person-centred approach that integrates ART with harm reduction, mental health, and social support services is key to improving outcomes in this population.

9.6.2 Screening and assessment for people who use substances

Screening for SUDs should be part of routine clinical care of all PLWH. Early identification supports timely intervention, optimises ART adherence, and reduces harm.

A simple initial screening can be done using the following questions:

- *“Have you ever used any substances or used a prescription medication for non-medical reasons?”*
- *“Have you ever had five (for men) or four (for women) or more drinks in a day?”*

A positive response to either question should prompt further assessment using validated screening tools, such as the Alcohol Use Disorders Identification Test (AUDIT), Alcohol, Smoking, and Substance Involvement Screening Test (ASSIST), CAGE questionnaires or others. Discussing results in a non-judgmental manner offers an opportunity for brief intervention, which is effective for individuals at moderate risk. Individuals at high risk should be referred for comprehensive assessment by trained SUD practitioners or addiction specialists.

All clients should also be screened for co-occurring mental health disorders and assessed for substance-related complications, including injection-site infections, STIs, viral hepatitis, tuberculosis, and malnutrition. Where appropriate, clients should be linked to harm-reduction services such as needle and syringe programmes, methadone maintenance therapy, and community-based peer support. Integrating these services into HIV care improves adherence, retention, and overall treatment outcomes.

9.6.3 Selecting, initiating and maintaining ART

ART should be initiated in all PLWH who use substances, regardless of ongoing drug or alcohol use. Active substance use is not a contraindication for ART. Effective treatment improves individual health, reduces transmission, and provides an entry point for engagement in broader health and harm-reduction services.

When selecting ART, clinicians should consider potential drug–drug interactions with substances or medications used to treat SUDs (e.g., methadone, buprenorphine, naltrexone). Comorbidities such as liver disease from alcohol or hepatitis C, psychiatric disorders, and adherence barriers should also be considered. Regimens should be simple, preferably once-daily single-tablet regimens, with a high barrier to resistance and low hepatotoxicity. Integrase strand transfer inhibitor (INSTI)-based regimens, such as dolutegravir-containing combinations, are preferred due to their favourable interaction profile and strong resistance barrier.

Maintaining ART in people who use substances requires a patient-centred approach that addresses adherence barriers and supports long-term engagement in care. Strategies include simplified regimens, peer or case-management support, flexible scheduling, community dispensing, and integration of ART with harm-reduction services such as opioid agonist therapy and needle–syringe programmes. Mental health and social services should also be incorporated to address depression, stigma, and unstable living conditions. Continuous monitoring of viral load and adherence should be performed at regular intervals, and any interruption in care such as hospitalisation or incarceration should trigger coordination to ensure uninterrupted ART supply and follow-up.

Table 9.9 • Commonly used substances and their impact on HIV and ART^{5,6,7}

Substance	Route of administration	Main effects	Specific harms / complications	ART considerations
Stimulants				
Cocaine (coke, crack)	Snorted, smoked, injected	Euphoria, hyper-alertness, sexual excitement	Vasospasm, arrhythmia, myocardial ischaemia, paranoia, psychosis, dependence	Risk of poor adherence; no major pharmacological interactions
MDMA (Ecstasy)	Oral	Energy, euphoria, empathy	Hyperthermia, dehydration, tachycardia, dilutional hyponatraemia and encephalopathy, serotonin syndrome, anxiety, mood changes, 'come down' fatigue	Increase level and toxicity of cobicistat (CYP2D6 inhibitor) and potential toxicity with ritonavir (CYP3A4 inhibitor); use with caution. Potential interaction with EFV, NVP
Mephedrone (meow meow, 4-MMC)	Smoked, snorted, injected, swallowed	Stimulation, euphoria, increased empathy	Psychosis, seizures, dystonia, hepatic/renal toxicity, sudden death	No major ART interactions; monitor for adherence issues
Methamphetamine (ice, meth, tina, crystal)	Smoked, snorted, injected, oral or anal (booty-bump)	Euphoria, increased energy, arousal, delayed ejaculation, prolonged sexual activity	Psychosis, hallucinations, agitation, hypertension, dehydration, seizures, hyperthermia, serotonin syndrome, depression during "crash", cardiovascular diseases	Caution with ritonavir/cobicistat (↑ stimulant levels) Risk of missed ART doses.
Opioids				
Heroin, Morphine, Fentanyl	Injected, inhaled, smoked	Euphoria, relaxation, analgesia	Overdose, respiratory depression, dependence, infection risk	EFV may ↓ methadone levels; monitor withdrawal; consider OAT

Substance	Route of administration	Main effects	Specific harms / complications	ART considerations
Depressants				
Alcohol	Oral	Relaxation, disinhibition	Liver disease, pancreatitis, poor adherence, intoxication	Avoid hepatotoxic ARVs (e.g., NVP); monitor LFTs
Benzodiazepine	Oral, injected	Sedation, anxiolysis	Dependence, overdose (esp. with opioids/ alcohol), cognitive impairment	Ritonavir and cobicistat may ↑ benzodiazepine levels; avoid long-acting forms
GHB / GBL (G-water, liquid E)	Oral	Euphoria, relaxation and sedation, social and sexual disinhibition	Coma, seizures, dependence, bradycardia and hypotension, respiratory depression, psychosis, severe withdrawal	Co-administration of GHB with CYP2D6 inhibitors (i.e. cobicistat) or CYP3A4 inhibitors (i.e. ritonavir, cobicistat) may lead to raised systemic exposures of GHB and increased toxicity. May precipitate seizure in patients with seizure-prone opportunistic infections e.g. toxoplasmosis, cryptococcal meningitis.
Hallucinogens				
Cannabis (Marijuana, Ganja, weed)	Smoked, oral	Relaxation, altered perception	Cognitive impairment, dependence, anxiety, poor adherence	Minimal ART interactions; may help appetite but impair adherence in heavy use
Ketamine (K, Kit-Kat, Vitamin K)	Snorted, oral, intramuscular	Dissociation, euphoria	Confusion, urinary tract and GI toxicity, dependence, cognitive impairment, neuropsychiatric issues	May interact with ritonavir and cobicistat (CYP3A4- and CYP2B6-inhibitor respectively). NNRTIs e.g. EFV and NVP may decrease ketamine effects through CYP3A4 and 2B6 induction. Monitor mental health, CV risk and adherence.

Substance	Route of administration	Main effects	Specific harms / complications	ART considerations
Hallucinogens				
LSD, magic mushrooms	Oral, sublingual/ buccal	Euphoria, mild stimulation, enhanced appreciation of music and surroundings, visually appealing distortions, intensification of sensual or sexual feelings	CNS, neuro-behavioural and psychiatric issues, dependence, Hallucinogen persisting perceptual disorder (HPPD)	No clinically significant ART interactions; counsel on impaired judgment leading to poor adherence or sexual risk
Others				
Amyl/Alkyl nitrites ('poppers')	Inhaled	Euphoria, vasodilation, sexual arousal	Hypotension, methaemoglobinemia, retinal toxicity	Contraindicated with PDE5 inhibitors (risk of severe hypotension)
Erectile dysfunction medications (sildenafil, tadalafil)	Oral	Erectile enhancement	Headache, flushing, hypotension, visual changes	Avoid co-administration with ritonavir/cobicistat (↑ drug levels, hypotension)
Kratom (Ketum)	Oral, rarely smoked	Stimulant effects at low doses; opioid-like sedation/ analgesia at higher doses	Dependence and withdrawal, nausea/ vomiting, constipation, hypertension, seizures (rare), liver injury, risk of adulterants	No major ART interactions known; monitor liver function if used frequently or with hepatotoxic ARVs (e.g., NVP, ritonavir)

Table 9.10 • Medications for treatment of substance use disorders⁸

Medication	Mechanism / main action	Clinical considerations	ART considerations
Opioid use disorder			
Methadone	Full opioid agonist; reduces cravings and withdrawal	First-line therapy; daily supervised dosing may be required; QT prolongation risk	EFV, NVP and PIs reduce methadone levels; monitor for withdrawal and adjust dose
Buprenorphine / Buprenorphine–Naloxone	Partial opioid agonist; ceiling effect for respiratory depression	Effective alternative to methadone; suitable for community-based treatment	Minimal ART interactions; monitor for additive CNS effects
Naltrexone (oral or depot injection)	Opioid antagonist; blocks euphoric effects	Must be opioid-free ≥ 7 –10 days before start; risk of poor adherence with oral form	May increase hepatotoxicity risk; monitor liver function (esp. with nevirapine or PIs)
Alcohol use disorder			
Naltrexone (oral or injectable)	Blocks reinforcing effects of alcohol	Reduces relapse; avoid in acute hepatitis or liver failure	Monitor LFTs; potential additive hepatotoxicity with ART
Acamprosate	Modulates glutamatergic transmission; reduces craving	Safe in liver disease; requires good adherence (TID dosing)	No known ART interactions; monitor renal function
Disulfiram	Causes unpleasant reaction with alcohol	Use only with motivated patients; avoid in hepatic impairment	Monitor for hepatotoxicity with ART (e.g. NVP)
Tobacco use disorder			
Nicotine Replacement Therapy (NRT)	Provides controlled nicotine to reduce withdrawal	Available as patch, gum, or lozenge; safe with ART	No ART interactions
Bupropion	Dopamine/ norepinephrine reuptake inhibitor; reduces cravings	Avoid in seizure risk; useful if concurrent depression	EFV may $\downarrow$ bupropion levels; monitor response
Varenicline	Partial nicotinic receptor agonist; reduces cravings	Start 1–2 weeks before quit date; monitor for mood changes	No significant ART interactions
Stimulant use disorder			
No approved pharmacotherapy	—	Behavioural therapies (CBT, contingency management) remain mainstay	Focus on psychosocial support and adherence counselling

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10.1 Pre-exposure prophylaxis (PrEP)

Pre-exposure prophylaxis (PrEP) is a highly effective HIV prevention strategy involving the use of antiretroviral medications by HIV-negative individuals at substantial risk of acquiring HIV. PrEP is recommended for populations at increased risk, including men who have sex with men (MSM), people who inject drugs (PWID), female sex workers (FSW) and transgender women (TGW).

When taken consistently, PrEP has been shown to reduce the risk of HIV acquisition by up to 99% among MSM, ~75% among heterosexual partners, and ~74% among PWID.^{5,6}

Initiation of PrEP requires confirmation of HIV-negative status and assessment for contraindications. PrEP should be provided as part of a comprehensive HIV prevention framework that includes consistent condom use, regular HIV testing, adherence counselling and screening for sexually transmitted infections (STIs).

10.1.1 Eligibility for PrEP

Individuals who may benefit from PrEP include:

- HIV-negative individuals without symptoms suggestive of acute HIV seroconversion
- Individuals at substantial risk of HIV acquisition, including:
 - Inconsistent or no condom use
 - Recent diagnosis of an STI
 - Repeated use of non-occupational post-exposure prophylaxis (nPEP)
 - Sexual partners with unsuppressed HIV viral load
- People who inject drugs (PWID)
- Individuals engaging in chemsex
- Individuals involved in transactional sex
- Individuals who self-request PrEP

10.1.2 Approved medications for HIV PrEP in Malaysia

Table 10.1 • PrEP options and its considerations

ART	Considerations
Tenofovir Disoproxil Fumarate/ Emtricitabine (TDF/FTC)	For all adults and adolescents > 35 kg, including PWID Requires CrCl > 60 mL/min
Injectable Long-Acting Cabotegravir	Once every 2 months For all adults and adolescents > 35 kg

10.1.3 Baseline and follow-up investigations for individuals on PrEP

Table 10.2 • Investigations for individuals on PrEP

Investigations	Screening considerations
HIV test (ideally 4th generation) and Rapid diagnostic test (RDT) – includes HIV self-testing (HIVST) / community-based testing (CBT)	Baseline screening and every 3 months
Syphilis test	Baseline screening and every 6 months
CT / NG NAAT test	Optional
Hepatitis C test (Anti-HCV or Hep C RDT)	Baseline screening and additional testing as indicated by risk factors or clinical suspicion
Hepatitis B test (HBsAg) (anti-HBs if available)	Baseline screening and additional testing as indicated by risk factors or clinical suspicion If anti-HBs < 10 mIU/mL, to vaccinate
Renal profile	Baseline screening and every 6 months for individuals aged 50 years and above Baseline screening and every 12 months for individuals with CrCl ≤ 90 mL/min, or more frequently if clinically indicated It is permissible to start PrEP in the absence of renal profile results

For further details on PrEP, refer to the National Guidelines on HIV PrEP Program Implementation 2025, Ministry of Health, Malaysia.

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10.2 Occupational post-exposure prophylaxis (PEP)

The risk for HIV transmission is approximately 0.3% after a percutaneous exposure to HIV-infected blood¹ and 0.09% after mucous membranes exposure.²

10.2.1 Types of exposure in which PEP is indicated

PEP is recommended following a high-risk injury if the index case is known to be HIV-positive and is not on ART for >6 months with a suppressed viral load within the last 6 months.³

Examples of high-risk injury:

- Percutaneous exposure: breach of skin by a sharp object (hollow-bore, solid-bore, cutting needles or broken glassware) that is contaminated with blood or visibly bloody fluids.^{1,4}
- Bites from a patient with visibly bleeding in the mouth and resulted in a severe, deep or multiple tissue injury.⁵
- Splash of blood, visibly bloody fluids or other potentially infectious materials (OPIM)⁶ to a mucosal surface (mouth, nose, or eyes).³
**Exposure to non-bloody saliva, tears, sweats, nasal secretions, vomitus, urine or feces does not require PEP.⁶*
- Non-intact skin: exposure of non-intact skin (abrasions, cuts, sores) to blood, visibly bloody fluids, or OPIM may be considered for PEP.

**Examples of other potentially infectious materials (OPIM) includes semen, vaginal secretions, breast milk, cerebrospinal fluid, synovial fluid, pleural fluid, peritoneal fluid, pericardial fluid, and amniotic fluid*

Table 10.3 • Summary table of PEP recommendations

Occupational exposure	Recommendation	
	Detectable or unknown HIV VL	Sustained undetectable HIV VL, on ART for at least 6 months with good adherence
Sharp injury	Recommended	Not recommended*
Mucosal splash injury	Recommended	Not recommended
Human bite with visibly bleeding in the mouth	Recommended	Not recommended
Non-intact skin	May be considered	Not recommended

**Consider PEP if viral suppression is uncertain or outdated, adherence is unreliable, drug resistance/treatment failure is suspected or the exposure is high-risk (e.g., deep puncture, hollow-bore needle, visible blood or intravascular needle use). Option not to start PEP should be a shared decision-making and consider to consult ID Physician as the evidence for occupational “U=U” is reassuring but less definitive than for sexual transmission*

10.2.2 Timing of initiation of PEP

PEP is most effective when initiated as soon as possible, ideally within 24 hours and no later than 72 hours after exposure.^{7,8}

Decisions regarding PEP beyond 72 hours should be made on a case-by-case basis.

10.2.3 Duration of PEP

The duration of PEP should be 28 days.

PEP should be discontinued immediately if an HIV test on the source is found to be negative, unless the source is considered to be at higher risk of recent infection (i.e. within the HIV window period).

10.2.4 PEP recommendation when exposed to an unknown source

Occupational exposure to an unknown source (e.g. a needle in a sharps disposal container) creates challenges for post-exposure management because the HIV risk is not known.

The decision to give PEP has to be individualised depending on the HIV epidemiology risk profile and nature of exposure.

10.2.5 PEP regimen

Providing three drugs for post-exposure prophylaxis is consistent with recommendations for ART, the standard for which is triple-drug combination therapy as shown in Table 10.4.⁹

If the HIV status of the source patient is unknown, consent for voluntary HIV testing of the patient has to be obtained. HIV testing using rapid diagnostic test is strongly recommended for the source patient and all positive rapid diagnostic test should be confirmed by confirmatory test as soon as possible.

Do not delay the first dose of PEP while waiting for this information.

Table 10.4 • Choice of ARV in PEP

Preferred regimen	TDF/FTC + DTG TDF/3TC/DTG
Alternative regimen	TDF/FTC + LPV/r AZT/3TC + LPV/r

10.2.6 Follow-up and monitoring after PEP initiation

All healthcare workers receiving PEP should be re-evaluated within 3 days of the exposure. The summary of monitoring parameters is shown in Table 10.5

This allows for further clarification of the nature of the exposure, review of available source patient data, evaluation of adherence to PEP and toxicities associated with the PEP regimen.

During the 12 week follow up period, HIV-exposed healthcare workers should be advised to use condoms to prevent potential sexual transmission, avoid pregnancy and breast feeding in female HCWs and refrain from donating blood, plasma, organ, tissue or semen.

Table 10.5 • Monitoring after PEP initiation

Test	Baseline	Week 2	Week 12	Month 6
HIV	4th gen laboratory assay		4th gen laboratory assay	
Hep B	HBsAb, HBsAg, HBcAb For immunocompetent adults who have completed Hep B vaccination and responded (HBsAb >10 IU at any time) no baseline or follow up HepB testing is required		If unvaccinated or HBsAb <10 IU at the time of exposure: HBsAb, HBsAg	Only advised if HBsAb remains <10 IU at 12-weeks: HBsAg
Hep C	HCV Ab		HCV Ab If high risk exposure e.g. HCV positive source, then HCV PCR or Hep C Ag is preferable as the window period is shorter for antigen-based tests and can be requested as early as 2 weeks post exposure	If high risk exposure e.g. HCV positive source and testing at week 12 negative: HCV Ab
FBC, RP, LFT	✓	✓		

Adapted from 2023 British HIV Association guidelines & 2025 US Public Health Service Guidelines for the Management of Occupational Exposures to Human Immunodeficiency Virus and Recommendations for Post-exposure Prophylaxis in Healthcare Settings.

10.2.7 Responsibilities of hospital administrators

All hospitals must have a comprehensive plan to manage exposed HCWs. The following details must be included in the plan:

- The person in-charge of performing counselling and post-exposure evaluation to determine the need for PEP during and after office hours.
- The availability of ARVs needed for PEP within 2 hours of an exposure during and after office hours.
- Funding for ART.

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10.3 Non-occupational post-exposure prophylaxis (nPEP)

10.3.1 Indications

Situations prompting a request for nPEP include:

- Unprotected vaginal or anal sex
- Protected sex with condom failure
- Unsafe needle sharing
- Non-consensual sex where a high probability exists that the source is HIV positive

Treatment should always be combined with education and counseling to prevent future exposures.

10.3.2 Immediate management of HIV exposure

- Do not douche the vagina or rectum after sexual exposure.
- After oral exposure, spit out blood/body fluids and rinse the mouth with water.
- Wash wounds and skin sites that have been in contact with blood or body fluids with soap and water.
- Irrigate mucous membranes and eyes (remove contact lenses) with water or saline.
- Do not inject antiseptics or disinfectants into wounds.

10.3.3 Initial assessment for nPEP

- A prompt risk assessment and the timely initiation of nPEP, if indicated, must occur in a clinical setting that can provide expert advice, comprehensive testing and prevention counseling.
- A risk-benefit analysis should be undertaken for every individual presenting following an exposure and the decision to initiate nPEP made on a case-by-case basis.
- If the source individual is of unknown HIV status, proactive attempts should be made to establish the HIV status of the source as early as possible.
- If the source individual is known to be HIV-positive, attempts should be made at the earliest opportunity to determine the HIV viral load resistance profile and treatment history.
- nPEP is not recommended if the source is a person living with HIV with a non-detectable VL, defined as <200 copies/mL.
- This requires the source to have been on ART for at least six months with documented good adherence.
- If there is uncertainty about the source's adherence, ART history or viral load status, nPEP should still be considered for high-risk exposures.

10.3.4 Factors that increase the risk of HIV transmission

- **Viral Load (VL):** Higher plasma VL (e.g., during seroconversion or advanced disease) increases transmission risk
- **Co-factors:** Sexually transmitted infections (STIs) in the exposed individual or source (especially genital ulcer disease or symptomatic gonococcal infections)
- **Exposure details:** Source ejaculation during receptive anal, vaginal or oral intercourse
- **Mucosal integrity:** Breach in genital or oral mucosal integrity (e.g., trauma, menstruation, genital piercing or genital tract infection)
- **Circumcision:** Uncircumcised status of the insertive HIV-negative partner practicing insertive anal intercourse (IAI) or insertive vaginal intercourse (IVI)

Table 10.6 • HIV risk following type of exposure

Type of exposure	Estimated risk of HIV transmission Type of exposure per exposure from a known HIV-positive individual not on ART
Receptive anal intercourse	1 in 90
Receptive anal intercourse with ejaculation	1 in 65
Receptive anal intercourse no ejaculation	1 in 170
Insertive anal intercourse	1 in 666
Insertive anal intercourse not circumcised	1 in 161
Insertive anal intercourse and circumcised	1 in 909
Receptive vaginal intercourse	1 in 1000
Insertive vaginal intercourse	1 in 1,219
Semen splash to eye	<1 in 10,000
Receptive oral sex (giving fellatio)	<1 in 10,000
Insertive oral sex (receiving fellatio)	<1 in 10,000
Mucocutaneous	1 in 1000
Blood transfusion (one unit)	1 in 1
Needlestick injury	1 in 333
Sharing injecting equipment (includes chemsex)	1 in 149
Human bite	< 1 in 10,000
*These figures are estimates that have been deduced from cohort and modelling studies	

Adapted from UK guideline for the use of HIV post-exposure prophylaxis 2021 (2023 amendment)

10.3.5 nPEP recommendations after non-occupational exposure

The decision to prescribe nPEP involves assessing the type of exposure and the HIV status of the source, as summarised in Table 10.7

Table 10.7 • nPEP recommendations based on source status

Exposure	Source HIV VL unknown or detectable	Source known to have HIV, VL undetectable (<200 c/mL)	Source HIV status unknown, from high prevalence population	Source HIV status unknown, from low prevalence population
Receptive anal sex (RAS)	Recommended	Not recommended	Recommended	Not recommended
Insertive anal sex (IAI)	Recommended	Not recommended	Consider	Not recommended
Receptive vaginal sex (RVS)	Recommended	Not recommended	Generally not recommended	Not recommended
Insertive vaginal sex (IVS)	Consider	Not recommended	Generally not recommended	Not recommended
Shared injecting equipment	Recommended	Not recommended	Generally not recommended, but may consider in localized outbreak context	Not recommended
Receptive fellatio w/ ejaculation	Not recommended	Not recommended	Not recommended	Not recommended

Adapted from UK guideline for the use of HIV post-exposure prophylaxis 2021 (2023 amendment)

Note on recommendations categories:

- Recommended: Benefits are likely to outweigh the risks.
- Consider: Risk is low; risk/benefit balance is less clear, assess case-by-case.
- Generally not recommended: Risk is very low; potential toxicity is likely to outweigh benefit unless a clear specific extenuating factor increases the risk.
- Not recommended: Risk is negligible.

10.3.6 Behavioural intervention and risk-reduction counselling

The clinician should provide risk-reduction counselling and primary prevention counselling whenever someone presents for nPEP.

- Clinicians should assess for emotional, psychological and social factors that can contribute to risk behaviour such as depression, history of sexual abuse and drug and alcohol use.
- Clinicians should refer patients to psychiatry and/or substance use programmes when indicated and should consider the need for intensive risk-reduction counselling services.
- Patients who present with repeated high-risk behaviour should be considered for intensive risk reduction counselling and initiation of pre-exposure prophylaxis (PrEP).

10.3.7 Timing and regimens for nPEP

- nPEP should be offered to individuals with suspected or known exposure to HIV and as soon as possible, ideally within 24 hours and not later than 72 hours.
- nPEP should generally not be prescribed after 72 hours, though it may be considered on a case-by-case basis in consultation with a specialist for very high-risk exposures (e.g., known high viral load).
- The recommended duration of the nPEP course is 28 days and supplying the full course at the first attendance is generally recommended to maximise adherence and completion rates as starter packs may negatively affect completion.
- The recommended nPEP regimens and ART to be avoided as nPEP are shown in Table 10.8 and 10.9 respectively.

Table 10.8 • Recommended nPEP regimens

Preferred regimen	TDF/FTC + DTG TDF/3TC/DTG
Alternative regimen	TDF/FTC + LPV/r TDF/FTC + ATV/r AZT/3TC + LPV/r AZT/3TC + ATV/r

Table 10.9 • Antiretrovirals to avoid as nPEP

ARV to be avoided	Reasons
Efavirenz	CNS side effects which may impact on adherence and ability to work Concerns around EFV resistance in community HIV isolates
Nevirapine	Contraindicated due to potential risk of severe hepatotoxicity and severe rash
Abacavir	Potential for hypersensitivity reaction

10.3.8 Special situations

Switching from oral PrEP to nPEP is recommended in:

- **Anal sex:** If less than 4 tablets have been taken within the last 7 days or if the last dose was taken more than 7 days ago.
- **Vaginal sex:** If more than 48 hours have elapsed since the last dosing or if less than 6 tablets have been taken within the last 7 days.
- For sub-optimal event-based PrEP, and the last sexual exposure is within 72 hours.

Further high-risk exposure while on nPEP:

If a high-risk exposure occurs while the individual is taking nPEP, the course duration should be adjusted based on the site of exposure:

- **Anal sex:** Continue nPEP for 48 hours after the last high-risk exposure.
- **Vaginal/neovaginal sex:** Continue nPEP for 7 days after the last high-risk exposure.

Transitioning from nPEP to PrEP:

- Individuals who have completed the 28-day nPEP regimen and wish to use PrEP can start PrEP without a gap if they have a negative HIV test result upon completion of nPEP.

10.3.9 Follow-up and monitoring after nPEP initiation

All individuals receiving nPEP should be monitored as per Table 10.10.

Table 10.10 • Monitoring after nPEP initiation

Test	Baseline	Week 2	Week 6	Week 12
HIV serology	✓		✓ ^a	✓
Hep B ^b (HBsAb, HBsAg, HBcAb)	✓			✓
Hep C ^c (HCV Ab)	✓			✓
STI screening ^d	✓	✓		✓
Syphilis serology ^d	✓		✓	✓
RP	✓ ^e		✓	
UPT ^f	✓		✓	

Adapted from ASHM guidelines for PEP after non-occupational and occupational exposure to HIV Fourth Edition 2025

^a HIV testing is best performed at week 6, which is 2 weeks after cessation of nPEP.

^b If unvaccinated or HBsAb <10 IU at the time of exposure, should offer immunisation and follow-up to 6 months. If HBsAb >10 IU, will require no further follow-up.

^c Where HCV ab positive, and reflex HCV PCR testing is not available, recall patients for HCV PCR testing. Patients potentially at risk of HCV acquisition require baseline and follow-up testing for HCV.

^d Only required for sexual exposure. Conduct a full STI screen from all relevant sites as per history.

^e Seek specialist input for recommendation of alternative PEP drugs if eGFR <60 mL/min.

^f If clinically indicated. Consider emergency contraception.

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2. British Association for Sexual Health and HIV. UK guideline for the use of HIV post-exposure prophylaxis 2021: 2023 amendment [Internet]. London: BASHH; 2023. Available from: <https://bvhiva.org/clinical-guideline/pep-guidelines/>

Annexe 3.1 • Initial ART counselling guide

I. Education & Goals (The “Why”)

Point	Key Message for the Patient	Counsellor Action
Purpose	ART is now recommended for everyone with HIV, regardless of how well you feel or your test results. It is lifelong and does not cure HIV.	Explain that ART reduces the amount of virus in the body (Viral Load) and allows the immune system (CD4 count) to rebuild.
The Goal	The main goal is to get your Viral Load to Undetectable (below the level of detection). This is the best way to stay healthy and prevents the virus from changing (resistance).	State the measurable goal: Viral Suppression (e.g., <50 copies/mL).
U=U	When your viral load is sustained undetectable, you cannot transmit HIV to your sexual partners. Taking care of your health protects others.	Emphasize Undetectable = Untransmittable. Stress that this requires consistent adherence.
Resistance	Missing doses allows the virus to multiply and mutate, making the drugs stop working. Adherence must be >95%.	Explain the concept of drug resistance simply and the need for high adherence.

II. Medication Specifics (The “How”)

Point	Key Message for the Patient	Counsellor Action
Regimen	This is your specific medicine: [Name of Drug/Combination] and you will take [X] pill(s) [X] time(s) a day.	Give the patient written instructions. Explain the pill burden and dosing frequency.
Timing & Food	You must take this medicine at [Exact Time Chosen] every day. [Specify if it must be taken with food, on an empty stomach, or if food doesn't matter].	Collaboratively choose a time that fits their routine (e.g., breakfast, bedtime). Simplify the schedule where possible.
Side Effects	The first few weeks may bring mild side effects like nausea or fatigue, but they usually pass. Do NOT stop taking the drug.	Explain common, temporary side effects of the chosen regimen. Provide a clear plan for when to contact the clinic about side effects.

Point	Key Message for the Patient	Counsellor Action
Interactions	Tell the doctor/pharmacist about ALL other medications, supplements, vitamins, and herbal products, as they can interfere with your ART.	Specifically ask about traditional or street medications and over-the-counter pain relievers.

III. Adherence Plan & Support (The “When”)

Point	Key Message for the Patient	Counsellor Action
Reminder System	We need a reliable way to ensure you take your dose every day. What works best for you? (Alarm, pillbox, routine link).	Help the patient select a specific, personalized reminder tool and practice setting it up.
Missed Dose	If you miss a dose, take it as soon as you remember, unless it is almost time for your next dose. Never take a double dose.	Give a clear, written rule for missed doses, as protocols can vary slightly by drug.
Contingency	How will you take your medicine when you travel, go out for the weekend, or are away from home? Always pack extra.	Encourage planning ahead. Discuss where they will store the medicine (cool, dry, private place).
Readiness & Barriers	What, if anything, makes you feel nervous or unsure about starting treatment today? We can solve problems together.	Use open-ended, nonjudgmental questions to assess for practical barriers (housing, transport, cost) and psychosocial barriers (stigma, depression).
Follow-up	Your first follow-up appointment is crucial, as we need to see how your body is adjusting to the medicine and check your viral load soon.	Schedule the next appointment (often within 2-4 weeks) and emphasize the importance of consistent clinic attendance.

Before the patient leaves, ask them to repeat the name of the drug, the dose, and the time they chose to take it to confirm understanding.

REFERENCES

1. World Health Organization. Consolidated guidelines on HIV prevention, testing, treatment, service delivery and monitoring: recommendations for a public health approach. Geneva: World Health Organization; 2021. WHO
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3. Miller WR, Rollnick S. Motivational interviewing: helping people change. 3rd ed. New York: Guilford Press; 2013.

Annexe 4.1 Special considerations when initiating ART

Special considerations when initiating ART

Characteristics	Clinical scenario	Considerations	Rationale/comments
Pre-ART	HIV VL > 100,000 copies/mL	Avoid ABC-based regimen	Higher rate of virologic failure has been observed.
	HIV VL > 500,000 copies/mL	Avoid 3TC/DTG	Inadequate data
	HLA-B*5701 positive	Avoid ABC-based regimen	ABC hypersensitivity is highly associated with the presence of HLA-B*5701 allele.
Presence of specific comorbidities	HBV infection	Consider TDF or TAF-based regimen	TDF, TAF, FTC, and 3TC are active against both HIV and HBV. However, both 3TC and FTC associated with HBV mutations can emerge rapidly when used without another active agent against HBV.
	Treating TB with rifampicin	If rifampicin is used, PI is not recommended	Rifampicin is a strong inducer of CYP3A4 and UGT1A1 enzymes, causing significant decrease in concentrations of PIs and INSTIs. If a patient needs to start on PI or dolutegravir, there will be a 2-week washout period after discontinuing rifampicin before the ART can be commenced. If used together, the dose of INSTIs needs to be doubled.
		EFV can be used at a dose of 400mg with anti-tuberculosis therapy as the plasma levels were shown to be maintained above the levels considered to be effective.	Rifampicin has a less significant effect on EFV.

Characteristics	Clinical scenario	Considerations	Rationale/comments
Presence of specific comorbidities	Treating TB with rifampicin	TAF is not recommended with any rifampicin-containing regimen.	Rifampicin may significantly reduce TAF exposures.
	Osteoporosis	Avoid TDF-based regimen	TDF is highly associated with reduction in BMD, alongside renal tubulopathy, osteomalacia, and urine phosphate wasting.
	High risk of CV events	Avoid ABC-based regimen	ABC is associated with increased risk of CV events.
	Chronic kidney disease	Generally, avoid TDF if eGFR < 50 mL/min/1.73m ³ . If TDF cannot be avoided, the dose adjustment is recommended.	TDF is associated with proximal renal tubulopathy and renal dysfunction.

Annexe 4.2 Recommended vaccines for adults PLWH

Routine and risk-based non-live vaccines in adults PLWH

Vaccine	Vaccine type	Indications	Recommendations	Comments
COVID-19	mRNA or protein based	All PLWH	If never received before: 2-3 doses (4-8 weeks apart) If previously received: revaccinate 6-12 months after the most recent dose	Follow national recommendation; immune-compromised schedules apply
Hepatitis A	Inactivated	Non-immune PLWH, especially MSM, PWID, chronic liver disease, travel to endemic areas	Two dose series (0 and 6-12 months)	
Hepatitis B	Recombinant	All PLWH if non-immune	Recombivax HB: 40mcg at 0, 1, 6 months (3 doses) or Engerix-B: 40mcg at 0, 1, 2, 6 months (4 doses)	Non immune: anti-HBs <10mIU/mL Check HBsAg / antiHBS/anti-HBc before Hep B series Check anti-HBs 1-2 months after completing series; revaccinate if non-responder
HPV	Recombinant	Male and female < 26 years 27–45 years: via shared decision-making if at risk (e.g. MSM, prior HPV)	Three dose series at 0, 1-2, and 6 months	
Meningococcal ACWY	Conjugate	HIV is a recognized risk condition	2 doses of MenACWY ≥2 months apart and booster every 5 years if risk persists	Often recommended for travel (e.g. Hajj), or complement deficiency
Influenza	Inactivated Conjugate (PCV)	Annual vaccination	Single dose yearly	Avoid intranasal live vaccine (LAIV) - contraindicated

Vaccine	Vaccine type	Indications	Recommendations	Comments
Pneumo-coccal	Polysaccharide (PPSV23)	All PLWH (adults at any age)	Single dose PCV20 or PCV13 / PCV15 followed by PPSV23 ≥ 8 weeks later	If previously received PCV13 + PPSV23, give PCV20 once ≥ 5 years after last PPSV23
RSV	Recombinant, protein-based	Age ≥ 60 years	Single dose	Better immune response expected when CD4 ≥ 200 cells/mm ³
Shingles/ Zoster (RZV-Shingrix)	Recombinant, adjuvanted	All PLWH >18 years	Two-dose series at 0 and 2–6 months	Do not give RZV (Shingrix) during an acute episode of herpes zoster
Tetanus-Diphtheria	Toxoid	Routine adult booster	1 dose Tdap if not previously received Td/Tdap every 10 years	To give Tdap during each pregnancy for women; no HIV-specific modification *Tdap: Tetanus-Diphtheria toxoids and acellular pertussis vaccine

Live vaccines (conditional use) in adult PLWH

Vaccine	Indications	Recommendations	Comments
MMR	Non-immune adults (no prior vaccination or seronegativity), CD4 ≥ 200 cells/mm ³ and CD4 $\geq 15\%$	2 doses ≥ 4 weeks apart	Contraindicated if CD4 < 200 cells/mm ³ Avoid during pregnancy
Varicella	Seronegative adults with HIV, CD4 ≥ 200 cells/mm ³ and CD4 $\geq 15\%$	2 doses 4-8 weeks apart	Contraindicated if CD4 < 200 cells/mm ³
Yellow fever	Travel to endemic area and no safer alternative	May be given if CD4 ≥ 200 cells/mm ³ and stable; otherwise avoid or issue medical waiver	Pre-travel consult recommended; mosquito avoidance if vaccine not given

Annexe 5.1 Common and/or severe adverse events of ART^{1,2,3}

ART	Adverse Event Type	Adverse Event Description	Management
NRTIs			
ABC	Hypersensitivity reaction (HSR)	Clinical features (often multisystem; symptoms typically worsen with continued dosing): • Constitutional: fever, malaise, fatigue • Dermatological: rash • Gastrointestinal: nausea, vomiting, diarrhoea, abdominal pain • Respiratory: dyspnoea, cough, pharyngitis • Musculoskeletal: myalgia, arthralgia • Neurological: headache Timing: • Median onset: ~9 days after initiation • >90% of reactions occur within the first 6 weeks	Immediately discontinue ABC if HSR is suspected Do not rechallenge, regardless of HLA-B*57:01 status Re-exposure may result in rapid, severe, and potentially fatal reactions Symptoms usually resolve within 48 hours after discontinuation Provide supportive symptomatic management Switch to an alternative NRTI-based regimen
	Cardiovascular disease	Associated with ↑ risk of myocardial infarction Absolute cardiovascular risk is greatest in individuals with: • Established cardiovascular disease • Multiple traditional risk factors (e.g. smoking, diabetes, hypertension, dyslipidaemia)	In people with high cardiovascular risk (e.g. 10-year CVD risk ≥10%), consider switching to: • TDF- or TAF-based regimens or • An NRTI-sparing regimen, if clinically appropriate Optimisation of modifiable cardiovascular risk factors
	Metabolic effects	Dyslipidaemia: ↑ TG and ↑ LDL	Lifestyle modifications If dyslipidaemia is clinically significant or persistent: • Consider switching to antiretroviral agents with a more favourable lipid profile (e.g. TDF- or TAF-based regimens) • Initiate lipid-lowering therapy where indicated, with careful review of potential ARV–drug interactions

ART	Adverse Event Type	Adverse Event Description	Management	
NRTIs				
AZT	Gastrointestinal effects	GI intolerance (e.g., diarrhoea, nausea, vomiting)	Symptomatic management.	
		Higher incidence than other NRTIs	If symptoms persist, evaluate for alternative causes	
		Symptoms are usually self-limiting and resolve within 4–6 weeks	If intolerance is persistent and severe, consider switching to an alternative antiretroviral agent, where feasible	
	Metabolic effects	Lactic acidosis	Lactate 2–5 mmol/L Asymptomatic	Repeat test; monitor closely
			Lactate 2–5 mmol/L Symptomatic	Exclude other causes; if none found, watchfully follow-up or switch to low-risk NRTI (ABC, TDF, TAF)
			Lactate >5 mmol/L Normal pH/ HCO ₃	Switch to low-risk NRTI (ABC, TDF, TAF) with close monitoring or stop NRTI
			Lactate >5 mmol/L ↓ pH and/or ↓ HCO ₃	Stop NRTIs immediately
		Lipoatrophy: Associated with history of AZT exposure	Switch to an alternative NRTI (ABC, TDF, TAF)	
		Rhabdomyolysis, myopathy	Switch to an alternative NRTI (ABC, TDF, TAF)	
		Bone marrow suppression	Anaemia Leukopaenia	Switch to an alternative NRTI (ABC, TDF, TAF), if: • Hb drops ≥25% from baseline or <8.0 g/dL, or • Anaemia is symptomatic, and/or • Leukopaenia is significant
3TC	Generally well tolerated; significant adverse events are rare.			

ART	Adverse Event Type	Adverse Event Description	Management
NRTIs			
FTC	Dermatologic effects	Skin hyperpigmentation (usually mild and cosmetic)	No treatment required, may continue FTC Consider switching only if cosmetic concern is significant
TAF	Metabolic effects	Bone disease: Associated with less bone mineral density (BMD) loss compared with TDF	No specific intervention required Continue routine bone health assessment
		Dyslipidaemia: ↑ TG, ↑ LDL, and ↑ HDL (no change in TC:HDL ratio)	If dyslipidaemia is clinically significant or persistent: - Consider switching to antiretroviral agents with a more favourable lipid profile (e.g. TDF-based regimens) - Initiate lipid-lowering therapy where indicated, with careful review of potential ARV–drug interactions
		Weight gain: Greater ↑ weight reported with TAF compared with TDF	Lifestyle modifications
	Renal effects	Less renal effect than TDF due to lower systemic exposure	Continue routine renal monitoring assessment
TDF	Gastrointestinal effects	GI intolerance (e.g., diarrhoea, nausea, vomiting)	Symptomatic management. If symptoms persist, evaluate for alternative causes If intolerance is persistent and severe, consider switching to an alternative antiretroviral agent, where feasible
	Metabolic effects	Bone disease: Associated with greater loss of BMD following initiation Osteomalacia may occur in association with proximal renal tubulopathy and renal phosphate wasting Risk is increased with concurrent boosted PI use	Consider switching to an alternative NRTI (e.g. TAF or ABC) if: - Osteoporosis or progressive bone loss is present, or - There is a history of fragility fracture

ART	Adverse Event Type	Adverse Event Description	Management
NRTIs			
TDF	Metabolic effects	Dyslipidaemia: Associated with lower lipid levels than ABC or TAF	Lifestyle modifications If dyslipidaemia is clinically significant or persistent: Initiate lipid-lowering therapy where indicated, with careful review of any drug interactions
	Renal effects	Progressive ↑ SrCr or ↓ eGFR Proximal renal tubulopathy / Fanconi syndrome, characterised by: - Proteinuria, phosphaturia - Hypophosphataemia, hypouricaemia - Renal tubular acidosis - Normoglycaemic glucosuria Risk is increased with concurrent boosted PI use	Prevention: - Maintain adequate hydration (≥1.5–2 L/day of non-caffeinated fluids) - Avoid concurrent nephrotoxic agents - Optimise modifiable lifestyle risk factors Consider switching to an alternative NRTI (e.g. TAF or ABC) if: - Urine protein/creatinine ratio (UP/C) 15–50 mg/mmol, or - eGFR >60 mL/min with: - Decline ≥5 mL/min/year for ≥3 consecutive years, or ≥25% decline from baseline with no other cause - Presence of comorbidities with high CKD risk - Body weight <60 kg - Concurrent use of boosted PI Discontinue TDF and switch to an alternative NRTI if: - eGFR ≤60 mL/min - UP/C >50 mg/mmol - Previous TDF-associated proximal tubulopathy - Confirmed renal hypophosphataemia with no alternative cause

ART	Adverse Event Type	Adverse Event Description	Management
NNRTIs			
EFV	Neuro-psychiatric effects	Somnolence, insomnia, abnormal dreams, dizziness, impaired concentration Depression, psychosis, suicidal ideation, ataxia, encephalopathy may also occur EFV > RPV May be exacerbated by alcohol or other psychoactive substances	Symptoms often improve or resolve within 2–4 weeks Management • Advise bedtime dosing and taking EFV on empty stomach • Caution patients to avoid driving or operating machinery during initiation • Consider ↓ EFV dose from 600mg to 400mg in adults with significant neuropsychiatric effect • Avoid EFV or consider switching to an alternative ARV in individuals with: – Current or past depression, psychosis, suicidal ideation or attempts – Increased risk of self-harm
	Dermatologic effects	Rash Stevens-Johnson Syndrome/Toxic Epidermal Necrosis	Mild–moderate rash without systemic features: • Continue EFV • Provide symptomatic treatment (e.g. antihistamines) Severe rash or SJS/TEN: • Immediately discontinue EFV • Do not rechallenge Switch to an INSTI-based regimen where appropriate
	Cardiac conduction effects	QTc prolongation	Use with caution in patients with underlying conduction defects or on QT-prolonging drugs
	Metabolic effects	Dyslipidaemia: ↑ TG, ↑ LDL, ↑ HDL Lipohypertrophy: Truncal fat accumulation and gynaecomastia may occur	Lifestyle modifications If dyslipidaemia is clinically significant or persistent: • Initiate lipid-lowering therapy where indicated, with careful review of potential ARV–drug interactions Consider switching to an alternative NNRTI or INSTI

ART	Adverse Event Type	Adverse Event Description	Management
NNRTIs			
EFV	Hepatic effects	Most commonly manifests as ↑ ALT/AST Rarely: fulminant hepatitis or hepatic failure	Assess for other causes: alcohol, herbal products, supplements, or other hepatotoxic drugs Symptomatic patients: discontinue EFV and any other potential hepatotoxic agents immediately Asymptomatic patients: monitor liver function closely; consider discontinuing EFV if there is a persistent rising trend in transaminases after excluding other causes Restart ART with an alternative agent once liver function normalises, ensuring regimen efficacy is maintained Advise patients to stop alcohol and monitor ALT/AST until improvement
NVP	Hypersensitivity reactions	Risk factors: - Females with baseline CD4 >250 cells/μL - Males with baseline CD4 >400 cells/μL	Dose escalation during initiation reduces risk If hypersensitivity occurs, discontinue NVP and switch to EFV or an INSTI
	Dermatologic effects	Rash Stevens-Johnson Syndrome/Toxic Epidermal Necrosis	Mild–moderate rash without systemic features: - Continue NVP - Provide symptomatic treatment (e.g. antihistamines) Severe rash or SJS/TEN: - Immediately discontinue NVP - Do not rechallenge Switch to an INSTI-based regimen where appropriate

ART	Adverse Event Type	Adverse Event Description	Management
NNRTIs			
NVP	Hepatic effects	Most commonly manifests as ↑ transaminases Rarely: fulminant hepatitis or hepatic failure Typically occurs within the first 2–3 months of therapy Higher risk: ARV-naïve females with CD4 >250 cells/μL and males with CD4 >400 cells/μL	Assess for other causes: alcohol, herbal products, supplements, or other hepatotoxic drugs Symptomatic patients: discontinue NVP and any other potential hepatotoxic agents immediately Asymptomatic patients: monitor liver function closely; consider discontinuing NVP if there is a persistent rising trend in transaminases after excluding other causes Restart ART with an alternative agent once liver function normalises, ensuring regimen efficacy is maintained Advise patients to stop alcohol and monitor ALT/AST until improvement
RPV	Neuro-psychiatric effects	Depression, suicidal ideation, sleep disturbances (insomnia, abnormal dreams)	Monitor and provide supportive care for mild symptoms Consider switching to an alternative ARV (e.g., INSTI) if symptoms are severe or persistent
	Cardiac conduction effects	QTc prolongation	Use with caution in patients with underlying conduction defects or on QT-prolonging drugs Mild–moderate rash without systemic features: • Continue RPV • Provide symptomatic treatment (e.g. antihistamines)
	Dermatologic effects	Rash Stevens-Johnson Syndrome/Toxic Epidermal Necrosis	Severe rash or SJS/TEN: • Immediately discontinue RPV • Do not rechallenge Switch to an INSTI-based regimen where appropriate

ART	Adverse Event Type	Adverse Event Description	Management
NNRTIs			
RPV	Hepatic effects	Risk may be increased in people with HBV or HCV co-infection	Assess for other causes: alcohol, herbal products, supplements, or other hepatotoxic drugs Symptomatic patients: discontinue RPV and any other potential hepatotoxic agents immediately Asymptomatic patients: monitor liver function closely; consider discontinuing RPV if there is a persistent rising trend in transaminases after excluding other causes Restart ART with an alternative agent once liver function normalises, ensuring regimen efficacy is maintained Advise patients to stop alcohol and monitor ALT/AST until improvement
	Renal effects	Mild ↑ SrCr or ↓ eGFR (10–15 mL/min or up to 25%) due to inhibition of proximal tubular creatinine secretion; glomerular filtration is not impaired	Monitor renal function and establish a new baseline after 1–2 months If SrCr continues ↑ persistently, consider switching to an alternative ARV

ART	Adverse Event Type	Adverse Event Description	Management
PIs			
ATV	Cardiac conduction effects	May cause PR prolongation	Use with caution in patients with underlying conduction defects or on other drugs that prolong the PR interval
	Dermatologic effects	Rash	Mild–moderate rash without systemic features: • Continue ATV • Provide symptomatic treatment (e.g. antihistamines) Severe rash or SJS/TEN: • Immediately discontinue ATV • Do not rechallenge Switch to an INSTI-based regimen where appropriate
	Gastrointestinal effects	GI intolerance (e.g., diarrhoea, nausea, vomiting)	Symptomatic management. If symptoms persist, evaluate for alternative causes If intolerance is persistent and severe, consider switching to an alternative antiretroviral agent, where feasible
		Cholelithiasis and nephrolithiasis may occur, sometimes concurrently Median onset: ~42 months after ARV initiation	Symptomatic treatment for mild cases Consider switching if stones are recurrent, symptomatic, or causing complications
	Hepatic effects	Jaundice due to indirect (unconjugated) hyperbilirubinemia Benign and asymptomatic, not associated with hepatotoxicity	Reassure patients; no intervention usually required Monitor liver function if other risk factors for hepatotoxicity are present Consider switching only if jaundice is severe or cosmetically unacceptable

ART	Adverse Event Type	Adverse Event Description	Management
PIs			
ATV	Metabolic effects	Dyslipidaemia: ↑ TG, ↑ LDL, ↑ HDL	Lifestyle modifications If dyslipidaemia is clinically significant or persistent: Initiate lipid-lowering therapy where indicated, with careful review of potential ARV–drug interactions
	Renal effects	Nephrolithiasis/Urolithiasis	Consider switching to another PI or an INSTI if: • Confirmed renal stones • Recurrent loin pain ± haematuria • Progressive ↓ eGFR with no other identifiable cause
DRV	Cardiovascular disease	Associated with ↑ risk of cardiovascular events	In people with high cardiovascular risk (e.g. 10-year CVD risk ≥10%), consider switching to an INSTI-based regimen Optimisation of modifiable cardiovascular risk factors
	Dermatologic effects	Rash	Mild–moderate rash without systemic features: • Continue DRV • Provide symptomatic treatment (e.g. antihistamines) Severe rash or SJS/TEN: • Immediately discontinue DRV • Do not rechallenge Switch to an INSTI-based regimen where appropriate
	Gastrointestinal effects	GI intolerance (e.g., diarrhoea, nausea, vomiting)	Symptomatic management. • If symptoms persist, evaluate for alternative causes If intolerance is persistent and severe, consider switching to an alternative antiretroviral agent, where feasible

ART	Adverse Event Type	Adverse Event Description	Management
PIs			
DRV	Hepatic effects	Most commonly manifests as mild ↑ ALT/AST Rarely: fulminant hepatitis or hepatic failure	Assess for other causes: alcohol, herbal products, supplements, or other hepatotoxic drugs Symptomatic patients: discontinue DRV and any other potential hepatotoxic agents immediately Asymptomatic patients: monitor liver function closely; consider discontinuing DRV if there is a persistent rising trend in transaminases after excluding other causes Restart ART with an alternative agent once liver function normalises, ensuring regimen efficacy is maintained Advise patients to stop alcohol and monitor ALT/AST
	Metabolic effects	Dyslipidaemia: ↑ TG, ↑ LDL, ↑ HDL Hyperglycaemia/insulin resistance	Lifestyle modifications If dyslipidaemia is clinically significant or persistent: Initiate lipid-lowering therapy where indicated, with careful review of potential ARV–drug interactions
LPV	Cardiovascular disease	Associated with ↑ risk of cardiovascular	In people with high cardiovascular risk (e.g. 10-year CVD risk ≥10%), consider switching to an INSTI-based regimen Optimisation of modifiable cardiovascular risk factors
	Gastrointestinal effects	GI intolerance (e.g., diarrhoea, nausea, vomiting)	Symptomatic management. If symptoms persist, evaluate for alternative causes If intolerance is persistent and severe, consider switching to an alternative antiretroviral agent, where feasible

ART	Adverse Event Type	Adverse Event Description	Management
PIs			
LPV	Hepatic effects	Most commonly manifests as mild ↑ ALT/AST Rarely: fulminant hepatitis or hepatic failure	Assess for other causes: alcohol, herbal products, supplements, or other hepatotoxic drugs Symptomatic patients: discontinue LPV and any other potential hepatotoxic agents immediately Asymptomatic patients: monitor liver function closely; consider discontinuing LPV if there is a persistent rising trend in transaminases after excluding other causes Restart ART with an alternative agent once liver function normalises, ensuring regimen efficacy is maintained Advise patients to stop alcohol and monitor ALT/AST until improvement
	Metabolic effects	Dyslipidaemia: ↑ TG, ↑ LDL, ↑ HDL Hyperglycaemia/insulin resistance.	Lifestyle modifications If dyslipidaemia is clinically significant or persistent: Initiate lipid-lowering therapy where indicated, with careful review of potential ARV–drug interactions
INSTIs			
CAB	Injection site reaction	Very common (>80% of patients) Localised pain, nodules, induration, swelling, erythema, haematoma	Usually mild and self-limiting Symptomatic care: cold packs, analgesics if needed Reassure patients; reactions generally resolve without discontinuation
	Metabolic effects	Weight gain	Lifestyle modifications

ART	Adverse Event Type	Adverse Event Description	Management
INSTIs			
CAB	Neuro-psychiatric effect	Insomnia, depression, suicidal ideation Primarily reported in patients with pre-existing psychiatric conditions Generally milder than EFV	Use with caution in patients with a history of psychiatric illness Monitor for signs of depression or other neuropsychiatric symptoms Provide supportive care; consider switching ART if severe or persistent
DTG	Dermatologic effects	Rash: Mild rash occurs in ~2–5% of patients Severe or systemic reactions are very rare	Mild or no systemic symptoms: continue DTG Symptomatic relief: antihistamines Severe or systemic rash: discontinue DTG and consider alternative ART
	Metabolic effects	Weight gain	Lifestyle modifications
	Neuro-psychiatric effects	Insomnia, depression, suicidal ideation Primarily reported in patients with pre-existing psychiatric conditions Generally milder than EFV	Use with caution in patients with a history of psychiatric illness Monitor for signs of depression or other neuropsychiatric symptoms Provide supportive care; consider switching ART if severe or persistent
	Renal effects	Mild ↑ SrCr or ↓ eGFR (10–15 mL/min or up to 25%) due to inhibition of proximal tubular creatinine secretion; glomerular filtration is not impaired	Monitor renal function and establish a new baseline after 1–2 months If SrCr continues ↑ persistently, consider switching to an alternative ARV

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3. British HIV Association (BHIVA). BHIVA guidelines for the treatment of HIV-1-positive adults with antiretroviral therapy 2024. London (UK): BHIVA; 2024. Available at <https://www.bhiva.org/HIV-1-treatment-guidelines>. Accessed on 15/12/2025

Annexe 5.2 • ART dosing recommendations in adults with renal or hepatic impairment¹⁻⁴

ART	Usual dose	Dosing in renal impairment			Dosing in hepatic impairment																		
NRTIs																							
ABC	300mg q12h or 600mg q24h	No dose adjustment			Child-Pugh Class A: 200mg q12h (if available) Child-Pugh Class B or C: Contraindicated																		
ABC/3TC	One tablet q24h	Not recommended if CrCl <30 mL/min due to the 3TC component			Not recommended																		
AZT/3TC	One tablet q12h	<table><tr><th>CrCl (mL/min)^a</th><th>Dosing</th></tr><tr><td>≥ 50</td><td>No dose adjustment</td></tr><tr><td>30 - 49</td><td>Not recommended</td></tr></table>			CrCl (mL/min) ^a	Dosing	≥ 50	No dose adjustment	30 - 49	Not recommended	Not recommended												
		CrCl (mL/min) ^a	Dosing																				
≥ 50	No dose adjustment																						
30 - 49	Not recommended																						
3TC	300mg q24h or 150mg q12h	<table><tr><th>CrCl (mL/min)^a</th><th>Manufacturer labelling^b</th><th>Alternative dosing^c</th></tr><tr><td>≥ 50</td><td>No dose adjustment</td><td>No dose adjustment</td></tr><tr><td>30 - 49</td><td>150mg q24h</td><td>No dose adjustment</td></tr><tr><td>15 - 29</td><td>150mg q24h on day 1, then 100mg q24h</td><td>100mg - 150mg q24h</td></tr><tr><td>5 - 14</td><td>150mg q24h on day 1, then 50mg q24h</td><td>100mg - 150mg q24h</td></tr><tr><td><5 or on HD or PD</td><td>50mg q24h on day 1, then 25mg q24h</td><td>100mg – 150mg q24h</td></tr></table>			CrCl (mL/min) ^a	Manufacturer labelling ^b	Alternative dosing ^c	≥ 50	No dose adjustment	No dose adjustment	30 - 49	150mg q24h	No dose adjustment	15 - 29	150mg q24h on day 1, then 100mg q24h	100mg - 150mg q24h	5 - 14	150mg q24h on day 1, then 50mg q24h	100mg - 150mg q24h	<5 or on HD or PD	50mg q24h on day 1, then 25mg q24h	100mg – 150mg q24h	No dose adjustment
		CrCl (mL/min) ^a	Manufacturer labelling ^b	Alternative dosing ^c																			
		≥ 50	No dose adjustment	No dose adjustment																			
		30 - 49	150mg q24h	No dose adjustment																			
		15 - 29	150mg q24h on day 1, then 100mg q24h	100mg - 150mg q24h																			
		5 - 14	150mg q24h on day 1, then 50mg q24h	100mg - 150mg q24h																			
<5 or on HD or PD	50mg q24h on day 1, then 25mg q24h	100mg – 150mg q24h																					
TAF	25mg q24h	<table><tr><th>CrCl (mL/min)^a</th><th>Dosing</th></tr><tr><td>≥ 15</td><td>No dose adjustment</td></tr><tr><td>< 15 and not on HD</td><td>Not recommended</td></tr><tr><td>HD</td><td>No dose adjustment. On HD days, administer after dialysis</td></tr><tr><td>PD</td><td>Limited data. Consider alternative</td></tr></table>			CrCl (mL/min) ^a	Dosing	≥ 15	No dose adjustment	< 15 and not on HD	Not recommended	HD	No dose adjustment. On HD days, administer after dialysis	PD	Limited data. Consider alternative	Child-Pugh Class A: No dose adjustment Child-Pugh Class B or C: Not recommended								
		CrCl (mL/min) ^a	Dosing																				
		≥ 15	No dose adjustment																				
		< 15 and not on HD	Not recommended																				
		HD	No dose adjustment. On HD days, administer after dialysis																				
PD	Limited data. Consider alternative																						

ART	Usual dose	Dosing in renal impairment		Dosing in hepatic impairment
NRTIs				
TAF/ FTC	1 tablet q24h	CrCl (mL/min) ^a	Dosing	Child-Pugh Class A: No dose adjustment Child-Pugh Class B or C: Not recommended
		≥ 30	No dose adjustment	
		< 30	Not recommended	
		HD	No dose adjustment. On HD days, administer after dialysis	
		PD	Limited data. Use not recommended	
TDF	300mg q24h	CrCl (mL/min) ^a	Dosing	No dose adjustment
		30 - 49	300mg q48h	
		10 - 29	300mg q48h-q72h	
		<10 and not on HD	No recommendation	
		HD / PD	Avoid if possible If use cannot be avoided: 300mg q7d (administer post HD)	
TDF/ FTC	One tablet q24h	HIV treatment		No dose recommendation
		CrCl (mL/min) ^a	Dosing	
		30 - 49	Not recommended. If use cannot be avoided: one tablet q48h	
		<30 or on HD	Not recommended	
		PrEP and PEP		
		CrCl (mL/min) ^a	Dosing	
		≥ 60	No dose adjustment	
		< 60 or HD	Not recommended	
		NNRTIs		
EFV	600 - 400mg q24h	No dose adjustment		Child-Pugh Class A: No dose adjustment Use with caution Child-Pugh Class B or C: Not recommended

ART	Usual dose	Dosing in renal impairment		Dosing in hepatic impairment
NNRTIs				
TDF/ FTC/ EFV	One tablet q24h	CrCl (mL/min) ^a	Dosing	Child-Pugh Class A: No dose adjustment Use with caution
		30 - 49	Not recommended. If use cannot be avoided: one tablet q48h	
		<30 or on HD	Not recommended	Child-Pugh Class B or C: Not recommended
NVP	200mg q24h for 14 days, then 200mg q12h	No dose adjustment. In HD, an additional dose should be given following dialysis.		Child-Pugh Class A: No dose adjustment Use with caution Child-Pugh Class B or C: Not recommended
RPV	25mg q24h	No dose adjustment		Child-Pugh Class A or B: No dose adjustment Child-Pugh Class C: No dose recommendation
PIs				
ATV	300mg q24h (in combination with RTV 100mg)	RRT	Dosing	Not recommended
		HD (ART naïve)	No dose adjustment	
		HD (ART experienced)	Not recommended	
DRV	ART naïve or no DRV resistance: 800mg q24h (in combination with RTV 100mg) ART experienced with ≥1 DRV resistance mutations: 600mg q12h (in combination with RTV 100mg)	No dose adjustment		Child-Pugh Class A or B: No dose adjustment Child-Pugh Class C: Not recommended

ART	Usual dose	Dosing in renal impairment		Dosing in hepatic impairment
PIs				
LPV/r	400/100mg q12h	No dose adjustment		No dose adjustment Use with caution
INSTIs				
CAB	PrEP: IM 600mg once monthly for 2 doses, then 600mg once every 2 months after the last initiation dose.	No dose adjustment		Child-Pugh Class A or B: No dose adjustment Child-Pugh Class C: No dose recommendation
DTG	INSTI naïve: 50mg q24h Presence/ suspected INSTI resistance or co-administered with enzyme-inducers: 50mg q12h	No dose adjustment		Child-Pugh Class A or B: No dose adjustment Child-Pugh Class C: Not recommended
TDF/ 3TC/ DTG ⁵	One tablet q24h	CrCl (mL/min) ^a	Dosing	Child-Pugh Class A or B: No dose adjustment Child-Pugh Class C: Not recommended
		≥ 50	No dose adjustment	
		<50	Not recommended	

^a FDA prescribing information, renal dosing recommendations for most ART are based on CrCl using the Cockcroft–Gault formula

^b Dosing adjustment as per the manufacturer's labelling as an individual component of ART.

^c May consider using the nearest available tablet strength (100 mg or 150 mg). There is insufficient evidence to recommend for or against the use of full-dose 3TC in people with CrCl <30 mL/min. Use of 3TC doses higher than those recommended by the FDA for people with CrCl <30 mL/min has been reported in clinical practice and endorsed in the Guidelines for Chronic Kidney Disease in People With HIV and limited published literature has supported the safety of this practice.

REFERENCES

1. Panel on Antiretroviral Guidelines for Adults and Adolescents. Guidelines for the Use of Antiretroviral Agents in Adults and Adolescents with HIV. Department of Health and Human Services. 2025. Available at <https://clinicalinfo.hiv.gov/en/guidelines/adult-and-adolescent-arv>. Accessed on 14/12/2025
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Annexe 5.3A • Common NRTI drug interactions

This annexe summarises clinically important drug–drug interactions involving HIV Nucleoside Reverse Transcriptase Inhibitors (NRTIs), grouped by concomitant drug class. Interactions and management recommendations are based primarily on the latest U.S. DHHS HIV guidelines and the University of Liverpool interaction resources (accessed 2024–2025)

Drug class	Affected NRTI	Interaction	Management	Severity
Antivirals				
Adefovir / Cidofovir	TDF, TAF, ZDV	Additive nephrotoxicity; ↑ tenofovir	Avoid co-administration. Entecavir** is generally preferred as an alternative. If adefovir must be used, consider an alternative backbone which is clinically appropriate.	Avoid
Ganciclovir / Valganciclovir	TDF, TAF	↑ tenofovir/ganciclovir	Monitor renal function & FBC.	Caution
Ribavirin (with DAAs)	TDF	Certain DAAs ↑ tenofovir exposure. Sofosbuvir alone has limited impact; however, certain sofosbuvir-based combinations (e.g., sofosbuvir/velpatasvir, ledipasvir/sofosbuvir) can increase tenofovir exposure.	Monitor renal function.	Caution
Ribavirin	ZDV	Inhibits ZDV phosphorylation, reducing ARV efficacy and increasing haematological toxicity.	Avoid it if possible. If unavoidable, monitor for anaemia and neutropenia and ensure virologic monitoring.	Avoid
INSTI				
Dolutegravir (DTG)	TDF, TAF	Minor PK changes; not significant.	No adjustment needed.	Safe
Raltegravir (RAL)	TDF	↑ RAL AUC ~49%; TDF unchanged.	No adjustment needed.	Safe

Drug class	Affected NRTI	Interaction	Management	Severity
Anticonvulsants (Inducers)				
Carbamazepine	TAF	↓ TAF exposure ~55%	Avoid; use alternatives	Avoid
Phenytoin	TAF	Expected ↓ TAF. Phenytoin is a potent enzyme inducer and is expected to markedly lower TAF plasma concentration	Not recommended	Avoid
Phenobarbital	TAF	Expected ↓ TAF. Phenobarbital strongly induces drug-metabolising enzymes and is expected to reduce TAF and tenofovir diphosphate exposure significantly.	Not recommended	Avoid
Oxcarbazepine	TAF	Expected ↓ TAF. Expected to reduce TAF exposure, risking suboptimal tenofovir levels	Not recommended	Avoid
Antimycobacterial Drugs				
Rifampicin	TAF	↓ TAF ~55%; ↓ TFV-DP. Lowers intracellular tenofovir diphosphate, which can result in reduced antiviral efficacy.	Not recommended	Avoid
	TDF	No significant effect	No adjustment required	Safe
Rifabutin	TAF	Expected ↓ TAF. Rifabutin induces CYP3A and transporters to a lesser extent than rifampicin but is still expected to reduce TAF exposure and tenofovir diphosphate levels.	Not recommended	Avoid
	TDF	Minimal interaction. Rifabutin has minimal effect on TDF exposure; no clinically relevant interaction is expected.	No adjustment needed	Safe

Drug class	Affected NRTI	Interaction	Management	Severity
Narcotics / Opiates				
Buprenorphine	3TC, FTC, TDF, TAF, ZDV	No significant interaction	No adjustment needed	Safe
Methadone	ABC	↑ methadone clearance ~22%	Monitor clinically	Caution
	ZDV	↑ ZDV ~30–40%	Monitor for toxicity	Caution
Protease Inhibitors				
LPV/r	TAF	↑ TAF ~45–50%	No adjustment needed	Safe
	TDF	↑ TDF ~30%. Lopinavir/ritonavir increases TDF exposure by about 30%, which may raise the risk of tenofovir-associated renal impairment, especially in patients with other risk factors.	Monitor renal function	Caution
DRV/r	TAF	↑ TAF (variable)	No adjustment needed	Safe
	TDF	↑ TDF ~20–25%. Darunavir/ritonavir increases TDF exposure by approximately 20–25%, which can add to the risk of renal toxicity in susceptible patients.	Monitor renal function	Caution
Atazanavir (unboosted)	TDF	↓ ATV ~25%. Tenofovir reduces unboosted atazanavir AUC by about 25% through transporter-mediated effects, which may compromise atazanavir efficacy.	Do not co-administer. If atazanavir is required with TDF, use ritonavir-boosted atazanavir rather than unboosted atazanavir.	Avoid

Drug class	Affected NRTI	Interaction	Management	Severity
Others				
St John's Wort	TAF (also other ARVs)	Strong inducer ↓ TAF. St John's Wort is a strong inducer of CYP3A and P-gp, leading to substantially reduced TAF and other antiretroviral levels, risking virologic failure.	Avoid co-administration	Avoid

Abbreviations: DAA = Direct-acting antiviral; PK = Pharmacokinetics; TFV-DP = Tenofovir diphosphate.

* Severity coding: Red = Avoid; Yellow = Use with caution; Green = No adjustment required.

**Entecavir has minimal pharmacokinetic drug–drug interactions with antiretroviral agents due to its renal elimination and lack of CYP450 involvement. However, it should not be used as monotherapy in individuals with untreated HIV infection due to the risk of selecting for M184V resistance. In patients receiving tenofovir-containing regimens, renal function should be monitored due to potential additive nephrotoxicity.

Annexe 5.3B • Common NNRTI drug interactions

This annexe summarises clinically important drug–drug interactions involving HIV Non-Nucleoside Reverse Transcriptase Inhibitors (NNRTIs), grouped by concomitant drug class. Interactions and management recommendations are based primarily on the latest U.S. DHHS HIV guidelines and the University of Liverpool interaction resources (accessed 2024–2025).

Drug class	Affected NNRTI	Interaction	Management	Severity
Acid-reducing agents				
Antacids	RPV	Antacids containing polyvalent cations increase gastric pH, reducing solubility and absorption of rilpivirine, causing significantly lower plasma concentrations and potential loss of efficacy.	Give ≥ 2 h before or ≥ 4 h after RPV.	Caution
H2RA	RPV	H2-receptor antagonists elevate gastric pH for extended periods, lowering rilpivirine exposure and risking virologic failure.	Give ≥ 12 h before or ≥ 4 h after RPV.	Caution
PPIs	RPV	PPI causes profound pH elevation leading to severe reduction in RPV AUC ($>80\%$), which may result in loss of its therapeutic effect.	Contraindicated.	Avoid
Antibiotics				
Azithromycin	All NNRTIs	Azithromycin has minimal CYP3A involvement; clinically meaningful interactions with NNRTIs are unlikely.	No adjustment needed.	Safe
Clarithromycin	EFV ETR RPV NVP	Shared CYP3A metabolic pathway increases clarithromycin levels and may unpredictably increase NNRTI concentrations, increasing toxicity risk.	Alternative azithromycin is preferred.	Caution

Drug class	Affected NNRTI	Interaction	Management	Severity
Anticoagulants / Antiplatelets				
Apixaban/ Rivaroxaban	EFV ETR NVP	CYP3A induction by EFV/NVP markedly lowers DOAC exposure, increasing thrombosis risk.	Avoid; use alternatives.	Avoid
Dabigatran	All NNRTIs	Dabigatran is not metabolised by CYP pathways; minimal interaction expected.	No adjustment.	Safe
Clopidogrel	EFV, ETR	↓ activation of clopidogrel possible	Consider alternative ARV or antiplatelet. ETR may prevent metabolism of clopidogrel to its active metabolite.	Caution
Edoxaban	All NNRTIs	Minimal interaction expected.	No adjustment.	Safe
Ticagrelor	EFV ETR NVP	EFV and NVP induce CYP3A, significantly reducing ticagrelor exposure and antiplatelet effect.	Avoid or use alternative agents.	Avoid
Warfarin	EFV ETR NVP	NNRTIs variably induce or inhibit CYP2C9/ CYP3A, causing unpredictable INR fluctuations.	Monitor INR closely.	Caution
Anticonvulsants				
Carbamazepine	EFV ETR NVP RPV	Strong CYP induction lowers NNRTI plasma levels significantly, increasing risk of virologic failure.	Avoid.	Avoid
Phenytoin	EFV ETR NVP RPV	Variable induction may significantly reduce NNRTI concentrations; PK is highly unpredictable.	Avoid in RPV & ETR -contraindicated. Others, monitor levels	Caution
Phenobarbital	EFV ETR NVP RPV	Potent CYP induction → marked reduction in NNRTI concentrations.	Avoid.	Avoid

Drug class	Affected NNRTI	Interaction	Management	Severity
Antidepressants				
Bupropion	EFV NVP	EFV induces CYP2B6, reducing bupropion AUC by ~55%, lowering antidepressant effect.	Titrate dose based on response.	Caution
Escitalopram	EFV ETR NVP	EFV may reduce escitalopram exposure through CYP2C19 induction.	Titrate dose according to clinical effect.	Caution
Fluoxetine/ Fluvoxamine	All NNRTIs	Minimal meaningful interactions; CYP effects offset each other.	No adjustment needed.	Safe
Paroxetine	EFV ETR	Clinically minimal interaction documented.	No adjustment.	Safe
Sertraline	EFV	EFV reduces sertraline AUC by ~39%, potentially reducing antidepressant effect.	Titrate clinically.	Caution
Antifungals				
Fluconazole	EFV	Minor effects on EFV; occasional increases in EFV concentrations reported.	Monitor CNS effects.	Caution
	ETR	Fluconazole increases ETR AUC by ~86% through CYP2C9/ CYP3A inhibition.	Use with caution.	Caution
	NVP	Fluconazole increases NVP AUC by ~110%; may increase hepatotoxicity.	Monitor liver enzymes.	Caution
	RPV	Potential mild increase in RPV; not clinically significant.	No adjustment.	Safe
Isavuconazole	EFV ETR NVP	EFV induces CYP3A, lowering isavuconazole exposure significantly.	Monitor antifungal treatment response.	Caution

Drug class	Affected NNRTI	Interaction	Management	Severity
Antifungals				
Itraconazole	EFV NVP	EFV and NVP induce metabolism of itraconazole, markedly reducing levels. EFV with itraconazole solution-itraconazole and OH-itraconazole AUC, C _{max} , and C _{min} ↓ 37% to 44% EFV with itraconazole capsules • itraconazole AUC ↓ 86% and OH-itraconazole AUC 84%	Avoid if possible. Failure to achieve therapeutic itraconazole concentrations has been reported. If coadministration is necessary, closely monitor itraconazole concentration and adjust the dose accordingly. In settings where efavirenz is still used and itraconazole remains necessary, therapeutic drug monitoring should be considered.	Avoid
	ETR	Possible ↓ itraconazole and ↑ ETR levels through CYP3A interaction.	Monitor levels.	Caution
	RPV	Potential increase in RPV due to CYP3A inhibition.	Monitor toxicity.	Caution
Posaconazole	EFV	EFV significantly induces metabolism, reducing posaconazole AUC by ~50%.	If posaconazole is required, consider therapeutic drug monitoring where available.	Avoid
	RPV	Posaconazole may increase RPV exposure.	Monitor toxicity.	Caution
Voriconazole	EFV	EFV reduces voriconazole AUC by 77% and increases EFV AUC by 44%.	Contraindicated unless dose-adjusted closely. Voriconazole and efavirenz were shown to prolong the QT interval.	Avoid
	ETR	Voriconazole may increase ETR levels through CYP3A inhibition.	Monitor.	Caution
	RPV	Potential ↑ RPV due to CYP3A inhibition.	Monitor closely.	Caution
	NVP	Voriconazole exposure reduced; NVP exposure increased.	Avoid or monitor toxicity.	Caution

Drug class	Affected NNRTI	Interaction	Management	Severity
Antimalarials				
Artemether/ Lumefantrine	EFV	EFV reduces artemether and lumefantrine levels through CYP3A induction, risking treatment failure.	Avoid or monitor efficacy.	Caution
	ETR	ETR variably reduces artemether exposure.	Monitor efficacy.	Caution
	NVP	NVP lowers artemether by 67–72%; lumefantrine also reduced.	Monitor; consider alternative regimen.	Caution
Antimycobacterials				
Rifampicin	RPV	Potent CYP induction reduces RPV AUC by ~80%.	Contraindicated.	Avoid
	EFV	EFV AUC reduced by ~26%; EFV auto-induction compensates.	No adjustment.	Safe
	ETR	Marked reduction in ETR exposure.	Avoid.	Avoid
	NVP	Reduces NVP exposure by 20–58%.	Avoid.	Avoid
Rifabutin	EFV	EFV increases rifabutin AUC by 38%.	Use rifabutin 450–600 mg/day.	Caution
	ETR	ETR increases rifabutin; ETR AUC also ↑37%.	Avoid ETR + PI/r + rifabutin.	Avoid
	RPV	↑ RPV exposure; RPV should be increased to 50 mg OD.	No rifabutin adjustment.	Caution
	NVP	↑ rifabutin and ↓ NVP C _{min} .	Monitor.	Caution
Rifapentine	EFV	Minimal effect; EFV auto-induction compensates.	No adjustment needed.	Safe
	RPV	Substantial reduction in RPV exposure.	Contraindicated.	Avoid

Drug class	Affected NNRTI	Interaction	Management	Severity
Antipsychotics				
Aripiprazole	EFV ETR NVP	CYP induction reduces aripiprazole levels.	Consider increase aripiprazole dose; monitor.	Caution
Benzodiazepines				
Alprazolam	EFV ETR NVP	Variable reduction in alprazolam; may reduce anxiolytic effect.	Monitor.	Caution
Diazepam	EFV NVP ETR	Unpredictable ↑ or ↓ levels.	Monitor.	Caution
Lorazepam	All NNRTIs	Not metabolised via CYP; no expected interaction.	No adjustment.	Safe
Midazolam	EFV ETR NVP	May ↑ or ↓ midazolam via CYP3A effects.	Monitor sedation.	Caution
Cardiac medications				
Calcium Channel Blockers (CCB)	EFV ETR NVP	CYP induction reduces CCB exposure and effect.	Titrate CCB dose.	Caution
Ivabradine	EFV, ETR	↓ ivabradine expected. Potential risk of a QT interval prolongation	May require an adjustment of the dose of ivabradine	Avoid
Corticosteroids				
Dexamethasone	RPV	Dexamethasone strong induction significantly lowers RPV levels.	Contraindicated if >1 dose.	Avoid
	EFV ETR NVP	Induction reduces NNRTI levels.	Monitor VL.	Caution
HCV DAAs				
Daclatasvir	EFV ETR NVP	NNRTI induction reduces daclatasvir C _{min} ; AUC ↑ depending on dosing.	Use 90 mg OD.	Caution
	RPV	Minimal interaction with RPV.	No adjustment.	Safe

Drug class	Affected NNRTI	Interaction	Management	Severity
Hormonal therapies				
Oral Contraceptives	EFV	EFV reduces oestrogen/progestin exposure via induction.	Use additional contraception.	Caution
DMPA	EFV	Depot medroxyprogesterone levels generally remain adequate.	No adjustment; counsel.	Safe
Etonogestrel implant	EFV	Marked reduction → implant failures reported.	Avoid or add barrier method.	Avoid
Statins				
Simvastatin/ Lovastatin	EFV ETR NVP	CYP induction reduces statin exposure.	Adjust dose.	Caution
Atorvastatin	RPV	Minimal effect expected.	No adjustment.	Safe
Rosuvastatin	All NNRTIs	Minimal interaction.	No adjustment.	Safe
Pravastatin	EFV ETR	Reduced pravastatin levels due to induction.	Adjust dose.	Caution
Immunosuppressants				
Tacrolimus/ Cyclosporine/ Sirolimus/ Everolimus	EFV ETR NVP	Induction markedly lowers immunosuppressant exposure, risking graft rejection.	Therapeutic drug monitoring required.	Caution
Narcotics				
Buprenorphine	EFV ETR NVP	NNRTIs cause variable ↑ or ↓; withdrawal or sedation possible.	Monitor clinical status.	Caution
Methadone	EFV NVP	EFV reduces methadone AUC; withdrawal possible.	Monitor and increase methadone dose as necessary	Caution
	RPV	RPV reduces R-methadone by ~16%; usually not clinically significant.	Monitor.	Caution

Drug class	Affected NNRTI	Interaction	Management	Severity
PDE5 inhibitors				
Sildenafil/ Tadalafil	EFV ETR NVP	↓ PDE5 inhibitor exposure; reduced efficacy.	Titrate dose.	Caution
	RPV	Minimal interaction expected.	No adjustment.	Safe
Others				
St John's Wort	All NNRTIs	Strong CYP induction causes dramatic ↓ NNRTI levels, risking virologic failure.	Avoid.	Avoid
Tecovirimat	All NNRTIs	Possible ↓ ARV concentrations via induction.	Monitor viral load.	Caution

Abbreviations: PK=Pharmacokinetics; VL=Viral Load.

* Severity coding: Red = Avoid; Yellow = Use with caution; Green = No adjustment required

Annexe 5.3C • Common PI Drug Interactions

This annexe summarises clinically important drug–drug interactions involving HIV protease inhibitors (PIs), grouped by concomitant drug class. Interactions and management recommendations are based primarily on the latest U.S. DHHS HIV guidelines and the University of Liverpool interaction resources (accessed 2024–2025).

Drug class	Affected PI	Interaction	Management	Severity
Acid-reducing agents				
Antacids (aluminium, magnesium, calcium)	ATV, ATV/r	Antacids increase gastric pH and reduce ATV solubility, leading to reduced ATV exposure and potential loss of virologic efficacy.	Separate administration of ATV and antacids by at least 2 hours; avoid chronic high-dose antacid use and monitor VL if prolonged coadministration is required.	Caution
H2-receptor antagonists (e.g. ranitidine, famotidine)	ATV, ATV/r	H2 blockers increase gastric pH and can reduce ATV absorption; the effect is less marked than PPIs but still clinically relevant, especially in PI-experienced patients.	If co-administered, give ATV with food and follow guideline-recommended dose and timing separation; avoid high-dose H2 blockers in treatment-experienced and monitor VL.	Caution
Proton pump inhibitors (PPIs)	ATV, ATV/r	PPIs markedly increase gastric pH and can almost abolish ATV absorption, resulting in very low drug levels and high risk of virologic failure.	Avoid coadministration of PPIs with ATV (boosted or unboosted). Use an alternative PI or acid-suppressive strategy if long-term PPI therapy is essential.	Avoid
Antibiotics				
Azithromycin	All PIs	Azithromycin has minimal CYP3A involvement and does not significantly alter PI exposure; PK changes are not clinically meaningful.	No dose adjustment is required; monitor for usual macrolide adverse effects.	Safe

Drug class	Affected PI	Interaction	Management	Severity
Antibiotics				
Clarithromycin	All boosted PIs	Clarithromycin is a strong CYP3A inhibitor and substrate; coadministration with boosted PIs can substantially increase clarithromycin and, to a lesser extent, PI exposure, raising the risk of QT prolongation and other toxicities.	Prefer azithromycin where possible. If clarithromycin is required, consider dose adjustment in renal impairment, monitor ECG and for clarithromycin toxicity, and avoid additional QT-prolonging drugs.	Caution
Anticoagulants / Antiplatelets				
Apixaban	All boosted PIs	Strong CYP3A4 and P-gp inhibition by boosted PIs markedly increases exposure to apixaban, substantially raising bleeding risk.	Avoid combination where possible. Use an alternative anticoagulant such as low-molecular-weight heparin or warfarin with careful monitoring.	Avoid
Clopidogrel	All boosted PIs	Clopidogrel active metabolite AUC ↓ 69% in people with HIV on RTV regimens compared with healthy volunteers without HIV. Impaired platelet inhibition observed in people with HIV.	Do not coadminister.	Avoid
Dabigatran	All boosted PIs	Boosted PIs may increase dabigatran levels through P-gp inhibition, although the magnitude is variable and often modest.	Use with caution, especially in renal impairment; monitor for bleeding and consider dose reduction per product information.	Caution

Drug class	Affected PI	Interaction	Management	Severity
Anticoagulants / Antiplatelets				
Edoxaban	All boosted PIs	Coadministration has not been studied. Edoxaban is a substrate for P-gp and coadministration is expected to increase edoxaban concentrations due to P-gp inhibition	Treatment of Nonvalvular Atrial Fibrillation • No dose adjustment needed Treatment of DVT and PE • Reduce edoxaban dose to 30 mg once daily. US DHHS: No dose adjustment. European guidance: Consider reducing edoxaban from 60 mg to 30 mg with strong P-gp inhibitors.	Caution
Rivaroxaban	All boosted PIs	Strong CYP3A4 and P-gp inhibition by boosted PIs markedly increases exposure to rivaroxaban, substantially raising bleeding risk.	Avoid combinations where possible. Use an alternative anticoagulant such as low-molecular-weight heparin or warfarin with careful monitoring.	Avoid
Ticagrelor	All boosted PIs	Boosted PIs can substantially increase ticagrelor exposure via CYP3A4 inhibition, leading to an elevated bleeding risk.	Avoid ticagrelor. If a P2Y ₁₂ inhibitor is required, prasugrel is preferred (unless contraindicated).	Avoid
Warfarin	All boosted PIs	Boosted PIs can inhibit or induce CYP enzymes involved in warfarin metabolism, leading to unpredictable increases or decreases in INR.	Check INR more frequently after starting, stopping, or changing a PI and adjust warfarin dose accordingly.	Caution
Anticonvulsants				
Enzyme-inducing anticonvulsants (carbamazepine, phenytoin, phenobarbital, primidone)	All PIs	These anticonvulsants strongly induce CYP3A4 and other enzymes, causing marked reductions in PI plasma concentrations and risking virologic failure and resistance.	Avoid coadministration. If no alternatives exist, consider switching to a non-PI-based ART regimen and monitor viral load closely.	Avoid

Drug class	Affected PI	Interaction	Management	Severity
Anticonvulsants				
Valproic acid	All PIs	Valproic acid has minimal CYP3A4 involvement; modest changes in levels may occur but are generally not clinically significant	No routine dose adjustment is required, but monitor clinical response and valproate levels if toxicity or loss of seizure control is suspected.	Safe
Antidepressants, anxiolytics and antipsychotics				
SSRIs (e.g. escitalopram, fluoxetine, paroxetine, sertraline)	All PIs	PIs may modestly increase or decrease SSRI exposure via CYP interactions, but effects are usually small and variable.	Start SSRIs at the lower end of the usual dose range and titrate according to clinical response and tolerability; monitor for serotonin toxicity and lack of effect.	Caution
TCAs (e.g., amitriptyline, imipramine, nortriptyline)	All boosted PIs	Boosted PIs inhibit CYP2D6 and CYP3A4, which can increase TCA concentrations and risk of anticholinergic, sedative, and cardiotoxic effects.	Use low initial TCA doses, titrate slowly, and monitor for toxicity, including ECG where appropriate.	Caution
Antipsychotics (e.g., aripiprazole, quetiapine, risperidone)	All boosted PIs	Many antipsychotics are metabolized by CYP3A4 and CYP2D6; boosted PIs may substantially increase their exposure, increasing the risk of sedation, extrapyramidal symptoms, and QT prolongation.	Use the lowest effective dose, titrate cautiously, and monitor for adverse effects and QT prolongation; consider agents with lower interaction potential where feasible.	Caution
Zolpidem and related sedative-hypnotics	All PIs	PIs may increase sedative-hypnotic exposure to a modest extent, enhancing sedative and psychomotor effects.	Use the lowest effective dose and advise patients about additive CNS depression and next-day impairment.	Caution

Drug class	Affected PI	Interaction	Management	Severity
Antifungals				
Fluconazole	All PIs	Fluconazole is a moderate CYP3A inhibitor and can increase PI concentrations slightly, while PIs may also increase fluconazole exposure; overall effects are usually modest.	No routine dose adjustment is required, but monitor for hepatic toxicity and additive QT prolongation if other risk factors are present.	Caution
Isavuconazole	All PIs	Isavuconazole is metabolised by CYP3A4 and may have reduced or increased exposure when combined with PIs, although limited data suggest changes are usually modest.	Use with caution and monitor for antifungal efficacy and toxicity; consider therapeutic drug monitoring if available.	Caution
Itraconazole	All boosted PIs	Itraconazole is a potent CYP3A and P-gp inhibitor; coadministration with boosted PIs can markedly increase PI and itraconazole levels.	Avoid high itraconazole doses; Itraconazole doses >200 mg/day are not recommended unless dosing is guided by itraconazole concentrations and monitor closely for hepatotoxicity and drug toxicity.	Caution
Ketoconazole	All boosted PIs	Ketoconazole strongly inhibits CYP3A4, and coadministration with boosted PIs can lead to excessive PI exposure and increased risk of QT prolongation and hepatotoxicity.	Avoid high-dose ketoconazole; many guidelines recommend avoiding this combination where alternatives exist.	Avoid
Posaconazole	All boosted PIs	Posaconazole is a strong CYP3A inhibitor and P-gp substrate; combined use with boosted PIs can increase PI exposure, while PI effects on posaconazole may vary.	Use with caution, monitor hepatic function and drug tolerability, and consider therapeutic drug monitoring for posaconazole where available.	Caution

Drug class	Affected PI	Interaction	Management	Severity
Antifungals				
Voriconazole	All boosted PIs	Voriconazole and boosted PIs have complex bidirectional interactions; PIs may substantially decrease voriconazole exposure or increase toxicity depending on the regimen.	Avoid coadministration where possible; if unavoidable, consult a specialist and consider alternative ART or antifungal therapy.	Avoid
Antimalarials				
Artemether / Lumefantrine	All PIs	Boosted PIs may alter artemether and lumefantrine levels through CYP3A modulation, potentially reducing antimalarial efficacy or increasing toxicity.	Avoid empirical dose changes; monitor for malaria treatment failure and consider alternative antimalarial regimens if concern for significant interaction exists.	Caution
Antimycobacterials				
Rifampicin	All PIs	Rifampicin is a very strong inducer of CYP3A4 and P-gp, causing profound reductions in PI concentrations and rendering PI-based regimens ineffective.	Do not coadminister rifampicin with PIs. Use an alternative ART regimen or an alternative rifamycin strategy that does not include PIs.	Avoid
Rifabutin	All boosted PIs	Boosted PIs significantly increase rifabutin and its active metabolites, raising the risk of uveitis and neutropenia.	If used with boosted PIs, reduce rifabutin to 150 mg once daily. Monitor for antimycobacterial activity and consider therapeutic drug monitoring. Monitor for rifabutin-related adverse events, including neutropenia and uveitis.	Caution

Drug class	Affected PI	Interaction	Management	Severity
Antiarrhythmics and cardiac medications				
Amiodarone	All boosted PIs	Amiodarone is metabolised by CYP3A4 and has significant QT-prolonging potential; boosted PIs can increase amiodarone exposure and additive arrhythmia risk.	Avoid combination if possible; if absolutely necessary, use with cardiology input, ECG monitoring, and consideration of alternative ART or rhythm control strategies.	Avoid
Other class I/III antiarrhythmics (e.g. flecainide, propafenone, disopyramide, quinidine)	All boosted PIs	Many antiarrhythmics are CYP3A and/or P-gp substrates; boosted PIs can increase their plasma levels and increase risk of serious arrhythmia toxicity as well as proarrhythmic risk.	Avoid coadministration or use only with specialist input and intensive ECG and drug level monitoring.	Avoid
Calcium channel blockers (e.g. amlodipine, diltiazem, verapamil)	All boosted PIs	Boosted PIs inhibit CYP3A4 and can substantially increase CCB exposure, causing hypotension, bradycardia, and oedema.	Start CCBs at low doses and titrate carefully while monitoring blood pressure, heart rate, and for peripheral oedema.	Caution
Benzodiazepines and sedatives				
Alprazolam, Clonazepam, Diazepam	All boosted PIs	These benzodiazepines are metabolised by CYP3A; boosted PIs can increase their levels, prolonging sedation and psychomotor impairment.	Use the lowest effective dose, avoid concurrent CNS depressants, and monitor for oversedation; consider lorazepam, oxazepam, or temazepam as alternatives.	Caution

Drug class	Affected PI	Interaction	Management	Severity
Corticosteroids				
Inhaled / intranasal corticosteroids (fluticasone, budesonide, mometasone, ciclesonide)	All boosted PIs	Boosted PIs markedly increase exposure to these corticosteroids, leading to iatrogenic Cushing's syndrome and adrenal suppression even at standard inhaled doses.	Avoid these agents with boosted PIs. Prefer inhaled beclomethasone or non-steroidal options where possible and monitor for adrenal suppression if exposure has occurred. Because inhaled corticosteroids may still be essential for asthma control, switching to a non-boosted regimen such as DTG may be considered when clinically appropriate.	Avoid
Systemic corticosteroids (e.g., prednisolone, methyl-prednisolone)	All boosted PIs	Systemic corticosteroid concentrations can be increased by boosted PIs, raising the risk of steroid-related adverse effects including hyperglycaemia, osteoporosis, and adrenal suppression.	Use the lowest effective dose and shortest duration; monitor for steroid toxicity and taper carefully if prolonged therapy is required.	Caution
Dexamethasone (repeated or high doses)	All	Dexamethasone is a CYP3A inducer; with repeated or high doses it can reduce PI concentrations and compromise virologic control.	Avoid prolonged or high-dose dexamethasone in patients on PIs; if unavoidable, monitor viral load and consider switching to a non-PI-based ART regimen.	Caution
Endocrine / metabolic agents				
SGLT2 inhibitors (e.g. empagliflozin, dapagliflozin, combinations with saxagliptin)	All PIs	SGLT2 inhibitors are not major CYP3A substrates and do not significantly alter PI exposure; modest changes in glycaemic control may occur.	Limit saxagliptin dose to 2.5 mg once daily. Others: No dose adjustment is usually required, but monitor renal function, volume status, and glycaemic control, particularly when initiating or changing ART.	Safe

Drug class	Affected PI	Interaction	Management	Severity
HCV DAA				
Sofosbuvir / Velpatasvir (SOF/VEL) with TDF + boosted PI	All boosted PIs (in TDF-containing regimens)	Coadministration of SOF/VEL with TDF and a boosted PI increases TDF exposure, potentially exacerbating renal and bone toxicity.	Monitor renal function closely during treatment and consider switching TDF to TAF or using a non-tenofovir-based backbone in patients with eGFR <60 mL/min or other renal risk factors.	Caution
Ravidasvir ± Sofosbuvir	All	Available pharmacokinetic data suggest no clinically meaningful effect of PIs on ravidasvir exposure and vice versa; ARV and DAA levels remain within therapeutic ranges.	No dose adjustment is required; use standard clinical and laboratory monitoring for both HIV and HCV therapies.	Safe
Hormonal therapies				
Combined oral contraceptives (ethinylestradiol / progestin)	All boosted PIs	Boosted PIs can increase or decrease hormone levels depending on the regimen, with a net effect that may modestly reduce contraceptive efficacy or increase adverse effects.	Use additional or alternative contraception (e.g. condoms or non-hormonal methods), especially if strict pregnancy prevention is required.	Caution
DMPA	All PIs	DMPA levels are generally preserved in the presence of PIs, and contraceptive efficacy is not significantly affected.	No dose adjustment is necessary; continue routine contraceptive practice and counselling.	Safe
Immunosuppressants				
Cyclosporine, Sirolimus, Tacrolimus	All boosted PIs	Boosted PIs are potent CYP3A inhibitors and can increase levels of these immunosuppressants many-fold, leading to nephrotoxicity, neurotoxicity, and other serious adverse effects.	Use only with intensive therapeutic drug monitoring and drastic dose reductions of the immunosuppressant; close collaboration with the transplant or rheumatology team is essential.	Caution

Drug class	Affected PI	Interaction	Management	Severity
Immunosuppressants				
Everolimus	All boosted PIs	↑ immunosuppressant expected	Do not coadminister.	Avoid
Narcotics and opioid substitution				
Buprenorphine	All PIs	Bidirectional PK effects between buprenorphine and PIs are usually modest; most patients maintain stable opioid substitution control.	No routine dose adjustment is necessary; monitor for sedation or withdrawal and adjust buprenorphine dose if clinically indicated.	Caution
Methadone	All PIs	Some PIs can induce methadone metabolism, leading to reduced methadone levels and possible opioid withdrawal symptoms.	Monitor for withdrawal and increase methadone dose as needed based on clinical assessment.	Caution
PDE5 inhibitors				
Sildenafil, Tadalafil, Vardenafil	All boosted PIs	Boosted PIs markedly increase PDE5 inhibitor exposure by inhibiting CYP3A4, raising the risk of severe hypotension and priapism.	Use reduced starting doses (e.g. sildenafil 25 mg) with extended dosing intervals, and avoid use for pulmonary arterial hypertension in combination with PIs.	Caution
Statins				
Simvastatin, Lovastatin	All boosted PIs	These statins are extensively metabolised by CYP3A4; boosted PIs can cause very large increases in statin levels, with a high risk of myopathy and rhabdomyolysis.	Contraindicated with boosted PIs. Use alternative statins such as pitavastatin, atorvastatin at low doses, or rosuvastatin with dose limits.	Avoid

Drug class	Affected PI	Interaction	Management	Severity
Statins				
Atorvastatin	All boosted PIs	Boosted PIs increase exposure to atorvastatin and rosuvastatin to a moderate degree, potentially increasing the risk of statin-related myopathy	Use the lowest effective dose and adhere to guideline-recommended maximum doses; monitor for muscle symptoms and creatine kinase elevation. LPV/r: Daily dose of 20 mg for atorvastatin. ATVr & DRV/r: Daily dose should not exceed 10 mg with careful safety monitoring.	Caution
Rosuvastatin	All boosted PIs	Boosted PIs increase exposure to atorvastatin and rosuvastatin to a moderate degree, potentially increasing the risk of statin-related myopathy.	DRV/r: Use the lowest effective dose and adhere to guideline-recommended maximum doses. LPV/r & ATV/r: Do not exceed rosuvastatin 10 mg/day. Monitor for muscle symptoms and creatine kinase elevation.	Caution
Pitavastatin	All boosted PIs	Pitavastatin has minimal CYP3A metabolism and is preferred in many guidelines for patients receiving PIs due to a lower interaction potential.	No dose adjustment is generally required; use as first-line statin in many patients with HIV on PI-based ART.	Safe

Drug class	Affected PI	Interaction	Management	Severity
Others				
St John's Wort	All PIs	St John's Wort is a strong inducer of CYP3A4 and P-gp, which can substantially reduce PI concentrations and lead to virologic failure.	Do not use St John's Wort in patients receiving PIs; advise patients to avoid over-the-counter herbal preparations that contain it.	Avoid
Salmeterol	All boosted PIs	Boosted PIs can increase salmeterol exposure, and combined use may raise the risk of cardiovascular events and QT prolongation.	Avoid routine coadministration; consider alternative inhaled bronchodilators or ART regimens.	Avoid
Tecovirimat	All PIs	Tecovirimat induces CYP3A and may modestly reduce PI concentrations, although clinical data remain limited.	Use with caution, monitor HIV viral load and clinical response, and consider an alternative smallpox/mpox treatment if concerns arise.	Caution

Abbreviations: PK=Pharmacokinetics, VL=Viral Load, DAA=Direct-Acting Antiviral, SOF/VEL=Sofosbuvir/Velpatasvir, RDV=Ravidasvir.

* Severity coding: Red = Avoid; Yellow = Use with caution; Green = No adjustment required.

Annexe 5.3D • Common INSTI drug interactions

This annexe summarises clinically important drug–drug interactions involving HIV integrase strand transfer inhibitors (INSTIs), grouped by concomitant drug class. Interactions and management recommendations are based primarily on the latest U.S. DHHS HIV guidelines and the University of Liverpool interaction resources (accessed 2024–2025)

Drug class	Affected INSTI	Interaction	Management	Severity
Acid-reducing agents and cations				
Antacids (Al/Mg ± Ca)	DTG	Chelation in the gut ↓ DTG absorption and ↓ AUC 74 % (can be marked if taken together, especially fasting).	Separate DTG by ≥2 hours before or ≥6 hours after antacids to avoid reduced exposure.	Caution
H2-receptor antagonists (ranitidine, famotidine)	DTG	↑ gastric pH but DTG exposure remains essentially ↔; no clinically relevant PK change.	No dose adjustment required; may be co-administered without timing restrictions.	Safe
Proton pump inhibitors (PPIs)	DTG	PPIs do not meaningfully alter DTG levels (↔ AUC and C _{min}).	No adjustment necessary; DTG may be given with or without PPIs.	Safe
Polyvalent supplements (Ca, Fe, Mg, Zn, multivitamins with minerals)	DTG	Cations can chelate DTG → ↓ DTG exposure under fasting conditions; effect is less when DTG is taken with food, only for Ca and Fe	Under fasted conditions, dolutegravir should be administered 2 hours before or 6 hours. Alternatively, dolutegravir and supplements containing Ca or Fe only can be taken together with food	Caution
Anticonvulsants				
Carbamazepine	DTG	Strong enzyme induction via CYP3A/UGT1A1 → ↓ DTG AUC by ~49%, risking subtherapeutic levels.	Increase DTG dose to 50 mg twice daily in INSTI-naïve patients; avoid use if there is suspected or documented INSTI resistance.	Caution
Phenytoin / Phenobarbital	DTG	Potent induction → marked ↓ DTG exposure and high risk of virologic failure.	Avoid co-administration; use alternative anticonvulsants where possible.	Avoid

Drug class	Affected INSTI	Interaction	Management	Severity
Anticonvulsants				
Valproic acid	DTG	No clinically important change in DTG levels ($\leftrightarrow$); only small PK variations observed.	No dose adjustment is required; monitor seizure control and virologic response as usual.	Safe
Antidepressants / Antipsychotics				
SSRIs (e.g., sertraline, fluoxetine, paroxetine, escitalopram)	DTG	Small $\uparrow$ or $\downarrow$ in SSRI or DTG levels can occur, but overall exposure remains within therapeutic range ($\leftrightarrow$ clinically).	Use standard starting doses and titrate to effect; no routine DTG adjustment needed.	Safe
TCA's (e.g. amitriptyline, nortriptyline)	DTG	DTG has minimal effect on TCA PK ($\leftrightarrow$), but TCAs themselves carry QT and anticholinergic risk.	Start at low doses and monitor for TCA-related adverse effects; consider ECG in high-risk patients.	Caution
Antipsychotics (e.g. quetiapine, risperidone, aripiprazole)	DTG	No major PK interaction expected ($\leftrightarrow$ DTG and antipsychotic levels); effects mainly pharmacodynamic.	Use usual doses and monitor for CNS side effects and efficacy; no DTG dose adjustment required.	Safe
Antifungals				
Fluconazole	DTG	Moderate CYP3A inhibitor but has little impact on DTG exposure ($\leftrightarrow$ AUC).	No dose adjustment of DTG required; monitor hepatic function as per usual fluconazole use.	Safe
Itraconazole / Voriconazole	DTG	$\leftrightarrow$ itraconazole expected $\leftrightarrow$ voriconazole expected	Use usual antifungal dosing; monitor for azole toxicity (LFTs, QT) and overall tolerability.	Safe
Antimycobacterials				
Rifampicin	DTG	Potent inducer of UGT1A1 and CYP3A $\rightarrow$ $\downarrow$ DTG AUC by ~54% and $\downarrow$ trough concentrations.	Increase DTG dose to 50 mg twice daily in patients without INSTI resistance and monitor viral load closely.	Caution

Drug class	Affected INSTI	Interaction	Management	Severity
Antimycobacterials				
Rifabutin	DTG	Weaker inducer; only modest ↓ DTG exposure (generally within therapeutic range).	No DTG dose adjustment required; monitor virologic response, particularly in high-risk patients.	Safe
Rifapentine (e.g., 3HP regimen or 1HP regimen)	DTG	Rifapentine 900 mg Once Weekly - DTG AUC ↓ 26% and C_{min} ↓ 47% Rifapentine 600 mg Once Daily With DTG 50 mg Twice Daily vs DTG 50 mg Once Daily Alone ↔ DTG AUC and C_{min}	For 3HP, DTG 50 mg once daily may be used for those in whom once-daily DTG is appropriate. For 1HP, for a person with virologic suppression while on a DTG 50 mg once-daily regimen, the DTG dose should be increased to 50 mg twice daily throughout the course of 1HP, continuing DTG 50 mg twice daily for 14 days after 1HP completion before switching back to once-daily DTG dosing	Caution
Glucose-lowering agents				
Metformin	DTG	DTG inhibits renal transporters OCT2/ MATE1 → ↑ metformin AUC by ~79%, increasing risk of GI intolerance and lactic acidosis in susceptible patients.	Start metformin at a low dose, titrate carefully, and limit total daily dose to ≤1000 mg when used with DTG; monitor renal function and tolerability.	Caution
Other oral agents (e.g., sulfonylureas, DPP-4 inhibitors, SGLT2 inhibitors)	DTG	No clinically meaningful PK interaction is expected	No dose adjustments are required; monitor glycemic control as per routine diabetes management.	Safe

Drug class	Affected INSTI	Interaction	Management	Severity
Statins				
Atorvastatin, Rosuvastatin, Pravastatin, Pitavastatin	DTG	DTG has minimal effect on statin metabolism ($\leftrightarrow$ / mild $\uparrow$ exposure) compared with boosted PIs.	Use standard statin starting doses and titrate to lipid targets; monitor for myalgia and CK elevation if symptoms occur.	Safe
Hormonal therapies				
Combined oral contraceptives (ethinylestradiol/progestin)	DTG	DTG does not significantly change oestrogen or progestin exposure ($\leftrightarrow$). Contraceptive efficacy is maintained.	No dose adjustment of DTG or contraceptive is required; provide routine contraceptive counselling.	Safe
Depot Medroxy-progesterone Acetate and progestin	DTG	DTG shows no meaningful effect on systemic progestin levels ($\leftrightarrow$).	Use standard dosing intervals; no DTG dose modification is needed.	Safe
Others				
St John's Wort (Hypericum perforatum)	DTG	Strong inducer of CYP3A and transporters $\rightarrow$ marked $\downarrow$ DTG exposure and high risk of virologic failure.	Contraindicated; advise patients to avoid St John's Wort and herbal products containing it.	Avoid
Antiarrhythmics with QT risk (e.g. amiodarone, flecainide, propafenone)	DTG	DTG has little direct PK effect ($\leftrightarrow$), but combined use with QT-prolonging antiarrhythmics may increase overall arrhythmia risk.	Use with cardiology input where needed and perform ECG monitoring in high-risk patients; no DTG dose adjustment is required.	Caution

Abbreviations: VL = Viral Load; OCT2/MATE1 = Renal Transporters; Ca/Fe/Mg/Zn = Polyvalent Cations. AUC = Area Under the Curve; C_{min} = Minimum Concentration; C_{max} = Maximum Concentration; CYP = Cytochrome P450; UGT = UDP-glucuronosyltransferase; P-gp = P-glycoprotein; OATP = Organic Anion Transporting Polypeptide; OCT2/MATE1 = Renal Transporters; DAA = Direct-Acting Antiviral; RBV = Ribavirin; SOF = Sofosbuvir; SOF/VEL = Sofosbuvir/Velpatasvir; RDV = Ravidasvir; CCB = Calcium Channel Blocker; DOAC = Direct Oral Anticoagulant; TCA = Tricyclic Antidepressant; SSRI = Selective Serotonin Reuptake Inhibitor.

* Severity coding: Red = Avoid; Yellow = Use with caution; Green = No adjustment required.

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Annexe 8.1 • Patient Health Questionnaire

Patient Health Questionnaire-2 (PHQ-2)			
Item	Questions	Response	Score
1	Little interest or pleasure in doing things?	0 – Not at all 1 – Several days 2 – More than half the days 3 – Nearly everyday	
2	Feeling down, depressed, or hopeless?	0 – Not at all 1 – Several days 2 – More than half the days 3 – Nearly everyday	
Total score			

If the PHQ-2 brief screen result is positive (score ≥ 3), proceed with Patient Health Questionnaire-9 (PHQ-9)

Patient Health Questionnaire-9 (PHQ-9)			
Item	Questions	Response	Score
1	Little interest or pleasure in doing things?	0 – Not at all 1 – Several days 2 – More than half the days 3 – Nearly everyday	
2	Feeling down, depressed, or hopeless?	0 – Not at all 1 – Several days 2 – More than half the days 3 – Nearly everyday	
3	Trouble falling or staying asleep, or sleeping too much?	0 – Not at all 1 – Several days 2 – More than half the days 3 – Nearly everyday	
4	Feeling tired or having little energy?	0 – Not at all 1 – Several days 2 – More than half the days 3 – Nearly everyday	
5	Poor appetite or overeating?	0 – Not at all 1 – Several days 2 – More than half the days 3 – Nearly everyday	
6	Feeling bad about yourself - or that you are a failure or have let yourself or your family down?	0 – Not at all 1 – Several days 2 – More than half the days 3 – Nearly everyday	

Patient Health Questionnaire-9 (PHQ-9)			
Item	Questions	Response	Score
7	Trouble concentrating on things, such as reading the newspaper or watching television?	0 – Not at all 1 – Several days 2 – More than half the days 3 – Nearly everyday	
8	Moving or speaking so slowly that other people could have noticed? Or so fidgety or restless that you have been moving a lot more than usual?	0 – Not at all 1 – Several days 2 – More than half the days 3 – Nearly everyday	
9	Thoughts that you would be better off dead, or thoughts of hurting yourself in some way? *	0 – Not at all 1 – Several days 2 – More than half the days 3 – Nearly everyday	
Total score			

* Regardless of total score, positive score (1-3) on item 9 warrants a suicide risk assessment according to the facility protocol.¹

Recommended treatment according to severity of depression: ^{2,3}

Score	Questions	Recommended treatment actions
0 – 4	None	Repeat annual screening
5 – 9	Mild	Psychoeducation
10 – 14	Moderate	Psychosocial + psychotherapy ± pharmacotherapy
15 – 19	Moderately severe	Active psychotherapy + psychosocial + pharmacotherapy
20 – 27	Severe	Immediate initiation of pharmacotherapy with active psychotherapy

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Annexe 8.2 • Generalized Anxiety Disorder Questionnaire

Generalized Anxiety Disorder-2 (GAD-2)			
Item	Questions	Response	Score
1	Feeling nervous, anxious or on edge	0 – Not at all 1 – Several days 2 – More than half the days 3 – Nearly everyday	
2	Not being able to stop or control worrying	0 – Not at all 1 – Several days 2 – More than half the days 3 – Nearly everyday	
Total score			

If the GAD-2 brief screen result is positive (score ≥ 3), proceed with Generalized Anxiety Disorder-7 (GAD-7)

Generalized Anxiety Disorder-7 (GAD-7)			
Item	Questions	Response	Score
1	Feeling nervous, anxious or on edge	0 – Not at all 1 – Several days 2 – More than half the days 3 – Nearly everyday	
2	Not being able to stop or control worrying	0 – Not at all 1 – Several days 2 – More than half the days 3 – Nearly everyday	
3	Worrying too much about different things	0 – Not at all 1 – Several days 2 – More than half the days 3 – Nearly everyday	
4	Trouble relaxing	0 – Not at all 1 – Several days 2 – More than half the days 3 – Nearly everyday	
5	Being so restless that it is hard to sit still	0 – Not at all 1 – Several days 2 – More than half the days 3 – Nearly everyday	
6	Becoming easily annoyed or irritable	0 – Not at all 1 – Several days 2 – More than half the days 3 – Nearly everyday	

Generalized Anxiety Disorder-7 (GAD-7)			
Item	Questions	Response	Score
7	Feeling afraid as if something awful might happen	0 – Not at all 1 – Several days 2 – More than half the days 3 – Nearly everyday	
		Total score	

Recommended treatment according to severity of anxiety: ¹

Score	Questions	Recommended treatment actions
0 – 4	None	Repeat annual screening
5 – 9	Mild	Psychoeducation
10 – 14	Moderate	Psychosocial + psychotherapy ± pharmacotherapy
15 – 21	Severe	Active psychotherapy + pharmacotherapy

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1. Spitzer RL, Kroenke K, Williams JBW, Löwe B. A brief measure for assessing generalized anxiety disorder: the GAD-7. Arch Intern Med. 2006;166(10):1092–7. doi: 10.1001/archinte.166.10.1092.

Annexe 9.1 • Intrapartum Zidovudine (AZT) prophylaxis

Clinical situation (near delivery)	Intrapartum management
HIV RNA > 50 copies/mL or unknown viral load	Give IV AZT infusion starting at labour onset or ≥ 3 hours before elective caesarean
Positive rapid test during labour	Start IV AZT immediately pending confirmation
On ART, HIV RNA < 50 copies/mL within 4 weeks of birth with good adherence	IV AZT not required
HIV RNA 50 - 1,000 copies/mL	IV AZT may be considered (case-by-case; consider adherence, recent viral trends)
Not on ART or poor adherence	Start IV AZT and begin/continue full ART regimen orally

IV AZT loading dose: 2mg/kg over 1 hour followed by a continuous IV infusion of 1mg/kg/hour until delivery (clamping of the umbilical cord).

For scheduled caesarean delivery, begin IV AZT at least 3 hours before surgery.

Annexe 9.2 • Antiretroviral therapy (ART) and contraceptive drug-drug interactions

ART	Condoms	Combined Oral Contraceptive Pill (e.g., Yasmin / Microgynon)	Progestogen only pill	Depot Medroxyprogesterone Acetate injection (Depo-Provera)
NRTIs (TDF, FTC, 3TC, ABC, AZT) (incl. FDC)	R	R	R	C*
NNRTIs EFV / NVP (incl. FDC)	R	C*	C*	R
NNRTI: RPV	R	R	R	R
INSTI: DTG (incl. TLD)	R	R	R	R
Boosted Pls (ATV/r, DRV/r, LPV/r)	R	NR	NR	R

ART	Implant (Implanon NXT)	Copper IUD (e.g., Nova-T, Multi Load)	Hormonal IUD Levonorgestrel (LNG) Intrauterine System (Mirena)	Emergency pill (LNG: Postinor-2 / Escapelle)
NRTIs (TDF, FTC, 3TC, ABC, AZT) (incl. FDC)	R	R	R	R
NNRTIs EFV / NVP (incl. FDC)	NR	R	R	C#
NNRTI: RPV	R	R	R	R
INSTI: DTG (incl. TLD)	R	R	R	R
Boosted Pls (ATV/r, DRV/r, LPV/r)	NR	R	R	C#

Key:

- R = recommended (no additional contraceptive protection needed for drug–drug interaction reasons)
- C = can use, but needs caution / additional advice (see notes)
- NR = not recommended (use alternative or add reliable additional method)

Notes:

- * TDF + DMPA (bone health): No major PK interaction, but both are associated with reduced bone mineral density. Consider counselling, lifestyle advice (calcium/vitamin D, weight-bearing exercise), and choose alternatives if high fracture risk.
- + EFV/NVP + oral hormonal methods: Enzyme induction can lower hormone levels → reduced contraceptive efficacy.
 - If a woman insists on OCP/POP: advise consistent condom use and/or consider switching to IUD (copper or Mirena) or DMPA.
 - Many guidelines treat implants as inadequate with EFV → hence NR.
- # Emergency contraception (levonorgestrel) with enzyme inducers (EFV/NVP; and boosted regimens): reduced contraceptive efficacy.
 - Preferred: Copper IUD (most effective EC, if feasible).
 - If using LNG EC: consider double-dose LNG (3 mg total) as per common guidance for enzyme inducers, and advise follow-up pregnancy tests if menses are delayed.

Annexe 9.3 • Menopause Rating Scale (MRS)

Which of the following symptoms apply to you at this time? Mana satu di antara tanda-tanda berikut berlaku kepada anda pada masa ini? 请问在最近一段时间您有无下列症状?	0 None Tiada 无	1 Mild Sedikit 轻度	2 Moderate Sederhana 中度	3 Severe Teruk 严重	4 Extremely severe Sangat teruk 非常严重
Somatic domain					
Hot flushes/sweating (episodes of sweating) Rasa panas, berpeluh (Episod berpeluh) 潮热, 冒汗 (阵发性出汗)					
Cardiac complaints (unusual awareness of heartbeat, tightness) Ketidakselesaan jantung (kesedaran luar biasa denyutan jantung, jantung melompat, lumba jantung, sesak) 心脏不适 (感觉心跳异常, 漏跳, 心跳过快, 胸口发紧)					
Joint/muscle complaints (pain in the joints, rheumatoid complaints) Ketidakselesaan sendi dan otot (sakit pada sendi, aduan rheumatoid) 关节和肌肉不适 (关节痛, 症状与类风湿关节炎相似)					
Sleep disturbance (difficulty in falling asleep, waking up early) Masalah tidur (kesukaran untuk tidur, kesukaran untuk tidur berterusan, bangun awal) 睡眠问题 (入睡困难, 睡眠中经常惊醒, 早醒)					
Psychological domain					
Depressive mood (feeling down, sad, mood swings) Mood kemurungan (rasa sedih, sedih, di ambang air mata, kekurangan semangat, perubahan emosi) 情绪低落 (情绪低落, 伤感, 想哭, 缺乏活力, 情绪波动)					
Irritability (feeling nervous, inner tension, feeling aggressive) Cepat marah (rasa saraf, ketegangan dalaman, perasaan agresif) 急躁易怒 (感觉紧张, 心里总是感觉有压力, 总想与别人争吵)					
Anxiety (inner restlessness, feeling panicky) Kebimbangan (kegelisahan dalaman, rasa panik) 焦虑 (烦躁不安, 恐惧感)					

Which of the following symptoms apply to you at this time? Mana satu di antara tanda-tanda berikut berlaku kepada anda pada masa ini? 请问在最近一段时间您有无下列症状?	0 None Tiada 无	1 Mild Sedikit 轻度	2 Moderate Sederhana 中度	3 Severe Teruk 严重	4 Extremely severe Sangat teruk 非常严重
Psychological domain					
Physically and mentally exhausted (forgetfulness, decrease in concentration, general decrease in performance) Keletihan fizikal dan mental (penurunan umum dalam prestasi, memori terjejas, pengurangan dalam menumpukan perhatian, terlupa) 体力和精神上感觉非常疲劳 (做事效率整体下降, 记忆力减退, 不能集中注意力, 容易忘事)					
Urogenital domain					
Sexual problems (change in sexual desire, in sexual activity, and satisfaction) Masalah seksual (berubah dalam keinginan seks, dalam aktiviti seksual dan kepuasan) 性问题 (性欲改变, 性活动改变, 性满足感改变)					
Urinary complaints (difficulty in urinating, increased need to urinate, bladder incontinence) Masalah pundi kencing (kesukaran dalam membuang air kecil, pundi kencing inkontinen) 膀胱问题 (排尿困难, 尿意增加, 尿失禁)					
Vaginal dryness (sensation of dryness or burning of the vagina, difficulty with sexual intercourse) Kekeringan vagina (sensasi kekeringan atau kesakitan dalam faraj, masalah dalam hubungan seks) 阴道干燥 (感觉阴道干燥或烧灼, 性生活困难)					

Somatic domain 0-16 points	Psychological domain 0-16 points	Urogenital domain 0-12 points	Total 0-44 points

Result Interpretation:

- MRS Score 0-4: no / little severity
- MRS Score 5-8: mild
- MRS score 9-15: moderate impairment; consider referral
- MRS score > 16: severe symptoms; refer for specialist assessment
- Moderate / severe urogenital symptoms (e.g. urogenital domain score ≥ 4) should also prompt referral

Source:

Heinemann LAJ, Potthoff P, Schneider HPG. International versions of the Menopause Rating Scale (MRS). *Health Qual Life Outcomes*. 2003;1:28.

Mathialagan S, Ramasamy S, Nagandla K, Siew WF, Sreeramareddy CT. Menopause Rating Scale (MRS) in the Malay language-translation and validation in a multiethnic population of Selangor, Malaysia. *BMC Womens Health*. Aug 17 2022;22(1):347. doi:10.1186/s12905-022-01922-8

The Malay version is based on a validated translation in a multiethnic Malaysian population.

The Chinese version was translated by the authors for local use and has not undergone formal validation.

Published by
The Malaysian Society for HIV Medicine
c/o Department of Medicine
Hospital Sg Buloh
Jalan Hospital
47000 Sungai Buloh
<https://www.mashm.net/>
A 2026 Publication

Version 1.0